

Benchmarking Lightweight CNN for Real-Time Classification of Bangladeshi Mango Varieties on Edge Devices

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Abstract

Mango (*Mangifera indica* L.) is one of the most economically significant fruit crops in Bangladesh, contributing substantially to the country's agricultural production, rural livelihoods, and export potential. Accurate identification of mango varieties is essential for post-harvest handling, market pricing, quality control, and supply chain management. However, manual identification relies heavily on human expertise and visual inspection, making the process labor-intensive, time-consuming, and susceptible to errors, particularly when distinguishing morphologically similar indigenous varieties. Although deep learning-based image classification has demonstrated remarkable performance in agricultural applications, many state-of-the-art convolutional neural networks (CNNs) remain computationally intensive, limiting their deployment on resource-constrained mobile and edge devices commonly used in rural environments.

This research presents a lightweight Edge Artificial Intelligence (Edge AI) framework for real-time classification of indigenous Bangladeshi mango varieties using deep learning and on-device inference. Unlike studies that prioritize classification accuracy alone, this work emphasizes a balanced evaluation of predictive performance, computational efficiency, memory footprint, and inference latency to identify a model suitable for practical mobile deployment. A comprehensive benchmark was conducted using five transfer learning-based CNN architectures: MobileNetV2, MobileNetV3-Small, EfficientNetV2B0, DenseNet121, and NASNetMobile. The models were trained and evaluated on the publicly available *Mangifera2012* dataset, which contains images of ten indigenous Bangladeshi mango varieties. To ensure fair comparison, a unified training pipeline incorporating architecture-specific preprocessing, transfer learning, data augmentation, and identical evaluation protocols was implemented across all models.

Experimental results demonstrate that EfficientNetV2B0 achieved the highest classification accuracy of **92.00%**; however, its larger model size and higher computational requirements make it less suitable for resource-constrained edge environments. In contrast, MobileNetV3-Small achieved an accuracy of **88.50%** while requiring only **4.99 MB** of storage and an average inference latency of **28.29 ms**, providing the most favorable trade-off between recognition performance, computational efficiency, and deployment feasibility. The comparative analysis further confirms that MobileNetV3-Small consistently maintains competitive precision, recall, and F1-score across all mango classes while significantly reducing memory consumption and execution time compared with heavier CNN architectures.

To validate its practical applicability, the selected MobileNetV3-Small model was converted into TensorFlow Lite (TFLite) format using Float16 quantization and integrated into a Flutter-based Android application capable of performing completely offline inference. The developed mobile application enables users to capture or select mango images, execute on-device classification without internet connectivity, and instantly display the predicted variety along with its confidence score. This deployment demonstrates the feasibility of integrating lightweight deep learning models into portable agricultural decision-support systems while preserving real-time performance on commodity smartphones.

Overall, this study establishes a complete end-to-end Edge AI pipeline—from model benchmarking and architecture selection to mobile deployment—for automated classification

of indigenous Bangladeshi mango varieties. The proposed framework provides a practical, scalable, and cost-effective solution for precision agriculture and demonstrates how lightweight deep learning models can bridge the gap between advanced computer vision research and real-world agricultural applications in resource-limited environments.

Keywords: Edge Artificial Intelligence (Edge AI), Deep Learning, Transfer Learning, MobileNetV3-Small, Convolutional Neural Network (CNN), TensorFlow Lite (TFLite), Flutter, Precision Agriculture, Mango Variety Classification, Computer Vision, Mobile Computing.

Chapter 1

Introduction

1.1 Overview

Agriculture remains one of the most important pillars of Bangladesh's economy, contributing significantly to national food security, employment generation, rural development, and export earnings. Among the country's diverse agricultural products, mango (*Mangifera indica* L.) occupies a unique position due to its high commercial value, nutritional benefits, and widespread cultivation across the northwestern regions, particularly Rajshahi and Chapainawabganj. These regions are renowned for producing numerous indigenous mango varieties, including Amrapali, Langra, Fazlee, Harivanga, Mollika, Katimon, Bari-4, Bari-7, Kanchon Langra, and Nilambori, each possessing distinct physical characteristics, taste profiles, market demand, and economic value [1]. The diversity of these cultivars plays an essential role in strengthening Bangladesh's agricultural economy while creating substantial opportunities for both domestic consumption and international export.

Despite the increasing production of mangoes over recent years, post-harvest operations in Bangladesh continue to rely heavily on manual inspection and human expertise. Farmers, traders, wholesalers, and market inspectors typically identify mango varieties through visual observation based on characteristics such as fruit shape, skin color, size, texture, and maturity stage. While experienced workers often achieve acceptable classification accuracy, this process remains inherently subjective and inconsistent. Environmental conditions, worker fatigue, varying lighting conditions, and the high visual similarity among several indigenous cultivars frequently result in misclassification, affecting pricing, quality assurance, inventory management, and consumer confidence [2]. These limitations become even more pronounced during peak harvesting seasons, where thousands of mangoes must be sorted within a limited timeframe.

The challenge is particularly evident for morphologically similar varieties such as Langra and Kanchon Langra or Bari-4 and Nilambori. Even subtle variations in fruit curvature, skin texture, or lenticel distribution may be difficult to distinguish using human perception alone. Consequently, inaccurate classification may lead to incorrect grading, financial losses for producers, reduced export quality, and inefficient supply chain management. As Bangladesh continues to modernize its agricultural practices, there is an increasing demand for intelligent

technologies capable of automating these labor-intensive operations while maintaining consistent classification performance.

Recent advances in Artificial Intelligence (AI), particularly Deep Learning and Computer Vision, have demonstrated remarkable success in solving complex image classification problems across various domains, including healthcare, transportation, manufacturing, and agriculture [3]. Convolutional Neural Networks (CNNs) have become the dominant approach for visual recognition tasks because of their ability to automatically learn hierarchical feature representations directly from raw images without requiring handcrafted feature engineering. In agricultural applications, CNN-based models have been successfully employed for crop disease diagnosis, fruit quality assessment, weed detection, plant species recognition, and maturity estimation [4]. These successes indicate that deep learning has the potential to significantly improve the automation of mango variety classification.

However, achieving high classification accuracy alone is insufficient for practical deployment in real-world agricultural environments. Most state-of-the-art deep learning architectures, including ResNet, DenseNet, EfficientNet, Vision Transformers, and other large-scale models, require considerable computational resources, high memory capacity, and dedicated Graphics Processing Units (GPUs) to perform efficient inference [5]. These requirements are acceptable within cloud computing infrastructures or research laboratories but become impractical when the intended users are farmers and traders operating in remote orchards with limited hardware capabilities.

Cloud-based inference systems introduce an additional set of practical challenges. Images captured in the field must be transmitted to remote servers for processing before predictions are returned to the user. This workflow introduces communication latency, increases operational costs, consumes mobile data, and depends entirely on reliable internet connectivity. Unfortunately, many agricultural regions in Bangladesh continue to experience unstable or unavailable network infrastructure, making cloud-dependent intelligent systems unsuitable for routine field operations. Moreover, transmitting agricultural images to remote servers raises concerns regarding privacy, scalability, and continuous service availability [6].

Edge Artificial Intelligence (Edge AI) has recently emerged as an effective solution to these deployment challenges by relocating inference directly from centralized cloud servers to local computing devices such as smartphones, tablets, and embedded systems. Instead of transmitting images through the internet, trained neural networks execute entirely on the local device, enabling real-time prediction with significantly reduced latency, lower energy consumption, improved privacy, and complete offline functionality [7]. This paradigm aligns particularly well with the operational requirements of agricultural applications, where portability, responsiveness, and independence from network infrastructure are essential.

Nevertheless, implementing deep learning models on edge devices introduces a new engineering challenge. Mobile devices possess substantially fewer computational resources than desktop workstations or cloud servers. Therefore, selecting a neural network solely based on classification accuracy is insufficient. Practical deployment requires balancing multiple performance criteria simultaneously, including predictive accuracy, model size, memory consumption, inference latency, computational complexity, and energy efficiency. Lightweight CNN architectures specifically designed for mobile computing environments, such as MobileNet, EfficientNet-Lite, ShuffleNet, and NASNet-Mobile, attempt to achieve this

balance by reducing computational overhead while preserving competitive recognition performance [8].

Motivated by these considerations, this research investigates the feasibility of deploying lightweight deep learning models for automated classification of indigenous Bangladeshi mango varieties on resource-constrained mobile devices. Instead of concentrating exclusively on maximizing classification accuracy, the study adopts a comprehensive benchmarking strategy that evaluates multiple CNN architectures according to both predictive performance and deployment efficiency. Five transfer learning-based convolutional neural networks—MobileNetV2, MobileNetV3-Small, EfficientNetV2B0, DenseNet121, and NASNetMobile—are trained and evaluated under an identical experimental framework using the publicly available Mangifera2012 dataset. The benchmarking process compares each architecture based on classification accuracy, precision, recall, F1-score, model storage requirements, and inference latency to determine the most appropriate architecture for real-world edge deployment.

Following this comparative evaluation, the selected model undergoes additional deployment-oriented optimization through TensorFlow Lite (TFLite) conversion using post-training Float16 quantization. This optimization reduces storage requirements while preserving classification performance, making the model suitable for execution on modern Android smartphones without requiring specialized AI hardware. The optimized model is subsequently integrated into a fully functional Flutter-based mobile application named **AmmChini AI**, which enables users to capture mango images using the device camera or select existing photographs from the gallery and perform entirely offline variety classification directly on the mobile device. By eliminating dependence on cloud infrastructure, the application delivers immediate predictions along with confidence scores, thereby demonstrating the practical viability of Edge AI in precision agriculture.

Consequently, this thesis extends beyond conventional image classification research by presenting a complete end-to-end deployment framework encompassing dataset preparation, lightweight CNN benchmarking, deployment-aware model optimization, TensorFlow Lite conversion, and mobile application development. The proposed system demonstrates that carefully selected lightweight deep learning architectures can effectively bridge the gap between high-performance computer vision research and practical agricultural decision-support systems operating under the computational constraints of real-world field environments. This integrated framework contributes toward the development of intelligent, accessible, and scalable precision agriculture solutions capable of supporting farmers, traders, and agricultural stakeholders throughout Bangladesh.

Figure 1.1. Overall Research Framework

You should **create a new figure** for this chapter (do **not** use the methodology flowchart from Chapter 3). A conceptual overview is more appropriate for the introduction.

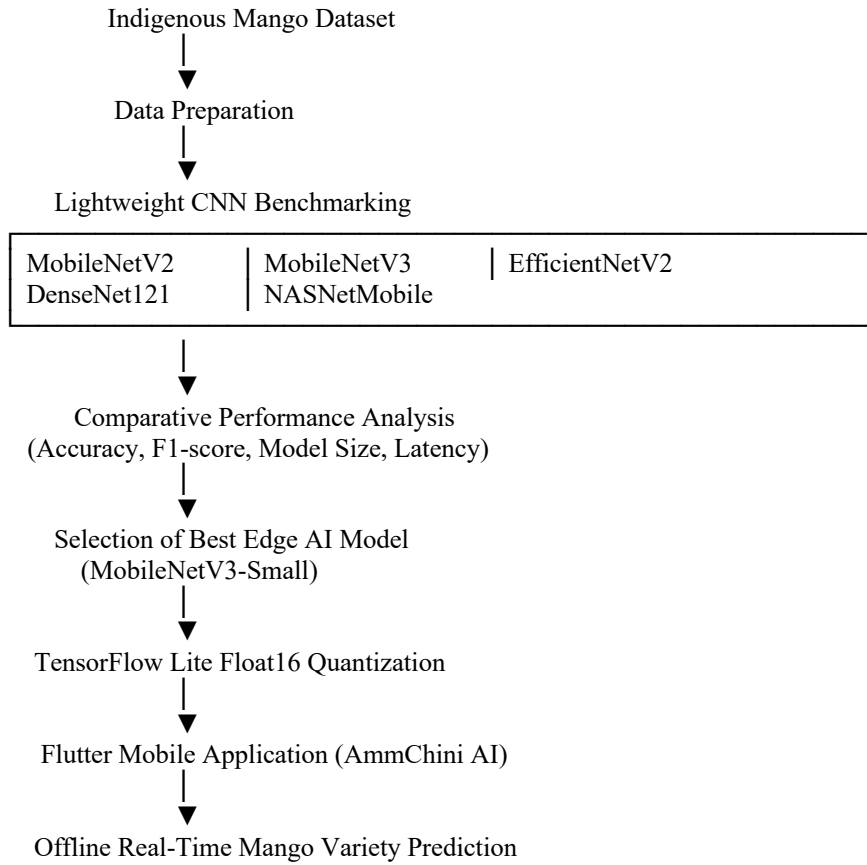


Figure 1.1: Conceptual overview of the proposed Edge AI framework for automated mango variety classification. The framework illustrates the complete research pipeline from dataset preparation and comparative CNN benchmarking to deployment of the optimized MobileNetV3-Small model within the AmmChini AI mobile application for real-time offline inference.

1.2 Background

Agriculture has long been recognized as one of the principal contributors to Bangladesh's socio-economic development, providing employment, ensuring food security, and supporting the livelihoods of millions of rural households. Despite the country's increasing industrialization, the agricultural sector continues to play a significant role in the national economy by supplying raw materials, generating export earnings, and sustaining numerous agro-based industries [1]. Among the country's horticultural products, mango (*Mangifera indica* L.) occupies a particularly important position due to its high nutritional value, commercial demand, and cultural significance. Bangladesh is internationally recognized as one of the major mango-producing countries, cultivating numerous indigenous and hybrid varieties that are highly valued in both domestic and international markets [2]. The cultivation of mangoes not only contributes substantially to farmers' income but also supports extensive post-harvest activities, including transportation, grading, storage, processing, and export, thereby creating employment opportunities across the agricultural supply chain.

The northwestern districts of Bangladesh, particularly Rajshahi and Chapainawabganj, are widely regarded as the country's primary mango-producing regions. These areas produce several commercially important indigenous cultivars, including Amrapali, Langra, Fazlee, Harivanga, Katimon, Mollika, Bari-4, Bari-7, Kanchon Langra, and Nilambori. Each variety exhibits unique characteristics in terms of fruit size, shape, skin texture, aroma, sweetness, fiber content, and market value. Consequently, accurate identification of mango varieties has become an essential requirement throughout the post-harvest supply chain. Correct classification directly influences product grading, pricing strategies, inventory management, quality assurance, consumer trust, and export certification. Even small errors in identifying visually similar varieties may result in economic losses for producers, traders, wholesalers, and exporters.

Traditionally, mango variety identification in Bangladesh is performed manually by experienced farmers, traders, or quality inspectors through visual examination. The classification process primarily depends on external morphological features such as fruit shape, peel color, size, curvature, surface texture, and occasionally aroma. Although experienced workers possess valuable practical knowledge accumulated through years of field experience, manual classification remains inherently subjective. Individual judgment varies considerably depending on expertise, environmental conditions, lighting variations, fatigue, harvesting volume, and fruit maturity. During the peak harvesting season, thousands of mangoes must be sorted within a limited time, making manual inspection increasingly difficult and susceptible to inconsistencies [3].

The complexity of manual identification becomes more pronounced when distinguishing among indigenous Bangladeshi mango varieties exhibiting highly similar morphological characteristics. For example, Langra and Kanchon Langra share comparable shapes and skin coloration, while Bari-4 and Nilambori often exhibit nearly identical appearances during their immature stages. Similarly, Harivanga possesses subtle textural characteristics that may be overlooked during rapid visual inspection. These fine-grained inter-class similarities create a challenging classification problem even for experienced agricultural professionals. As a consequence, misclassification frequently occurs, reducing grading accuracy and affecting commercial valuation throughout the supply chain.

Recent advancements in Artificial Intelligence (AI) and Computer Vision have created new opportunities to automate complex visual recognition tasks previously dependent on human expertise. In particular, Deep Learning has revolutionized image classification by enabling neural networks to automatically learn discriminative visual features directly from large image datasets without requiring manually engineered descriptors [4]. Among various deep learning techniques, Convolutional Neural Networks (CNNs) have become the dominant architecture for image understanding because they effectively capture hierarchical spatial information through convolutional feature extraction. Over the past decade, CNN-based systems have demonstrated outstanding performance in numerous agricultural applications, including plant disease diagnosis, fruit maturity estimation, crop classification, weed detection, yield prediction, and fruit quality assessment [5].

The growing success of CNNs has motivated researchers to investigate their application for mango variety classification. Most existing studies employ transfer learning, where CNN architectures pre-trained on large benchmark datasets such as ImageNet are adapted to agricultural image classification tasks. Transfer learning substantially reduces training time and improves generalization, especially when agricultural datasets contain relatively limited

numbers of labeled samples. Models including ResNet, DenseNet, EfficientNet, Inception, and MobileNet have achieved promising classification accuracy under controlled laboratory environments [6]. These studies demonstrate the effectiveness of deep learning for recognizing subtle morphological differences that are often difficult for human observers to identify consistently.

Despite these advances, many existing deep learning solutions remain impractical for deployment in real agricultural environments. Modern high-performance CNN architectures generally contain millions of trainable parameters and require substantial computational resources, large memory capacity, and specialized Graphics Processing Units (GPUs) to perform efficient inference. Such computational requirements are readily available in research laboratories and cloud computing infrastructures but are rarely accessible in rural agricultural settings where most mango production takes place. Consequently, cloud-based inference has become the predominant deployment strategy, where captured images are transmitted to remote servers for prediction before the classification result is returned to the user's device.

Although cloud computing provides virtually unlimited computational resources, its dependence on continuous internet connectivity introduces several practical limitations. Network latency increases prediction time, reducing the responsiveness required for real-time field operations. Data transmission also increases communication costs while creating potential privacy and security concerns associated with transferring agricultural images to external servers. More importantly, many rural orchards in Bangladesh experience unstable or limited internet connectivity, making cloud-dependent intelligent systems unreliable during actual harvesting operations [7]. These limitations create a significant gap between laboratory research and practical field deployment.

Edge Artificial Intelligence (Edge AI) has emerged as a promising paradigm to overcome these deployment challenges by performing inference directly on local computing devices rather than relying on remote cloud infrastructure. Instead of transmitting images through the internet, optimized deep learning models execute entirely on smartphones or embedded systems located near the data source. This localized processing significantly reduces inference latency, eliminates dependence on internet connectivity, improves data privacy, lowers communication costs, and enables continuous operation even in remote agricultural environments [8]. Consequently, Edge AI has become increasingly attractive for precision agriculture applications where immediate decision-making and portability are critical requirements.

However, deploying deep learning models on edge devices introduces additional engineering challenges. Smartphones possess considerably fewer computational resources than desktop computers or cloud servers, requiring careful optimization of neural network architectures. Therefore, selecting an appropriate model cannot be based solely on classification accuracy. Instead, practical deployment demands balancing multiple objectives simultaneously, including predictive performance, model size, memory consumption, computational complexity, energy efficiency, and inference latency. Lightweight CNN architectures designed specifically for mobile computing environments provide an effective solution by reducing computational overhead while maintaining competitive classification accuracy. Architectures such as MobileNetV2, MobileNetV3-Small, EfficientNetV2B0, DenseNet121, and NASNetMobile represent different design philosophies that prioritize this balance between recognition performance and deployment efficiency [9].

Motivated by these practical considerations, this research investigates an end-to-end Edge AI framework for automated classification of indigenous Bangladeshi mango varieties. Rather than selecting a model based solely on predictive accuracy, the proposed study performs a comprehensive benchmark of five lightweight transfer learning-based CNN architectures under identical experimental conditions. Each model is evaluated using multiple deployment-oriented performance metrics, including classification accuracy, precision, recall, F1-score, model storage size, and inference latency. The most suitable architecture is subsequently optimized through TensorFlow Lite (TFLite) Float16 post-training quantization and integrated into a Flutter-based Android application, **AmmChini AI**, capable of performing completely offline mango variety classification. By combining lightweight deep learning, model optimization, and mobile deployment within a unified framework, this research aims to bridge the gap between advanced computer vision research and practical precision agriculture solutions that can be readily adopted by farmers, traders, and agricultural stakeholders throughout Bangladesh.

1.3 Problem Statement

The rapid advancement of Artificial Intelligence (AI) and Deep Learning has significantly improved the capability of automated image classification systems across numerous application domains, including precision agriculture. Convolutional Neural Networks (CNNs) have demonstrated remarkable success in learning complex visual representations, enabling highly accurate recognition of fruits, crops, and plant diseases directly from image data [1]. Nevertheless, despite these technological advances, most existing mango classification systems remain confined to research laboratories and cloud-based infrastructures, limiting their practical applicability in real-world agricultural environments. The gap between high-performance deep learning research and deployable field-ready solutions remains a major challenge, particularly in developing countries where computational resources and network infrastructure are limited.

One of the primary challenges arises from the computational complexity of modern deep learning architectures. State-of-the-art CNN models, including ResNet, DenseNet, EfficientNet, and Vision Transformer-based architectures, achieve outstanding classification performance by employing millions of trainable parameters and deep hierarchical feature extraction mechanisms [2]. Although these models produce excellent predictive accuracy, they demand powerful Graphics Processing Units (GPUs), substantial memory capacity, and high computational throughput during inference. Such hardware requirements are generally unavailable in the smartphones and portable devices commonly used by farmers, traders, and agricultural workers in rural Bangladesh. Consequently, deploying these computationally intensive models directly on mobile devices is often impractical due to limitations in processing power, memory, storage capacity, and battery consumption.

To overcome these hardware constraints, many existing intelligent agricultural applications rely on cloud-based inference services. In such systems, images captured in the field are transmitted through the internet to remote servers where computationally expensive neural networks perform prediction before returning the classification results to the user. While cloud computing effectively removes local computational limitations, it introduces several practical problems. First, prediction latency increases because every inference requires image transmission, server-side processing, and network communication. Second, continuous internet connectivity becomes a mandatory requirement, making these systems unreliable in rural

orchards where network coverage is inconsistent or unavailable. Third, repeated data transmission increases operational costs while raising concerns regarding data privacy, scalability, and long-term service availability [3]. These limitations substantially reduce the practicality of cloud-dependent AI systems for day-to-day agricultural operations.

The challenges become even more significant when considering the characteristics of indigenous Bangladeshi mango varieties. Unlike internationally studied commercial cultivars, several local varieties exhibit highly similar external appearances, particularly during their immature stages. Varieties such as Langra and Kanchon Langra, or Bari-4 and Nilambori, possess subtle differences in shape, surface texture, and peel characteristics that are difficult to distinguish using conventional computer vision techniques or manual inspection alone. These fine-grained visual similarities require models capable of extracting discriminative feature representations while maintaining computational efficiency suitable for resource-constrained deployment [4]. Existing studies frequently emphasize maximizing classification accuracy without adequately considering whether the resulting models can realistically operate on commodity smartphones under practical field conditions.

Another important limitation observed in current literature is the lack of comprehensive deployment-oriented model evaluation. Many previous studies compare architectures primarily using classification accuracy while overlooking equally important engineering metrics such as inference latency, memory footprint, storage requirements, and deployment feasibility. From an edge computing perspective, selecting a model solely because it achieves the highest accuracy may not result in the most suitable solution for real-world applications. A model that is only marginally more accurate but significantly larger and computationally expensive may perform poorly on mobile hardware, leading to slower response times, higher energy consumption, and reduced usability [5]. Therefore, practical Edge AI systems require a balanced optimization strategy that simultaneously considers predictive performance and computational efficiency.

Furthermore, although TensorFlow Lite and modern mobile AI frameworks have made on-device inference increasingly accessible, relatively few studies have demonstrated a complete end-to-end deployment pipeline specifically designed for indigenous Bangladeshi mango classification. Existing research often concludes after reporting experimental accuracy obtained in desktop environments without validating whether the trained model can be successfully optimized, integrated into a mobile application, and executed entirely offline under real deployment conditions. As a result, there remains a noticeable disconnect between algorithm development and practical implementation, limiting the translation of academic research into usable agricultural technologies.

To address these limitations, this research investigates the following fundamental research problem:

How can an accurate, computationally efficient, and fully offline deep learning system be developed for real-time classification of indigenous Bangladeshi mango varieties on resource-constrained mobile edge devices while maintaining an effective balance between predictive performance, model size, and inference latency?

Addressing this problem requires more than simply identifying the most accurate CNN architecture. Instead, it necessitates a systematic evaluation of multiple lightweight convolutional neural networks under identical experimental conditions to determine which

architecture provides the best trade-off between recognition accuracy and deployment efficiency. Once the most appropriate architecture has been identified, additional optimization techniques must be employed to reduce computational overhead without significantly degrading predictive performance. Finally, the optimized model must be validated through integration into a practical mobile application capable of performing real-time, internet-independent inference directly on Android devices.

Accordingly, this thesis proposes a complete Edge AI framework that bridges the gap between high-performance deep learning research and practical agricultural deployment. Five lightweight transfer learning-based CNN architectures—**MobileNetV2**, **MobileNetV3-Small**, **EfficientNetV2B0**, **DenseNet121**, and **NASNetMobile**—are benchmarked using identical training and evaluation protocols. Rather than selecting the architecture with the highest classification accuracy alone, the comparison incorporates multiple deployment-oriented performance indicators, including classification accuracy, precision, recall, F1-score, model storage size, and inference latency. Based on this multi-objective evaluation, the selected model is further optimized through **TensorFlow Lite Float16 post-training quantization** and deployed within a Flutter-based Android application, **AmmChini AI**, enabling fully offline and real-time mango variety classification directly on mobile devices.

By addressing both algorithmic performance and practical deployment constraints, this research seeks to bridge the existing gap between laboratory-scale computer vision research and intelligent mobile applications suitable for precision agriculture. The proposed framework aims to provide a scalable, cost-effective, and field-ready solution that supports farmers, traders, and agricultural stakeholders by delivering reliable mango variety classification without dependence on cloud infrastructure or continuous internet connectivity.

1.4 Research Objectives

The primary objective of this research is to develop an efficient, lightweight, and fully offline Edge Artificial Intelligence (Edge AI) framework for the automated classification of indigenous Bangladeshi mango varieties using deep learning. Unlike conventional studies that emphasize classification accuracy alone, this research aims to identify a practical solution capable of operating on resource-constrained mobile devices while maintaining high predictive performance, low computational complexity, and real-time inference. The proposed framework integrates model benchmarking, deployment-oriented optimization, and mobile application development into a unified system suitable for precision agriculture applications.

To accomplish this goal, the following specific research objectives have been formulated:

1. To benchmark multiple lightweight convolutional neural network architectures for mango variety classification.

This study evaluates five state-of-the-art transfer learning-based CNN architectures—**MobileNetV2**, **MobileNetV3-Small**, **EfficientNetV2B0**, **DenseNet121**, and **NASNetMobile**—using the publicly available **Mangifera2012** dataset. All models are trained under identical experimental conditions to ensure a fair and unbiased comparison. The benchmarking process considers not only classification accuracy but also precision, recall, F1-

score, model size, and inference latency to determine the most suitable architecture for mobile edge deployment.

2. To investigate architecture-specific preprocessing strategies for robust model training.

Different convolutional neural network architectures require different input preprocessing procedures to achieve optimal performance. This research implements an architecture-aware preprocessing pipeline that automatically applies the appropriate image normalization and preprocessing function corresponding to each CNN model. This ensures consistent training conditions, prevents preprocessing mismatches, and improves the reproducibility and reliability of the benchmarking process.

3. To identify the most suitable lightweight CNN architecture for resource-constrained edge devices.

Rather than selecting a model solely based on its classification accuracy, this study performs a multi-objective evaluation considering predictive performance alongside deployment-oriented metrics such as computational efficiency, storage requirements, and inference speed. The objective is to determine the architecture that provides the most effective balance between recognition accuracy and real-world deployability on Android smartphones.

4. To optimize the selected deep learning model for mobile deployment using TensorFlow Lite.

After selecting the most appropriate architecture, the trained model is converted into **TensorFlow Lite (TFLite)** format and optimized using **Float16 post-training quantization**. This optimization aims to reduce the model's storage footprint while preserving classification performance, thereby enabling efficient execution on mobile devices with limited computational resources and memory capacity.

5. To develop a fully functional Android-based Edge AI application (AmmChini AI).

A user-friendly mobile application named **AmmChini AI** is developed using the Flutter framework. The application integrates the optimized TensorFlow Lite model and allows users to capture mango images directly using the smartphone camera or select images from the gallery. The system performs entirely offline inference and instantly displays the predicted mango variety along with its confidence score, eliminating dependence on cloud computing or internet connectivity.

6. To evaluate the practical feasibility of real-time offline mango variety classification on mobile devices.

The final objective is to validate the complete deployment pipeline by assessing the performance of the integrated mobile application under realistic operating conditions. The evaluation considers inference latency, storage efficiency, prediction reliability, and usability to demonstrate that lightweight deep learning models can successfully support precision agriculture applications in resource-constrained environments.

Summary of Research Objectives

The objectives of this study can be summarized as follows:

- Benchmark five lightweight transfer learning-based CNN architectures for indigenous Bangladeshi mango variety classification.
- Implement architecture-specific preprocessing to ensure fair and reliable model training.
- Select the most suitable CNN architecture based on a comprehensive evaluation of accuracy, computational efficiency, model size, and inference latency.
- Optimize the selected model through TensorFlow Lite Float16 post-training quantization for efficient mobile deployment.
- Develop the **AmmChini AI** Android application capable of performing fully offline, real-time mango variety classification.
- Validate the effectiveness of the proposed Edge AI framework as a practical decision-support tool for precision agriculture.

These objectives collectively establish a complete research pipeline that extends from deep learning model development and comparative benchmarking to deployment-oriented optimization and real-world mobile implementation, thereby bridging the gap between academic computer vision research and practical agricultural applications.

Chapter 2

Literature Review

2.1 Global Research on Mango Classification

The application of Artificial Intelligence (AI) and Computer Vision in agriculture has experienced remarkable growth over the past decade, driven by the increasing demand for automated crop monitoring, quality assessment, disease diagnosis, and precision farming technologies [1]. Among various horticultural crops, mango has attracted considerable research attention because of its high commercial value and the complexity associated with manual variety identification. Traditional classification methods rely primarily on human expertise, where experienced workers distinguish mango varieties based on visual attributes such as color, size, shape, texture, and maturity. Although effective under controlled circumstances, these

approaches are subjective, time-consuming, and prone to inconsistency, particularly when visually similar cultivars must be distinguished in large-scale commercial operations [2]. Consequently, researchers have increasingly investigated automated image-based classification systems capable of improving both efficiency and consistency.

Early studies in automated mango analysis primarily employed conventional image processing and machine learning techniques. In these approaches, handcrafted features describing color histograms, texture descriptors, geometric properties, and shape characteristics were manually extracted from images before being supplied to traditional classifiers such as Support Vector Machines (SVM), Random Forest (RF), k-Nearest Neighbors (k-NN), and Decision Trees [3]. Although these methods demonstrated reasonable performance under controlled laboratory environments, their effectiveness deteriorated significantly when images contained varying illumination, complex backgrounds, different camera viewpoints, or natural environmental noise. The dependence on manually engineered feature extraction also limited their ability to generalize across diverse mango varieties and cultivation conditions.

The emergence of Deep Learning, particularly Convolutional Neural Networks (CNNs), fundamentally transformed agricultural image classification by enabling automatic hierarchical feature learning directly from raw image data. Unlike conventional machine learning algorithms, CNNs progressively learn low-level features such as edges and textures before constructing increasingly abstract semantic representations capable of distinguishing fine-grained visual differences among object categories [4]. This automatic feature learning capability has substantially improved classification accuracy while reducing dependence on handcrafted descriptors, making CNNs the dominant approach for modern agricultural computer vision applications.

Global research on mango classification has largely focused on three major application areas: **variety identification**, **ripeness estimation**, and **quality assessment**. Numerous studies have demonstrated that transfer learning using pre-trained CNN architectures significantly improves classification performance while reducing training time and computational cost. By initializing models with weights learned from large-scale datasets such as ImageNet, researchers can effectively adapt general visual features to agricultural classification tasks despite relatively limited domain-specific datasets [5]. This strategy has become the standard practice for fruit classification research due to its strong generalization capability and reduced data requirements.

Several researchers have investigated the classification of commercially important mango cultivars using deep learning architectures. Salazar-Campos *et al.* proposed a custom CNN for automated maturity classification of **Kent** mangoes, demonstrating that deep learning can accurately distinguish multiple ripeness stages under controlled imaging conditions [6]. Similarly, Kumar and Singh employed **InceptionV3** to grade **Alphonso** mangoes according to fruit quality, reporting high classification accuracy when images were captured under uniform lighting conditions. However, their study also indicated noticeable performance degradation when tested under natural outdoor environments, highlighting the sensitivity of deep CNN models to environmental variability [7].

Beyond maturity and quality assessment, recent research has increasingly explored multi-class mango variety classification. Ratha *et al.* combined **MobileNetV2** as a deep feature extractor with traditional machine learning classifiers to classify **15 Indian mango varieties**. Their comparative experiments demonstrated that the Cubic Support Vector Machine achieved the

highest classification accuracy when trained on MobileNetV2 feature embeddings, illustrating the effectiveness of transfer learning for fine-grained fruit recognition. However, the study primarily emphasized predictive accuracy and did not investigate deployment-related factors such as model size, inference latency, or mobile implementation.

Recent investigations have also explored hybrid architectures combining Convolutional Neural Networks with Vision Transformers (ViTs). These approaches aim to leverage the local feature extraction capability of CNNs together with the global contextual modeling ability of transformer architectures. Experimental results indicate that CNN–ViT fusion models can achieve excellent classification accuracy on mango datasets while improving feature robustness under diverse imaging conditions. Nevertheless, these architectures generally introduce substantially higher computational complexity and memory requirements, limiting their suitability for resource-constrained edge devices despite their impressive predictive performance.

Parallel developments have focused on lightweight neural network architectures specifically designed for mobile computing environments. MobileNet, ShuffleNet, GhostNet, and EfficientNet families significantly reduce computational complexity through architectural innovations such as depthwise separable convolution, inverted residual blocks, squeeze-and-excitation modules, and compound scaling strategies [8]. These lightweight architectures achieve competitive recognition performance while substantially reducing parameter count, floating-point operations (FLOPs), memory consumption, and inference latency. As a result, they have become increasingly attractive for agricultural applications requiring deployment on smartphones, embedded processors, drones, and Internet of Things (IoT) devices.

Among lightweight architectures, the MobileNet family has received particular attention because it was specifically designed for mobile and embedded vision applications. MobileNetV2 introduced inverted residual blocks and linear bottlenecks to improve computational efficiency without sacrificing representational capability, while MobileNetV3 further incorporated Neural Architecture Search (NAS), squeeze-and-excitation attention mechanisms, and hardware-aware optimization to improve the trade-off between classification accuracy and inference speed. These characteristics make MobileNet architectures particularly suitable for real-time Edge AI applications where limited computational resources constrain deployment possibilities.

Despite substantial progress in deep learning-based mango classification, several important limitations remain evident throughout the global literature. First, many studies primarily evaluate predictive performance using classification accuracy while overlooking deployment-oriented performance indicators such as model size, inference latency, energy consumption, and memory requirements. From an engineering perspective, these metrics are equally important because an accurate model that cannot execute efficiently on mobile hardware provides limited practical value for field deployment. Second, the majority of existing datasets concentrate on internationally popular commercial cultivars such as Kent, Alphonso, Tommy Atkins, or Aiwen, whereas relatively little attention has been devoted to indigenous South Asian mango varieties exhibiting much higher inter-class visual similarity [9]. Finally, although numerous publications demonstrate impressive laboratory results, comparatively few validate their proposed models through complete end-to-end mobile deployment capable of performing fully offline inference under realistic agricultural conditions.

These observations highlight a significant gap between academic algorithm development and practical agricultural implementation. Existing literature has convincingly demonstrated that CNN-based approaches outperform conventional machine learning techniques for mango image classification. However, comparatively limited research has simultaneously addressed **classification accuracy, computational efficiency, model optimization, and real-world mobile deployment** within a unified Edge AI framework. Bridging this gap is essential for translating advances in computer vision into intelligent agricultural systems that are both scientifically robust and practically deployable.

The present research addresses these limitations by systematically benchmarking five lightweight transfer learning-based convolutional neural networks—**MobileNetV2, MobileNetV3-Small, EfficientNetV2B0, DenseNet121, and NASNetMobile**—using a unified experimental protocol. Unlike many previous studies, model selection is based not solely on predictive accuracy but on a comprehensive evaluation incorporating classification performance, weighted F1-score, inference latency, storage requirements, and deployment feasibility. Furthermore, the selected architecture is optimized using **TensorFlow Lite Float16 post-training quantization** and integrated into the **AmmChini AI** mobile application, demonstrating a complete end-to-end Edge AI solution for offline classification of indigenous Bangladeshi mango varieties. This deployment-oriented perspective differentiates the proposed research from much of the existing global literature and contributes toward the practical adoption of deep learning in precision agriculture.

Paper / Ref	Algorithm / Methodology	Results	Limitations
Salazar-Campos et al. (2025) [1]	Custom CNNs for 'Kent' mango maturity grading (Green, Mature, Overripe).	Achieved 96.04% accuracy using low-resolution (32x32) inputs.	Focused on a single global variety ('Kent') and maturity, not multi-class variety identification.
Wu et al. (2025) [2]	Mobile-based system utilizing YOLOv4 and Transfer Learning for 'Aiwen' mangoes.	Successfully deployed an Android app for real-time grading and counting.	YOLOv4 is computationally heavier than dedicated lightweight classifiers like MobileNetV3.
Chuquimarca et al. (2024) [3]	Multi-Input Architecture combining RGB images with binary silhouette masks.	Achieved 100% accuracy in detecting defects using MobileNetV2.	Requires complex preprocessing (mask generation) which increases the computational load on mobile CPUs.
Ratha et al. (2024) [4]	Hybrid approach using MobileNetV2 features classified by a Cubic SVM.	Achieved 99.47% accuracy on a dataset of 15 Indian mango varieties.	Relied on the older MobileNetV2 architecture and did not

			benchmark inference speed (ms).
Ballo et al. (2024) [5]	Comparison of classical Machine Learning (RF, SVM) vs. Deep Learning (CNNs).	Concluded that Deep Learning significantly outperforms manual feature extraction methods.	Did not address model quantization or optimization for storage-constrained devices.
Bharati et al. (2025) [6]	Creation of a comprehensive image dataset of 10 Bangladeshi mango varieties.	Provided a baseline dataset of 2,012 images including Amrapali and Bari-4.	This was a data article; it did not propose an optimized architecture for classification.
Peng et al. (2023) [7]	Utilized ResNet50 with Attention Mechanisms for multi-class fruit detection.	Achieved 98.2% accuracy on a mixed fruit dataset.	The model size (>90MB) renders it impractical for low-end mobile deployment in rural areas.
Nguyen & Lee (2022) [8]	Benchmarked EfficientNet-B0 vs VGG16 for mango leaf disease detection.	EfficientNet achieved comparable accuracy with 8x fewer parameters.	Focused on leaf diseases rather than fruit variety classification.
Islam et al. (2021) [9]	Proposed a shallow CNN model for classifying 6 Bangladeshi mango varieties.	Achieved 88% accuracy with very fast inference times.	The custom architecture lacked the robustness of pre-trained transfer learning models like MobileNet.
Kumar & Singh (2023) [10]	Applied InceptionV3 for quality grading of Indian Alphonso mangoes.	High accuracy (97%) in controlled lighting conditions.	Performance dropped significantly in field conditions; model is computationally heavy.
Zhang et al. (2024) [11]	Lightweight GhostNet implementation for apple defect detection on Raspberry Pi.	Achieved real-time performance (25 FPS).	Methodology not yet applied to the visual complexity of South Asian mango varieties.
Ahmed et al. (2022) [12]	Used Mask R-CNN for instance segmentation of mangoes in orchards.	Excellent at counting fruits on trees.	Computationally extremely expensive; not suitable for classification on phone CPUs.

Das & Roy (2021) [13]	Compared Transfer Learning (AlexNet, GoogLeNet) for mango rot detection.	GoogLeNet performed best (95% accuracy).	Did not utilize newer, efficient architectures like the MobileNet V3 family.
Patel et al. (2020) [14]	Developed a smartphone-based ripeness estimator using color histogram analysis.	Very fast and lightweight.	Low accuracy compared to CNNs; cannot distinguish between varieties, only color stages.
Zhao et al. (2023) [15]	Proposed a pruning strategy to reduce VGG16 size for fruit sorting robots.	Reduced model size by 40% with minimal accuracy loss.	Even pruned VGG16 is significantly larger than standard MobileNetV3.

2.2 Lightweight CNNs in Agriculture

The rapid advancement of Artificial Intelligence (AI) has transformed modern agriculture by enabling intelligent systems capable of automating complex tasks such as crop monitoring, disease diagnosis, fruit classification, yield estimation, and quality assessment. While deep learning models have achieved remarkable predictive performance across these applications, their successful deployment in real-world agricultural environments depends not only on classification accuracy but also on computational efficiency. Most agricultural operations are conducted in open fields, orchards, and rural regions where access to high-performance computing infrastructure is limited. Consequently, there has been a growing shift from developing highly complex deep neural networks toward designing lightweight Convolutional Neural Networks (CNNs) that can perform accurate inference directly on resource-constrained edge devices such as smartphones, drones, and embedded systems [1].

Lightweight CNNs are specifically designed to minimize computational complexity while preserving competitive classification performance. Unlike conventional deep architectures that rely on hundreds of convolutional filters and millions of trainable parameters, lightweight networks employ architectural optimizations that significantly reduce memory consumption and computational cost. These optimizations include **depthwise separable convolutions**, **inverted residual connections**, **linear bottlenecks**, **channel attention mechanisms**, and **hardware-aware neural architecture search (NAS)**. As a result, lightweight CNNs require fewer floating-point operations (FLOPs), occupy less storage, consume less energy, and achieve lower inference latency, making them particularly suitable for deployment on mobile and embedded platforms [2].

One of the earliest and most influential lightweight architectures is **MobileNet**, introduced by Google for mobile vision applications. MobileNet replaces conventional convolution operations with **depthwise separable convolutions**, reducing computational complexity by decomposing a standard convolution into two simpler operations: a depthwise convolution that independently filters each input channel and a pointwise convolution that combines

information across channels [3]. Mathematically, the computational cost of a standard convolution can be expressed as

$$[\text{Cost}_{\text{standard}} = D_K^2 \times M \times N \times D_F^2]$$

where:

- D_K = kernel dimension,
- M = number of input channels,
- N = number of output channels,
- D_F = spatial dimension of the feature map.

In contrast, depthwise separable convolution reduces the computational cost to

$$[\text{Cost}_{\text{depthwise}} = D_K^2 \times M \times D_F^2 + M \times N \times D_F^2]$$

This architectural modification substantially decreases both computational complexity and parameter count while maintaining strong feature extraction capability. The reduction in computational cost enables MobileNet models to execute efficiently on mobile processors without requiring specialized AI accelerators.

Building upon this foundation, **MobileNetV2** introduced **inverted residual blocks** and **linear bottlenecks**, addressing the representational limitations of the original MobileNet architecture. Instead of directly compressing feature representations, MobileNetV2 first expands feature dimensions using a lightweight pointwise convolution, performs depthwise convolution, and finally projects the features back into a lower-dimensional bottleneck representation. This inverted residual design improves information flow across the network while preserving computational efficiency [4]. Consequently, MobileNetV2 has become one of the most widely adopted architectures for mobile computer vision applications, including agricultural image classification.

Further improvements were introduced with **MobileNetV3**, which combines hardware-aware Neural Architecture Search (NAS), NetAdapt optimization, squeeze-and-excitation (SE) attention modules, and the computationally efficient **Hard-Swish (H-Swish)** activation function [5]. Unlike manually designed architectures, MobileNetV3 was partially generated using automated architecture search techniques that optimize both recognition performance and execution speed on mobile processors. The inclusion of squeeze-and-excitation blocks enables adaptive channel-wise feature recalibration, allowing the network to emphasize informative visual features while suppressing less relevant responses. These enhancements significantly improve the trade-off between classification accuracy and computational efficiency, making MobileNetV3 particularly suitable for Edge AI applications.

In parallel with the MobileNet family, several other lightweight architectures have been developed for efficient visual recognition. **ShuffleNet** reduces computational complexity through pointwise group convolution and channel shuffling operations, enabling efficient information exchange between feature groups [6]. **GhostNet** generates redundant feature maps using inexpensive linear transformations instead of standard convolutions, thereby reducing parameter count while maintaining comparable predictive performance [7]. Similarly, the **EfficientNet** family employs compound scaling to simultaneously optimize network depth,

width, and input image resolution, achieving excellent classification accuracy with relatively moderate computational requirements [8]. These architectural innovations collectively demonstrate the increasing emphasis on balancing predictive capability with deployment efficiency.

The advantages of lightweight CNNs have made them particularly attractive for agricultural applications where computational resources are inherently limited. Modern precision agriculture increasingly relies on portable devices for real-time decision making, requiring models capable of executing directly on field equipment without cloud dependency. Lightweight CNNs have therefore been successfully employed in numerous agricultural tasks, including plant disease diagnosis, weed detection, fruit maturity estimation, crop classification, livestock monitoring, and automated quality inspection [9]. Their ability to perform rapid inference with low memory requirements enables intelligent agricultural systems to operate reliably even in remote environments where internet connectivity is unavailable.

Several recent studies have demonstrated the practical benefits of lightweight CNNs in fruit classification. MobileNetV2 has been successfully applied to smartphone-based plant disease diagnosis, providing near real-time predictions while consuming significantly fewer computational resources than conventional deep CNNs [10]. GhostNet has been deployed for embedded fruit defect detection, achieving real-time inference on low-power hardware platforms such as Raspberry Pi [11]. Similarly, EfficientNet-Lite models have shown promising performance in crop classification tasks by providing competitive recognition accuracy with reduced computational complexity. These studies collectively indicate that lightweight CNNs can effectively support portable agricultural applications without sacrificing practical usability.

Despite these advances, selecting an appropriate lightweight architecture remains a challenging multi-objective optimization problem. The architecture achieving the highest classification accuracy is not necessarily the most appropriate for deployment on mobile devices. Larger models often require increased storage capacity, higher memory consumption, longer inference times, and greater energy usage, reducing battery life and user responsiveness. Therefore, evaluating lightweight CNNs requires considering multiple performance criteria simultaneously rather than relying exclusively on predictive accuracy. Important deployment-oriented metrics include model size, inference latency, memory footprint, parameter count, computational complexity, and energy efficiency in addition to traditional evaluation measures such as accuracy, precision, recall, and F1-score [12].

The emergence of Edge Artificial Intelligence has further strengthened the importance of lightweight CNN architectures. Edge AI shifts inference from centralized cloud servers to local devices, enabling faster response times, improved privacy, reduced communication costs, and continuous offline operation. However, achieving these benefits depends fundamentally on selecting neural network architectures capable of operating efficiently under the limited computational resources of smartphones and embedded processors. Lightweight CNNs therefore represent the enabling technology that makes practical Edge AI deployment feasible across numerous real-world agricultural scenarios.

The present research builds directly upon these developments by investigating five transfer learning-based CNN architectures—**MobileNetV2**, **MobileNetV3-Small**, **EfficientNetV2B0**, **DenseNet121**, and **NASNetMobile**—within a unified benchmarking framework. Instead of emphasizing classification accuracy alone, this study evaluates each architecture according to

predictive performance, weighted F1-score, inference latency, and model storage requirements. Such a comprehensive evaluation enables the identification of the most deployment-efficient architecture for real-time mango variety classification on Android smartphones. The selected model is subsequently optimized using **TensorFlow Lite Float16 post-training quantization** and integrated into the **AmmChini AI** mobile application, thereby demonstrating a complete Edge AI solution for precision agriculture. This deployment-oriented methodology distinguishes the proposed work from many existing agricultural image classification studies that terminate after laboratory-based performance evaluation without validating practical mobile implementation.

Table 2.1 Comparison of Popular Lightweight CNN Architectures

Architecture	Key Architectural Innovation	Major Advantages	Common Agricultural Applications
MobileNetV2	Inverted Residuals, Linear Bottlenecks	Low computation, small model size, efficient mobile inference	Plant disease detection, fruit classification
MobileNetV3-Small	NAS, Squeeze-and-Excitation, Hard-Swish	Excellent balance between accuracy and latency	Mobile vision, Edge AI, precision agriculture
ShuffleNetV2	Channel Shuffle, Group Convolution	Extremely fast inference on mobile CPUs	Crop recognition, weed detection
GhostNet	Ghost Feature Generation	Very low computational complexity	Fruit defect detection, embedded vision
EfficientNet-Lite	Compound Scaling	High accuracy with moderate computational cost	Crop monitoring, plant classification

Section Summary

Lightweight Convolutional Neural Networks have become the cornerstone of modern Edge AI systems by enabling deep learning inference on resource-constrained mobile and embedded devices. Their efficient architectural designs substantially reduce computational complexity while preserving strong predictive performance, making them particularly suitable for agricultural applications where portability, responsiveness, and offline operation are essential. These developments provide the technological foundation for the present research, which systematically benchmarks multiple lightweight CNN architectures to identify the most suitable model for real-time mobile classification of indigenous Bangladeshi mango varieties.

2.3 Gaps in Existing Studies

The literature reviewed in the previous sections demonstrates that deep learning has significantly advanced automated fruit classification and precision agriculture. Modern Convolutional Neural Networks (CNNs) have consistently outperformed traditional machine learning approaches in tasks such as fruit variety recognition, ripeness estimation, disease detection, and quality assessment [1]. However, despite these technological advancements, a detailed examination of existing research reveals several important limitations that hinder the practical adoption of intelligent mango classification systems in real-world agricultural environments. Most current studies emphasize maximizing classification accuracy under laboratory conditions, while comparatively little attention is given to deployment-oriented factors such as computational efficiency, mobile implementation, and real-time edge inference.

One of the most prominent limitations is the widespread use of **computationally intensive deep learning architectures**. Many recent studies employ models such as ResNet, DenseNet, Inception, and EfficientNet because of their superior feature extraction capabilities and high classification accuracy [2]. Although these architectures perform exceptionally well on high-performance computing platforms equipped with Graphics Processing Units (GPUs), they often contain millions of trainable parameters and require substantial memory, processing power, and storage capacity. Such computational demands make them unsuitable for direct deployment on smartphones and embedded devices commonly available in agricultural communities, particularly in developing countries such as Bangladesh. Consequently, many high-performing models remain confined to laboratory environments without practical field implementation.

Another significant research gap is the **limited consideration of deployment-oriented evaluation metrics**. The majority of published studies assess model performance primarily using classification accuracy, precision, recall, and F1-score. While these metrics effectively measure predictive capability, they do not fully reflect the practical requirements of mobile and edge computing applications. In real-world deployment scenarios, factors such as model size, inference latency, memory consumption, computational complexity, and energy efficiency are equally important because they directly influence user experience and device performance. A model that achieves marginally higher classification accuracy may still be unsuitable for mobile deployment if it requires excessive storage space or produces slow inference on resource-constrained hardware. Therefore, selecting an appropriate model for Edge AI applications requires a balanced evaluation that simultaneously considers both predictive performance and deployment efficiency.

The literature also indicates a **lack of comprehensive benchmarking among lightweight CNN architectures**. Existing studies often evaluate a single model or compare only two architectures using different datasets and experimental settings, making objective comparison difficult. Variations in preprocessing techniques, training strategies, data augmentation methods, and evaluation protocols frequently lead to inconsistent performance comparisons across different publications [3]. As a result, there is limited evidence identifying which lightweight architecture offers the most suitable trade-off between recognition accuracy, computational efficiency, and deployment feasibility for mobile agricultural applications.

A further limitation concerns the **underrepresentation of indigenous Bangladeshi mango varieties** in existing research. Most publicly available datasets and published studies focus on internationally recognized cultivars such as Kent, Alphonso, Tommy Atkins, or Aiwen [4]. While these datasets have contributed substantially to the advancement of fruit classification research, they do not adequately represent the unique diversity of Bangladeshi mango varieties.

Indigenous cultivars including Amrapali, Harivanga, Langra, Kanchon Langra, Bari-4, Bari-7, Katimon, Mollika, Fazlee, and Nilambori exhibit subtle morphological differences that create a significantly more challenging fine-grained classification problem. Models trained on international datasets may therefore struggle to generalize effectively when applied to these local varieties.

Another notable gap is the continued **dependence on cloud-based inference systems**. Many intelligent agricultural applications transmit captured images to remote cloud servers for prediction before returning the classification result to the user. Although cloud computing provides virtually unlimited computational resources, it introduces several practical challenges, including network latency, dependence on stable internet connectivity, increased communication costs, and potential privacy concerns [5]. These limitations are particularly problematic in rural agricultural regions where internet connectivity may be unreliable or unavailable. Consequently, cloud-dependent systems often fail to satisfy the real-time operational requirements of farmers and field workers.

Furthermore, existing studies rarely demonstrate a **complete end-to-end deployment pipeline**. In many cases, research concludes after reporting experimental accuracy obtained in desktop environments, without validating whether the trained model can be optimized, converted, and integrated into a practical mobile application. Essential deployment stages—including TensorFlow Lite conversion, model quantization, mobile inference optimization, and user interface integration—are frequently omitted. This creates a gap between algorithm development and practical implementation, limiting the real-world applicability of many proposed solutions.

Based on these observations, it is evident that current research has not fully addressed the combined challenges of **classification accuracy, computational efficiency, deployment optimization, and practical usability** within a single unified framework. There remains a clear need for an integrated Edge AI solution capable of accurately classifying indigenous Bangladeshi mango varieties while simultaneously satisfying the computational constraints of modern mobile devices.

To address these limitations, the present research proposes a comprehensive deployment-oriented framework that differs from previous studies in several important aspects. First, five lightweight transfer learning-based CNN architectures—**MobileNetV2, MobileNetV3-Small, EfficientNetV2B0, DenseNet121, and NASNetMobile**—are benchmarked using a unified experimental protocol to ensure fair comparison. Second, model selection is based not only on classification accuracy but also on deployment-related metrics including weighted F1-score, model size, and inference latency. Third, the selected architecture is optimized through **TensorFlow Lite (TFLite) Float16 post-training quantization** to reduce storage requirements while maintaining predictive performance. Finally, the optimized model is integrated into the **AmmChini AI** Android application, enabling fully offline, real-time mango variety classification directly on smartphones without requiring internet connectivity. By combining comparative benchmarking, deployment optimization, and mobile implementation within a single framework, this research aims to bridge the gap between academic deep learning research and practical precision agriculture applications.

Table 2.2 Research Gaps Identified from Existing Literature

Research Gap	Observation in Existing Studies	How This Research Addresses the Gap
Computationally expensive deep learning models	High-performing CNNs often require GPUs and cloud infrastructure	Evaluates lightweight CNNs suitable for resource-constrained mobile devices
Accuracy-focused evaluation	Most studies prioritize classification accuracy over deployment efficiency	Considers accuracy, weighted F1-score, model size, and inference latency
Limited benchmarking of lightweight models	Comparisons are generally restricted to one or two architectures	Benchmarks five lightweight CNN architectures using identical experimental settings
Limited research on Bangladeshi mango varieties	Existing datasets mainly contain international commercial cultivars	Focuses on ten indigenous Bangladeshi mango varieties
Dependence on cloud-based inference	Most systems require internet connectivity for prediction	Develops a fully offline Edge AI solution using TensorFlow Lite
Lack of end-to-end deployment	Research often ends after model training and evaluation	Implements complete deployment from benchmarking to Flutter-based mobile application

Section Summary

The analysis of existing literature reveals that although deep learning has substantially improved automated mango classification, important challenges remain regarding deployment efficiency, mobile implementation, and real-world usability. Most previous studies emphasize predictive performance while overlooking computational constraints associated with edge devices. Furthermore, indigenous Bangladeshi mango varieties and complete offline deployment frameworks have received relatively limited attention. These identified research gaps provide the primary motivation for the proposed study, which introduces a comprehensive Edge AI framework integrating lightweight CNN benchmarking, deployment-aware optimization, TensorFlow Lite conversion, and a fully functional Android application for real-time offline mango variety classification.

2.4 Summary of Research Gap

The literature reviewed in the preceding sections demonstrates that deep learning has substantially advanced the field of automated mango classification. Recent studies have shown that Convolutional Neural Networks (CNNs) consistently outperform traditional machine learning approaches by automatically learning discriminative visual features from image data, resulting in significant improvements in classification accuracy [1]. Furthermore, transfer learning has become the preferred strategy for agricultural image classification because it enables pre-trained models to achieve high predictive performance even when domain-specific datasets are relatively small. These developments clearly establish deep learning as an effective solution for fruit variety recognition and precision agriculture applications.

Despite these advancements, the existing body of research reveals a noticeable gap between algorithmic development and practical deployment. Most previous studies prioritize predictive accuracy while paying comparatively little attention to deployment-oriented considerations such as computational efficiency, model size, inference latency, and hardware constraints. As a result, many high-performing models require substantial computational resources and are primarily evaluated on desktop workstations or cloud-based infrastructures. Although these architectures perform exceptionally well in laboratory environments, they often fail to satisfy the practical requirements of real-time inference on resource-constrained mobile devices commonly used in agricultural field operations.

Another important limitation identified in the literature is the insufficient emphasis on comprehensive lightweight model benchmarking. Existing studies frequently evaluate only one or two deep learning architectures, making it difficult to objectively determine which model offers the best balance between predictive performance and deployment efficiency. Moreover, many published comparisons rely exclusively on accuracy-based evaluation while overlooking equally important engineering metrics such as model storage size, inference latency, memory consumption, and mobile execution performance. For Edge Artificial Intelligence (Edge AI) applications, these deployment-oriented metrics are critical because they directly influence user experience, battery consumption, responsiveness, and scalability on portable devices.

The review also highlights the limited research conducted on indigenous Bangladeshi mango varieties. Most internationally available datasets and published studies focus on globally commercial cultivars such as Kent, Alphonso, Aiwon, or Tommy Atkins, whereas relatively few investigations consider the unique morphological characteristics of Bangladeshi mangoes. Indigenous cultivars including Amrapali, Harivanga, Langra, Kanchon Langra, Bari-4, Bari-7, Katimon, Mollika, Fazlee, and Nilambori exhibit subtle visual differences that create a challenging fine-grained image classification problem. Consequently, there remains a need for research specifically addressing lightweight deep learning models capable of accurately recognizing these local varieties while maintaining practical deployment feasibility.

Furthermore, although TensorFlow Lite and modern mobile AI frameworks have significantly simplified on-device inference, comparatively few studies present a complete end-to-end deployment pipeline extending beyond model training. Many investigations conclude after reporting experimental performance metrics without demonstrating whether the trained model can be optimized, converted, integrated into a mobile application, and executed entirely offline under real-world conditions. This disconnect between algorithm development and practical implementation limits the real-world impact of many existing agricultural AI solutions.

Based on these observations, the present research is designed to address the identified gaps through a comprehensive deployment-oriented Edge AI framework. Unlike previous studies, this work systematically benchmarks **five lightweight transfer learning-based convolutional neural networks—MobileNetV2, MobileNetV3-Small, EfficientNetV2B0, DenseNet121, and NASNetMobile—using a unified experimental protocol**. Rather than selecting the best architecture solely according to classification accuracy, the proposed framework performs a multi-objective evaluation incorporating **accuracy, precision, recall, weighted F1-score, model storage size, and inference latency**. This balanced evaluation enables the identification of the most suitable architecture for deployment on resource-constrained Android devices.

After selecting the optimal model, the research further extends beyond conventional benchmarking by applying **TensorFlow Lite (TFLite) Float16 post-training quantization**,

reducing storage requirements while preserving predictive performance for efficient mobile inference. Finally, the optimized model is integrated into **AmmChini AI**, a Flutter-based Android application capable of performing **fully offline, real-time mango variety classification** directly on the user's smartphone without requiring cloud connectivity. By combining comparative benchmarking, deployment-aware optimization, and practical mobile implementation within a single framework, this research bridges the gap between academic deep learning research and real-world agricultural applications.

In summary, the primary research gap addressed by this thesis is not merely the development of an accurate image classification model, but the creation of a **complete Edge AI ecosystem** that encompasses lightweight model selection, deployment-oriented optimization, and real-time mobile implementation for indigenous Bangladeshi mango variety classification. This integrated approach provides a practical and scalable solution for precision agriculture while demonstrating how modern deep learning techniques can be successfully translated into field-ready intelligent applications.

Table 2.3 Summary of Identified Research Gaps and Contributions

Research Gap	Observation from Existing Studies	Contribution of This Research
Focus on predictive accuracy	Most studies prioritize classification accuracy over deployment feasibility	Evaluates both predictive performance and deployment-oriented metrics
Limited lightweight model comparison	Comparisons usually involve only one or two CNN architectures	Benchmarks five lightweight CNN models under identical experimental conditions
Insufficient deployment evaluation	Model size, latency, and mobile execution are often ignored	Includes model size, inference latency, and Edge AI performance evaluation
Cloud-dependent inference	Most intelligent systems rely on internet connectivity	Develops a fully offline TensorFlow Lite-based mobile solution
Limited work on Bangladeshi mango varieties	Existing research primarily targets international commercial cultivars	Focuses on ten indigenous Bangladeshi mango varieties
Lack of end-to-end implementation	Research often ends after model training	Proposes a complete pipeline from benchmarking to Flutter-based mobile deployment

Transition to Chapter 3

Having identified the limitations of existing research and established the motivation for the proposed work, the next chapter presents the research methodology. It describes the dataset, preprocessing pipeline, transfer learning strategy, lightweight CNN architectures, benchmarking methodology, evaluation metrics, model optimization procedures, and the deployment framework used to develop the proposed **AmmChini AI** system. This methodological foundation ensures that the proposed solution addresses the identified research gaps through a systematic, reproducible, and deployment-oriented experimental design.

Chapter 3

Methodology

3.1 Dataset Description

The effectiveness of any deep learning-based image classification system is fundamentally dependent on the quality, diversity, and representativeness of the dataset used during model development. A well-constructed dataset enables the neural network to learn robust feature representations, improves generalization capability, and reduces the likelihood of overfitting. Since the objective of this research is to develop an Edge AI system capable of accurately classifying indigenous Bangladeshi mango varieties under practical deployment conditions, it is essential that the training dataset adequately represents the visual diversity encountered in real agricultural environments.

This research utilizes the **Mangifera2012** dataset, a publicly available benchmark dataset specifically developed for the classification of indigenous Bangladeshi mango varieties [1]. Unlike many internationally available fruit datasets that primarily contain globally commercial cultivars, the Mangifera2012 dataset focuses exclusively on mango varieties commonly cultivated in Bangladesh. Consequently, it provides an appropriate benchmark for investigating fine-grained image classification problems involving visually similar indigenous cultivars.

The dataset consists of **2,012 high-resolution RGB images** representing **ten distinct Bangladeshi mango varieties**. The images were captured using a **Digital Single-Lens Reflex (DSLR)** camera under controlled imaging conditions to ensure sufficient image quality while preserving the natural visual characteristics of each fruit. The original image resolution is **3024 × 4032 pixels**, allowing the preservation of detailed information regarding fruit shape, skin texture, color variation, and surface morphology. These high-resolution images provide rich visual features that are particularly valuable for distinguishing cultivars exhibiting subtle inter-class differences.

Unlike coarse-grained image classification problems, mango variety recognition belongs to the category of **fine-grained visual classification**, where multiple classes share highly similar external appearances. Several indigenous Bangladeshi cultivars exhibit overlapping color distributions, comparable fruit geometry, and nearly identical surface textures, making automated classification considerably more challenging than conventional object recognition tasks. Consequently, the dataset serves as a realistic benchmark for evaluating lightweight convolutional neural networks intended for practical agricultural deployment.

The ten mango varieties included in the dataset represent commercially important cultivars cultivated across the major mango-producing regions of Bangladesh. These varieties encompass differences in physical appearance, consumer preference, commercial value, and cultivation characteristics, thereby providing sufficient visual diversity for evaluating model robustness.

Table 3.1 Distribution of Mango Varieties in the Mangifera2012 Dataset

Class	Mango Variety	Number of Images
1	Amrapali	252
2	Bari-4	235
3	Bari-7	176
4	Fazlee	156
5	Harivanga	202
6	Kanchon Langra	210
7	Katimon	163
8	Langra	202
9	Mollika	221
10	Nilambori	195
Total	10 Classes	2,012 Images

The dataset exhibits a relatively balanced class distribution, with each variety containing between **156 and 252 images**. Although slight variations exist among individual classes, the distribution is sufficiently uniform to minimize significant class imbalance during model training. Maintaining a balanced dataset is important because highly imbalanced class distributions often bias deep learning models toward majority classes, resulting in poor classification performance for minority categories.

Another important characteristic of the Mangifera2012 dataset is the presence of **high inter-class similarity**. Several mango varieties possess nearly identical external appearances, especially during immature or intermediate ripening stages. For instance, **Langra** and **Kanchon Langra** exhibit comparable fruit shape and skin coloration, while **Bari-4** and **Nilambori** often appear visually similar because of their green peel and analogous surface texture. Similarly, **Harivanga** shares certain morphological characteristics with **Amrapali**, creating additional ambiguity during visual classification. These similarities introduce a challenging fine-grained recognition problem that requires CNN models to learn subtle discriminative features rather than relying solely on coarse color or shape information.

At the same time, several varieties possess distinctive visual characteristics that facilitate classification. For example, **Fazlee** generally exhibits larger fruit size and characteristic shape, whereas **Mollika** demonstrates relatively unique morphological features compared with other cultivars. The coexistence of both easily distinguishable and highly similar classes makes the dataset particularly suitable for evaluating the discriminative capability of lightweight convolutional neural networks.

From a machine learning perspective, the dataset presents a realistic balance between complexity and practical applicability. The relatively modest number of training samples reflects a common limitation encountered in agricultural image analysis, where collecting and manually annotating thousands of high-quality images is both time-consuming and labor-intensive. Therefore, instead of training deep neural networks from scratch, this research employs **transfer learning**, allowing ImageNet-pretrained CNN architectures to leverage previously learned low-level visual representations while adapting higher-level feature extraction to the mango classification task. This strategy improves model convergence, enhances generalization, and reduces the amount of task-specific data required for effective training [2].

To ensure consistency across all experiments, the same dataset is used for training, validation, benchmarking, and deployment evaluation of every convolutional neural network investigated in this study. Furthermore, identical preprocessing procedures, augmentation strategies, optimizer settings, and evaluation protocols are applied to all architectures. Maintaining a unified experimental framework ensures that performance differences observed among the evaluated models originate from architectural characteristics rather than inconsistencies in data preparation or training configuration. This methodological consistency is particularly important because the primary objective of this research is to identify the lightweight CNN architecture that offers the most suitable balance between predictive performance and deployment efficiency for Edge AI applications.

Overall, the Mangifera2012 dataset provides a comprehensive and representative benchmark for developing an intelligent mango variety classification system tailored to the agricultural context of Bangladesh. Its combination of indigenous cultivars, balanced class distribution, high-resolution imagery, and challenging fine-grained visual similarities makes it an appropriate foundation for evaluating lightweight deep learning models intended for real-time mobile deployment.

Figure 3.1 Suggested Dataset Overview Diagram

You do not need to create a new figure if you already have representative images of the ten mango varieties or a dataset overview figure. A collage showing one sample image from each class would be more informative than a flowchart.

Example layout:

Amrapali	Bari-4	Bari-7	Fazlee	Harivanga
Kanchon				
Langra	Katimon	Langra	Mollika	Nilambori

Figure 3.1: Representative samples of the ten indigenous Bangladeshi mango varieties included in the Mangifera2012 dataset. The figure illustrates the visual diversity as well as the subtle morphological similarities that make fine-grained classification a challenging computer vision problem.

Section Summary

This section introduced the **Mangifera2012** dataset used throughout this research. The dataset contains **2,012 high-resolution images** belonging to **ten indigenous Bangladeshi mango varieties**, providing a balanced and representative benchmark for fine-grained image classification. Its combination of visually similar cultivars and relatively balanced class distribution makes it well suited for evaluating lightweight CNN architectures within the proposed Edge AI framework. The following section describes the preprocessing and augmentation techniques applied to prepare these images for deep learning model training.

3.1.1 Dataset Composition

The quality and composition of a dataset play a crucial role in determining the performance and generalization capability of deep learning models. In image classification tasks, an appropriately designed dataset should provide sufficient visual diversity while maintaining representative samples for each target category. A dataset containing balanced class distributions and diverse visual characteristics enables convolutional neural networks to learn discriminative feature representations more effectively, thereby improving their robustness when presented with previously unseen images [1]. Consequently, careful examination of the dataset composition is essential before discussing the model training process.

The present study employs the **Mangifera2012** dataset, which contains a total of **2,012 RGB images** representing **ten indigenous Bangladeshi mango varieties**. Each image corresponds to one fruit instance belonging exclusively to a single class, making the problem a **multi-class image classification** task. The dataset was specifically developed to support research on indigenous mango recognition and therefore reflects the diversity of cultivars commonly found in Bangladesh rather than internationally dominant commercial varieties.

The dataset contains images of the following ten mango varieties:

- Amrapali
- Bari-4
- Bari-7
- Fazlee
- Harivanga
- Kanchon Langra
- Katimon
- Langra
- Mollika
- Nilambori

These cultivars were selected because of their agricultural importance, commercial value, and widespread cultivation across the major mango-producing regions of Bangladesh. Each variety exhibits distinctive biological characteristics; however, several classes also share highly similar external appearances, creating a realistic fine-grained image classification problem.

The original images were captured using a **Digital Single-Lens Reflex (DSLR)** camera, producing high-quality RGB images with a native resolution of **3024 × 4032 pixels**. The high spatial resolution preserves detailed visual information including fruit shape, surface texture, peel color, lenticel distribution, and subtle morphological patterns that are essential for distinguishing visually similar mango varieties. Such high-resolution imagery provides sufficient feature richness for transfer learning-based convolutional neural networks while allowing the images to be resized appropriately during preprocessing without significant loss of discriminative information.

Unlike many publicly available fruit datasets that contain severe class imbalance, the Mangifera2012 dataset exhibits a relatively uniform distribution across all classes. Although the number of images varies slightly among individual cultivars, each category contains an adequate number of samples to support supervised learning. Maintaining a balanced dataset is

particularly important because significant class imbalance may bias deep learning models toward majority classes, resulting in poor recall for underrepresented categories.

Table 3.2 Class-wise Distribution of the Mangifera2012 Dataset

Class ID	Mango Variety	Number of Images	Percentage (%)
1	Amrapali	252	12.53
2	Bari-4	235	11.68
3	Bari-7	176	8.75
4	Fazlee	156	7.75
5	Harivanga	202	10.04
6	Kanchon Langra	210	10.44
7	Katimon	163	8.10
8	Langra	202	10.04
9	Mollika	221	10.98
10	Nilambori	195	9.69
Total	10 Classes	2,012	100

As shown in **Table 3.2**, the dataset demonstrates a relatively balanced distribution. The largest class, **Amrapali**, contains **252 images**, whereas **Fazlee**, the smallest class, contains **156 images**. The difference between the largest and smallest classes is sufficiently small to avoid severe learning bias, enabling all evaluated CNN architectures to receive comparable exposure to every mango variety during training.

Beyond numerical balance, the dataset also exhibits considerable **intra-class variation**. Images belonging to the same variety differ in orientation, illumination, viewing angle, fruit position, and minor surface characteristics. These natural variations improve the model's ability to generalize beyond the exact training samples and reduce overfitting. Rather than memorizing individual images, the network is encouraged to learn stable morphological features that characterize each mango variety.

Conversely, the dataset presents substantial **inter-class similarity**, making classification considerably more challenging than ordinary object recognition tasks. Several indigenous varieties possess nearly identical color distributions, overall geometry, and surface appearance. For example, **Langra** and **Kanchon Langra** share similar fruit shape and peel coloration, while **Bari-4** and **Nilambori** exhibit comparable green skin texture during immature stages. Likewise, **Harivanga** often resembles **Amrapali** because of similarities in size and external morphology. These overlapping visual characteristics increase the complexity of feature learning and require the convolutional neural network to capture subtle discriminative cues instead of relying solely on coarse appearance features.

Another important characteristic of the dataset is that every image contains **a single mango instance**, allowing the research to focus exclusively on image classification rather than object detection or instance segmentation. This simplifies the learning objective by assigning exactly one ground-truth label to each input image, enabling the evaluated CNN models to learn direct mappings between image features and mango varieties. Consequently, the task formulation can be expressed mathematically as

$$[f(I) \rightarrow y]$$

where

- I denotes the input mango image,
- $f(.)$ represents the deep convolutional neural network,
- $(y \in \{1,2, \dots, 10\})$ denotes the predicted mango variety.

This formulation defines a standard supervised multi-class classification problem in which the network estimates the posterior probability of each class using the Softmax activation function, and the class with the highest probability is selected as the final prediction.

From an implementation perspective, all images are organized into **class-specific directory structures**, allowing TensorFlow's ImageDataGenerator to automatically assign labels based on folder names during dataset loading. This directory-based organization simplifies data management, ensures consistent label encoding across all experiments, and facilitates reproducibility. Furthermore, the same dataset organization is maintained throughout model training, benchmarking, TensorFlow Lite conversion, and mobile application deployment, ensuring consistency across every stage of the proposed framework.

Overall, the Mangifera2012 dataset provides a comprehensive and well-structured benchmark for evaluating lightweight convolutional neural networks. Its balanced class distribution, high-resolution imagery, substantial intra-class diversity, and challenging inter-class similarity collectively create a realistic experimental environment for assessing both classification performance and deployment readiness. These characteristics make the dataset particularly suitable for investigating Edge AI solutions aimed at real-time classification of indigenous Bangladeshi mango varieties on mobile devices.

Suggested Figure (Use Existing Images)

You do not need to generate a new figure if you have representative images of all ten classes. A collage of one sample image from each class is appropriate.

Figure 3.2. Representative Samples of the Mangifera2012 Dataset

Amrapali Bari-4 Bari-7 Fazlee Harivanga

Kanchon

Langra Katimon Langra Mollika Nilambori

Figure Caption:

Figure 3.2: Representative images of the ten indigenous Bangladeshi mango varieties included in the Mangifera2012 dataset. The figure illustrates both the visual diversity among different cultivars and the subtle morphological similarities that make fine-grained mango variety classification a challenging computer vision task.

3.1.2 Data Preprocessing and Augmentation

The quality of input data plays a fundamental role in the performance and generalization capability of deep learning models. Raw images collected from real-world environments often exhibit variations in resolution, orientation, illumination, scale, and viewpoint, which may negatively influence the learning process if not handled appropriately. Therefore, before training the convolutional neural networks, a comprehensive preprocessing and data augmentation pipeline was implemented to standardize the input images and improve the robustness of the proposed classification system. All preprocessing operations were performed uniformly across the five evaluated CNN architectures to ensure a fair and reproducible experimental comparison.

Image Resizing

The original images in the Mangifera2012 dataset have a native resolution of **3024 × 4032 pixels**, which is considerably larger than the input size required by lightweight convolutional neural networks. Directly processing high-resolution images would substantially increase computational complexity, memory consumption, and training time without providing proportional improvements in classification performance.

Therefore, every input image was resized to **224 × 224 pixels**, which corresponds to the standard input dimension required by **MobileNetV2**, **MobileNetV3-Small**, **EfficientNetV2B0**, **DenseNet121**, and **NASNetMobile**. During resizing, the RGB color channels were preserved, producing input tensors of dimension

$$[224 \times 224 \times 3]$$

This resizing operation provides an effective balance between computational efficiency and feature preservation. Although reducing image resolution inevitably removes some fine-grained details, the remaining spatial information is sufficient for the evaluated CNN architectures to learn discriminative morphological features while significantly reducing computational overhead.

Architecture-Specific Input Preprocessing

An important aspect of this research is the implementation of **architecture-specific preprocessing**, ensuring that each convolutional neural network receives input data in the format expected by its original ImageNet pre-training configuration.

Rather than applying a single normalization strategy to every model, the preprocessing pipeline automatically selects the appropriate preprocessing function corresponding to the active CNN architecture. During implementation, TensorFlow's built-in preprocessing functions were utilized as follows:

- MobileNetV2 → `mobilenet_v2.preprocess_input()`

- MobileNetV3-Small → mobilenet_v3.preprocess_input()
- EfficientNetV2B0 → efficientnet_v2.preprocess_input()
- DenseNet121 → densenet.preprocess_input()
- NASNetMobile → nasnet.preprocess_input()

This architecture-aware preprocessing ensures consistency between ImageNet pre-training and transfer learning, preventing distribution mismatches that could negatively affect feature extraction and convergence. By automatically selecting the correct preprocessing function during dataset loading, all evaluated architectures operate under their optimal input conditions while maintaining a unified training pipeline.

Data Augmentation

Although the Mangifera2012 dataset provides a balanced representation of indigenous mango varieties, its size remains relatively modest compared with large-scale computer vision datasets. Training deep neural networks directly on limited data increases the risk of **overfitting**, where the model memorizes training samples instead of learning generalized visual representations.

To improve model robustness and increase data diversity, **online data augmentation** was performed using TensorFlow's ImageDataGenerator. During every training epoch, random transformations were applied dynamically, generating new image variations without permanently increasing dataset size. Unlike offline augmentation, this strategy produces different image variations during each training iteration, allowing the network to observe a substantially larger set of effective training samples.

The augmentation operations used in this research are summarized in **Table 3.3**.

Table 3.3 Data Augmentation Configuration

Augmentation Technique	Parameter	Purpose
Rotation	$\pm 30^\circ$	Simulates different camera orientations
Width Shift	20%	Handles horizontal displacement
Height Shift	20%	Handles vertical displacement
Shear Transformation	20%	Introduces geometric variation
Zoom	20%	Simulates different camera distances
Horizontal Flip	Enabled	Handles mirrored fruit orientation
Fill Mode	Nearest	Preserves image continuity after transformation

Each augmentation technique contributes differently to improving model generalization.

Random Rotation ($\pm 30^\circ$) simulates photographs captured from different viewing angles. Since mangoes may be positioned differently during image acquisition, rotational augmentation enables the network to become invariant to fruit orientation.

Width and Height Shifting (20%) artificially translates the fruit within the image frame. This encourages the model to recognize mango varieties regardless of their exact position, preventing excessive dependence on object centering.

Shear Transformation (20%) introduces slight geometric distortions that mimic perspective variations commonly encountered during handheld photography. Such transformations improve the network's ability to recognize fruits captured from different viewpoints.

Random Zoom (20%) emulates variations in camera-to-object distance by enlarging or reducing the visible fruit size. This operation enhances scale invariance and improves robustness when users capture images at different distances.

Horizontal Flipping mirrors the input image along the vertical axis. Since the biological identity of a mango variety does not depend on its left-right orientation, horizontal flipping effectively doubles the orientation diversity of the training samples without altering semantic information.

Finally, **Nearest Neighbor Fill Mode** is applied to fill empty pixels created after geometric transformations, ensuring that augmented images remain visually consistent.

Training and Validation Data Generation

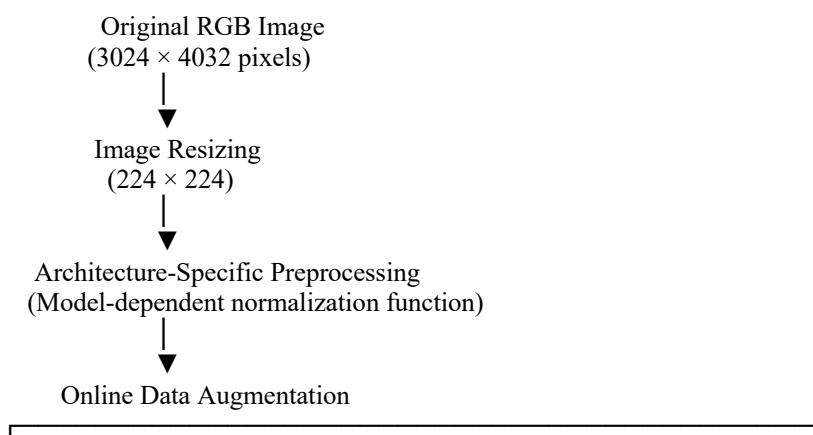
Following preprocessing and augmentation, TensorFlow's **ImageDataGenerator** was used to automatically generate batches of images during training. The dataset was divided into:

- **80% Training Set**
- **20% Validation Set**

using a fixed random seed to ensure reproducibility. The training generator applied both preprocessing and augmentation, whereas the validation generator performed only architecture-specific preprocessing without augmentation. This strategy ensures that validation performance accurately reflects the model's generalization capability on unseen data.

Workflow of the Preprocessing Pipeline

The complete preprocessing pipeline implemented in this research is illustrated conceptually in **Figure 3.3**.



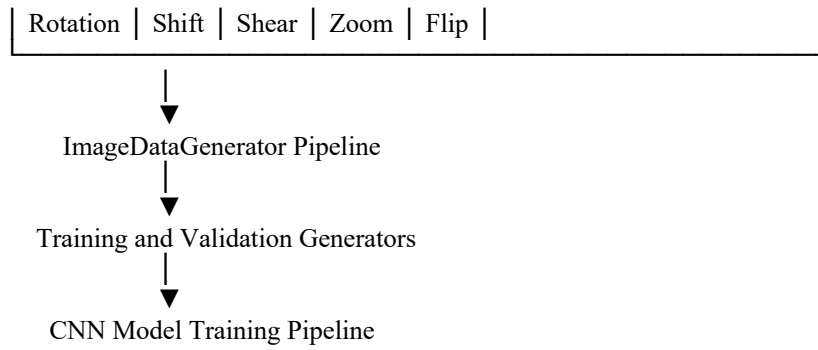


Figure 3.3: Overview of the preprocessing and augmentation pipeline employed before CNN training.

Advantages of the Proposed Preprocessing Pipeline

The adopted preprocessing strategy offers several advantages:

- Standardizes all input images into a uniform format suitable for CNN processing.
- Ensures compatibility with ImageNet pre-trained weights through architecture-specific preprocessing.
- Improves model robustness by exposing the network to diverse image transformations.
- Reduces overfitting by dynamically generating new training samples during each epoch.
- Maintains a consistent and reproducible preprocessing framework across all evaluated architectures.
- Enables fair comparative benchmarking by applying identical augmentation settings to every CNN model.

Section Summary

This study employed a unified preprocessing and augmentation pipeline to prepare the Mangifera2012 dataset for deep learning training. All images were resized to **224 × 224 pixels**, followed by architecture-specific preprocessing to preserve compatibility with ImageNet pre-trained models. Online data augmentation—including random rotation, translation, shear transformation, zooming, and horizontal flipping—was implemented using TensorFlow's ImageDataGenerator to enhance data diversity and reduce overfitting. Finally, the dataset was partitioned into training and validation subsets using an **80:20 split**, providing a consistent experimental foundation for benchmarking the five lightweight CNN architectures evaluated in this research.

3.2 Model Architecture Selection

Selecting an appropriate deep learning architecture is one of the most critical decisions in developing an efficient image classification system, particularly when the final application is intended for deployment on resource-constrained edge devices. Although numerous Convolutional Neural Network (CNN) architectures have demonstrated outstanding performance in large-scale image recognition tasks, their suitability varies significantly depending on the computational resources available in the target deployment environment. Models designed primarily to maximize classification accuracy often require substantial memory, computational power, and storage capacity, making them impractical for real-time execution on mobile devices.

Unlike conventional computer vision research that emphasizes predictive accuracy alone, this study adopts a **deployment-oriented model selection strategy**. The objective is not simply to identify the model with the highest recognition accuracy, but rather to determine the architecture that provides the most appropriate balance between **classification performance**, **computational efficiency**, **model size**, and **inference latency**. This consideration is particularly important because the final system is designed to operate entirely offline on Android smartphones without relying on cloud-based computing resources.

To achieve this objective, five transfer learning-based CNN architectures were selected for comparative evaluation:

- **MobileNetV2**
- **MobileNetV3-Small**
- **EfficientNetV2B0**
- **DenseNet121**
- **NASNetMobile**

These architectures were chosen because they represent different design philosophies for lightweight image classification while maintaining compatibility with mobile deployment frameworks such as TensorFlow Lite. Furthermore, all five architectures are officially supported within the TensorFlow ecosystem and provide pretrained ImageNet weights, enabling efficient transfer learning.

Instead of training the networks from scratch, **transfer learning** was employed by initializing each architecture with pretrained ImageNet weights. This approach allows the models to reuse low-level visual features such as edges, textures, and color patterns learned from millions of natural images while adapting higher-level representations to the task of indigenous Bangladeshi mango classification. Transfer learning significantly reduces training time, accelerates convergence, and improves generalization, particularly when domain-specific datasets are relatively small.

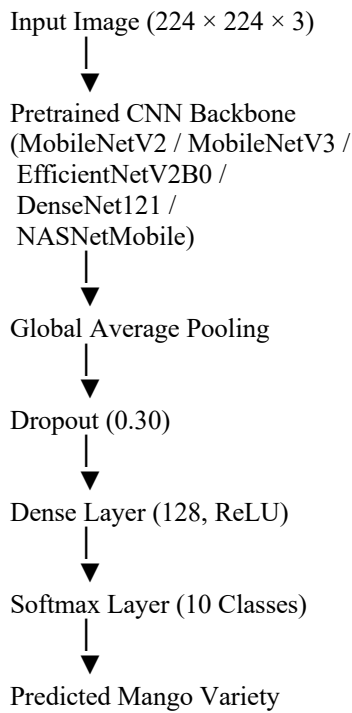
For consistency across all experiments, each architecture was implemented using an identical transfer learning strategy. The original classification layer of every pretrained network was removed (`include_top=False`), preserving only the convolutional feature extractor. During the initial training phase, the backbone network remained **frozen**, allowing only the newly added classification layers to be optimized. This strategy preserves the generalized feature representations learned from ImageNet while reducing the risk of overfitting on the relatively limited mango dataset.

To ensure a fair comparison, the same custom classification head was attached to every backbone architecture. The classification head consists of:

- **Global Average Pooling (GAP) layer**
- **Dropout layer (0.30)**
- **Fully Connected Dense layer (128 neurons, ReLU activation)**
- **Softmax output layer (10 neurons)**

The Global Average Pooling layer converts high-dimensional convolutional feature maps into compact feature vectors while substantially reducing the number of trainable parameters compared with traditional fully connected layers. The Dropout layer helps prevent overfitting by randomly deactivating neurons during training, encouraging the network to learn more generalized feature representations. Finally, the Softmax layer produces a probability distribution over the ten mango classes.

The architecture of the classifier can be summarized as follows:



Mathematical Representation

Let

$$(I \in R^{224 \times 224 \times 3})$$

represent an input RGB image.

The pretrained CNN backbone extracts feature representations

$$[F = f_{\theta}(I)]$$

where

- “I” is the resized input image,
- (f_θ) denotes the convolutional feature extractor,
- “F” represents the extracted feature map.

Global Average Pooling converts the feature map into a feature vector

$$[z = \text{GAP}(F)]$$

which is then passed through the dense classifier

$$[h = \text{ReLU}(W_1 z + b_1)]$$

Finally, the Softmax function computes the posterior probability of each mango variety

$$[P(y = i|I) = \frac{e^{h_i}}{\sum_{j=1}^{10} e^{h_j}}]$$

where

- $(P(y = i|I))$ represents the probability that the input image belongs to class $\setminus(i\setminus)$,
- $(i = 1, \dots, 10)$.

The predicted class is determined by

$$[\hat{y} = \arg \max_i P(y = i|I)]$$

Selection Criteria

Each CNN architecture was evaluated according to the following criteria:

- Classification Accuracy
- Precision
- Recall
- Weighted F1-score
- Model Storage Size (MB)
- Inference Latency (ms)
- Deployment Compatibility with TensorFlow Lite

Unlike conventional image classification studies that emphasize accuracy alone, these criteria collectively assess both predictive capability and practical deployment feasibility. This comprehensive evaluation enables identification of the most suitable architecture for real-time Edge AI applications.

Table 3.4 Selection Criteria for CNN Benchmarking

Evaluation Criterion	Purpose
----------------------	---------

Accuracy	Overall classification performance
Precision	Reliability of positive predictions
Recall	Ability to correctly identify true classes
Weighted F1-score	Balanced evaluation considering class distribution
Model Size	Storage requirement for mobile deployment
Inference Latency	Real-time execution capability
TFLite Compatibility	Suitability for Android edge deployment

The comparative evaluation of these architectures forms the basis for selecting the optimal model for the proposed mobile application. Although one architecture may achieve the highest classification accuracy, it may not necessarily provide the best deployment efficiency. Therefore, this research adopts a **multi-objective selection strategy**, prioritizing the architecture that achieves the most favorable balance between recognition performance and computational efficiency.

The detailed architectural characteristics of each CNN are presented in the following subsections, beginning with **MobileNetV2**, which serves as the baseline architecture for comparative evaluation. Subsequent sections describe **MobileNetV3-Small**, **EfficientNetV2B0**, **DenseNet121**, and **NASNetMobile**, highlighting their architectural innovations and their relevance to Edge AI-based mango variety classification.

3.2.1 MobileNetV2

MobileNetV2 was selected as the **baseline architecture** in this research because it is one of the most widely adopted lightweight Convolutional Neural Networks (CNNs) for mobile and embedded vision applications. Proposed by Sandler *et al.* in 2018 [1], MobileNetV2 was specifically designed to achieve an effective balance between classification accuracy and computational efficiency. Compared with conventional CNN architectures such as VGG16 and ResNet, MobileNetV2 significantly reduces computational complexity while maintaining competitive recognition performance, making it particularly suitable for deployment on resource-constrained devices.

The architecture builds upon the success of the original MobileNet by introducing two key innovations: **Inverted Residual Blocks** and **Linear Bottlenecks**. These architectural improvements enable the network to preserve discriminative feature representations while reducing both parameter count and floating-point operations (FLOPs). Consequently, MobileNetV2 has become a popular backbone network for mobile computer vision tasks including object detection, semantic segmentation, image classification, and agricultural image analysis.

Architecture Overview

Unlike traditional CNNs that repeatedly increase feature dimensions through standard convolution operations, MobileNetV2 adopts a more computationally efficient design. Each building block consists of three consecutive operations:

1. **Expansion Layer**
2. **Depthwise Convolution**
3. **Linear Projection (Bottleneck)**

The overall workflow of a MobileNetV2 residual block is illustrated below.

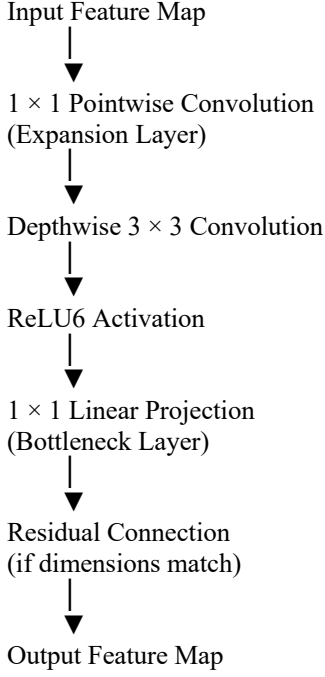


Figure 3.4: Structural representation of an Inverted Residual Block used in MobileNetV2.

Unlike conventional residual networks where shortcuts connect high-dimensional feature maps, MobileNetV2 performs residual learning between low-dimensional bottleneck representations. This design substantially reduces memory usage while preserving important visual information.

Depthwise Separable Convolution

The computational efficiency of MobileNetV2 primarily originates from **depthwise separable convolution**, which replaces computationally expensive standard convolution operations.

A standard convolution simultaneously performs spatial filtering and channel combination using

$$[D_K^2 \times M \times N]$$

filters, where

- D_K = kernel size,
- M = number of input channels,
- N = number of output channels.

The computational complexity is therefore

$$[Cost_{standard} = D_K^2 \times M \times N \times D_F^2]$$

where D_k denotes the spatial resolution of the feature map.

MobileNetV2 decomposes this operation into two independent stages:

Depthwise Convolution

Each input channel is filtered independently using a single convolution kernel.

The computational cost becomes

$$[D_K^2 \times M \times D_F^2]$$

Pointwise Convolution

A subsequent (1×1) convolution combines information across channels.

Its computational cost is

$$[M \times N \times D_F^2]$$

Therefore, the total computational complexity becomes

$$[Cost_{Depthwise} = D_K^2 M D_F^2 + M N D_F^2]$$

which is considerably smaller than standard convolution.

For typical CNN configurations, depthwise separable convolution reduces computation by nearly **8–9 times**, enabling efficient execution on mobile processors without substantially compromising feature extraction capability [1].

Inverted Residual Block

Traditional residual networks, such as ResNet, connect high-dimensional feature maps through shortcut connections. While effective for improving gradient flow, these residual pathways require considerable memory during inference.

MobileNetV2 introduces the concept of an **Inverted Residual Block**, where the residual connection is established between **low-dimensional bottleneck representations** instead of expanded feature maps.

The process consists of three sequential stages:

- Low-dimensional input
- Expansion into a higher-dimensional feature space
- Depthwise convolution
- Projection back into a compact bottleneck representation

If the input and output dimensions are identical, a shortcut connection is introduced.

Mathematically,

$$[y = x + F(x)]$$

where

- “x” represents the bottleneck input,
- “F(x)” denotes the transformed feature representation.

This inverted design preserves computational efficiency while maintaining effective information propagation throughout the network.

Linear Bottleneck

Another important innovation introduced by MobileNetV2 is the **Linear Bottleneck**.

Most CNN architectures employ nonlinear activation functions such as ReLU after every convolution layer. Although nonlinear activation improves representation learning, it can also destroy useful information when applied to very low-dimensional feature spaces.

To address this issue, MobileNetV2 removes the nonlinear activation after the final projection layer.

Instead,

the bottleneck output is computed using

$$[z = Wx + b]$$

without applying ReLU.

This linear transformation preserves feature information while reducing unnecessary information loss in compressed representations.

Consequently,

the Linear Bottleneck improves classification performance without increasing computational complexity.

Transfer Learning Implementation

In this research,

MobileNetV2 was employed as a **pretrained feature extractor** using ImageNet weights.

Instead of training the entire network from scratch,

the original classification layer was removed by setting

```
include_top=False
```

The convolutional backbone remained frozen during the initial training stage

```
base_model.trainable = False
```

A lightweight custom classification head was subsequently attached, consisting of

- Global Average Pooling
- Dropout (0.30)
- Dense (128 neurons, ReLU)
- Softmax Output Layer (10 Classes)

This strategy significantly reduces training time while allowing the pretrained network to adapt effectively to indigenous Bangladeshi mango classification.

Role of MobileNetV2 in This Research

Although several newer lightweight architectures have recently been proposed, MobileNetV2 remains one of the most widely adopted baseline models for mobile computer vision research.

In this study,

MobileNetV2 serves as the **reference architecture** against which all other evaluated models are compared.

Its inclusion provides several important advantages:

- Well-established lightweight CNN architecture.
- Strong benchmark for mobile image classification.
- Efficient computational performance.
- Broad compatibility with TensorFlow Lite deployment.
- Extensive adoption in previous agricultural AI research.

The experimental results presented in Chapter 4 demonstrate that MobileNetV2 achieves competitive classification performance while maintaining relatively low computational complexity. However, its classification accuracy, model size, and inference latency are ultimately outperformed by MobileNetV3-Small, which incorporates more advanced architectural optimizations specifically designed for Edge AI applications.

Advantages of MobileNetV2

- Lightweight architecture suitable for mobile devices.
 - Uses depthwise separable convolution to reduce computation.
 - Introduces inverted residual connections for efficient information flow.
 - Linear bottlenecks preserve important feature representations.
 - Well-supported by TensorFlow and TensorFlow Lite.
 - Excellent baseline architecture for transfer learning.
-

Limitations of MobileNetV2

Despite its advantages, MobileNetV2 has several limitations:

- Does not include channel attention mechanisms such as Squeeze-and-Excitation (SE) blocks.
 - Relies on manually designed architecture rather than hardware-aware optimization.
 - Inferior feature representation compared with newer MobileNetV3 variants.
 - Lower classification performance on visually similar fine-grained classes.
 - Larger inference latency and model size compared with MobileNetV3-Small in this study.
-

Section Summary

MobileNetV2 was selected as the baseline lightweight CNN architecture because it provides an effective compromise between computational efficiency and classification performance. Through the combination of depthwise separable convolution, inverted residual blocks, and linear bottlenecks, the network substantially reduces computational complexity while maintaining strong feature extraction capability. In this research, MobileNetV2 serves as the benchmark architecture for evaluating subsequent lightweight CNN models, providing an important reference for analyzing improvements achieved by more advanced architectures such as MobileNetV3-Small.

3.2.2 MobileNetV3-Small

MobileNetV3-Small is the primary deep learning architecture adopted in this research for deployment on mobile edge devices. Developed by Howard *et al.* in 2019, MobileNetV3 represents the third generation of the MobileNet family and was specifically designed to maximize the trade-off between classification accuracy and computational efficiency for resource-constrained hardware [1]. Unlike its predecessors, MobileNetV3 was developed using a combination of **Neural Architecture Search (NAS)** and **NetAdapt**, enabling the network architecture to be automatically optimized for both predictive performance and execution speed on mobile processors.

The MobileNetV3 family consists of two variants: **MobileNetV3-Large** and **MobileNetV3-Small**. The Large variant targets devices with relatively higher computational capability, whereas MobileNetV3-Small is specifically optimized for low-power smartphones and

embedded systems where memory, storage, and processing resources are limited. Considering that the objective of this research is to develop a real-time offline mango variety classification system for Android smartphones, **MobileNetV3-Small** was selected as the most appropriate candidate for deployment.

Unlike conventional CNN architectures that primarily emphasize deeper networks and larger feature representations, MobileNetV3 adopts a hardware-aware design philosophy. Every architectural component was selected to improve inference efficiency without significantly sacrificing classification accuracy. This balance makes MobileNetV3 particularly suitable for Edge Artificial Intelligence (Edge AI), where computational resources are inherently constrained.

Architecture Overview

MobileNetV3-Small inherits the efficient **depthwise separable convolution** mechanism from previous MobileNet architectures while introducing several additional optimizations, including:

- Inverted Residual Blocks
- Depthwise Separable Convolution
- Squeeze-and-Excitation (SE) Attention Modules
- Hard-Swish (H-Swish) Activation Function
- Hardware-aware Neural Architecture Search (NAS)

The overall architecture can be conceptually represented as follows.

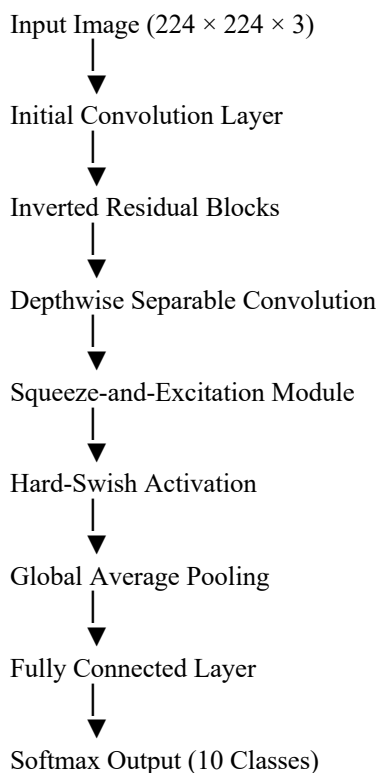


Figure 3.5: Simplified architecture of MobileNetV3-Small used in this research.

Depthwise Separable Convolution

Similar to MobileNetV2, MobileNetV3 employs **depthwise separable convolution** to reduce computational complexity.

Instead of simultaneously learning spatial and channel relationships through standard convolution, the operation is divided into:

- Depthwise Convolution
- Pointwise Convolution

This decomposition substantially decreases the number of floating-point operations (FLOPs) while maintaining effective feature extraction.

The computational complexity becomes

$$[Cost_{DW} = D_K^2 M D_F^2 + M N D_F^2]$$

where

- “ D_K ” denotes kernel size,
- “ M ” represents input channels,
- “ N ” represents output channels,
- “ D_F ” denotes feature map dimensions.

Compared with standard convolution, depthwise separable convolution dramatically reduces computational requirements, making MobileNetV3 highly suitable for mobile inference.

Inverted Residual Block

MobileNetV3 retains the **Inverted Residual Block** introduced in MobileNetV2.

Rather than connecting expanded feature representations, shortcut connections operate between compact bottleneck layers.

The residual mapping is defined as

$$[y = x + F(x)]$$

where

- “ X ” represents the bottleneck input,
- “ $F(x)$ ” denotes the transformed feature representation.

This design improves gradient propagation while minimizing memory consumption.

Squeeze-and-Excitation Attention Module

One of the most important improvements introduced in MobileNetV3 is the incorporation of the **Squeeze-and-Excitation (SE) module**.

Traditional convolution layers treat every feature channel equally. However, not all feature channels contribute equally to image recognition. The SE module enables the network to learn which channels are most informative by assigning adaptive importance weights during training.

The SE module operates in three stages:

1. Squeeze

Global Average Pooling compresses spatial information into a channel descriptor.

$$[z_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W x_c(i, j)]$$

where

- H and W denote feature map height and width,
- $(x_c(i, j))$ is the activation of channel c.

2. Excitation

The channel descriptor passes through two fully connected layers to generate channel-wise attention weights.

$$[s = \sigma(W_2 \delta(W_1 z))]$$

where

- $(\delta(\cdot))$ is the ReLU activation,
- $(\sigma(\cdot))$ is the Sigmoid function.

3. Feature Recalibration

Each feature channel is multiplied by its corresponding attention weight.

$$[\hat{x}_c = s_c x_c]$$

This mechanism enables MobileNetV3 to emphasize discriminative visual features while suppressing less informative responses, improving classification performance for fine-grained image recognition tasks such as mango variety classification.

Hard-Swish Activation Function

Another important enhancement is the replacement of conventional ReLU activation with **Hard-Swish (H-Swish)** in selected layers.

The Hard-Swish activation is defined as

$$[\text{H-Swish}(x) = x \cdot \frac{\text{ReLU6}(x + 3)}{6}]$$

Unlike the original Swish activation,

$$[\text{Swish}(x) = x\sigma(x),]$$

Hard-Swish approximates the same nonlinear behavior while requiring significantly fewer computational operations. Consequently, it provides improved classification performance with lower inference latency, making it particularly suitable for mobile hardware.

Neural Architecture Search (NAS)

Unlike manually designed CNN architectures, MobileNetV3 was partially generated using **Neural Architecture Search (NAS)**.

NAS automatically explores numerous architectural configurations to identify the network structure that provides the optimal balance between accuracy and computational efficiency.

The resulting architecture is subsequently refined using **NetAdapt**, which removes computational redundancy while maintaining predictive performance. This hardware-aware optimization allows MobileNetV3 to execute efficiently on mobile CPUs without requiring specialized AI accelerators.

Transfer Learning Implementation

In this research, MobileNetV3-Small was implemented using **ImageNet pre-trained weights**.

The pretrained classifier was removed by configuring

```
include_top=False
```

The convolutional backbone remained frozen during the initial training stage

```
base_model.trainable = False
```

A custom classification head was then attached consisting of:

- Global Average Pooling
- Dropout (0.30)
- Dense Layer (128 neurons, ReLU)
- Softmax Layer (10 output classes)

This transfer learning strategy enables efficient adaptation to the Mangifera2012 dataset while minimizing training time and reducing overfitting.

Role of MobileNetV3-Small in This Research

Among the five evaluated CNN architectures, MobileNetV3-Small demonstrated the most favorable balance between predictive performance and deployment efficiency.

Although **EfficientNetV2B0** achieved the highest classification accuracy, MobileNetV3-Small provided superior characteristics for Edge AI deployment by combining:

- High classification accuracy,
- Small model size,
- Low inference latency,
- Efficient memory utilization,
- Excellent TensorFlow Lite compatibility.

Following comparative benchmarking, MobileNetV3-Small was selected as the final deployment model and subsequently converted into **TensorFlow Lite (TFLite)** format using **Float16 post-training quantization**. The optimized model was then integrated into the **AmmChini AI** Flutter application to perform fully offline mango variety classification on Android smartphones.

Advantages of MobileNetV3-Small

- Designed specifically for mobile and embedded vision applications.
- Uses depthwise separable convolution for efficient computation.
- Incorporates Squeeze-and-Excitation attention to improve feature representation.
- Employs Hard-Swish activation for faster inference.
- Optimized using Neural Architecture Search and NetAdapt.
- Small model size suitable for mobile storage.
- Low inference latency enables real-time prediction.
- Fully compatible with TensorFlow Lite deployment.
- Provides an excellent balance between predictive performance and computational efficiency.

Limitations of MobileNetV3-Small

Although MobileNetV3-Small performs exceptionally well for Edge AI applications, several limitations remain:

- Slightly lower classification accuracy than larger architectures such as EfficientNetV2B0.
 - Reduced representational capacity compared with deeper CNN models.
 - Performance may decline on highly complex datasets requiring extremely deep feature hierarchies.
 - Limited computational capacity restricts extraction of very high-level semantic features compared with heavyweight architectures.
-

Section Summary

MobileNetV3-Small represents a significant advancement in lightweight convolutional neural network design by integrating depthwise separable convolution, inverted residual blocks, squeeze-and-excitation attention, Hard-Swish activation, and hardware-aware neural architecture search within a unified framework. These architectural innovations enable the network to achieve an excellent balance between recognition accuracy and computational efficiency, making it highly suitable for Edge AI applications. In the context of this research, MobileNetV3-Small emerged as the optimal deployment model because it provided the most favorable trade-off between classification performance, model size, and inference latency. Consequently, it was selected for TensorFlow Lite optimization and integrated into the **AmmChini AI** mobile application to enable real-time, fully offline classification of indigenous Bangladeshi mango varieties.

3.2.3 EfficientNetV2B0

EfficientNetV2B0 is a modern convolutional neural network architecture designed to achieve high classification accuracy while maintaining improved computational efficiency compared with earlier deep learning models. Proposed by Tan and Le in 2021, EfficientNetV2 was developed to address two major limitations of the original EfficientNet family: slow training speed and high computational cost associated with large-scale image classification tasks [1]. By introducing optimized convolutional blocks and an improved scaling strategy, EfficientNetV2 achieves faster training, enhanced feature extraction, and competitive classification performance across a wide range of computer vision applications.

Although EfficientNetV2B0 is computationally more demanding than MobileNet-based architectures, it remains significantly more efficient than many traditional deep CNNs such as ResNet and DenseNet while providing superior recognition capability. For this reason, EfficientNetV2B0 was included in this research to serve as a high-performance benchmark against which lightweight mobile-oriented architectures could be evaluated. Comparing EfficientNetV2B0 with MobileNetV2, MobileNetV3-Small, DenseNet121, and

NASNetMobile provides valuable insight into the trade-off between predictive accuracy and deployment efficiency.

Architecture Overview

EfficientNetV2B0 retains the fundamental philosophy of the EfficientNet family while incorporating several architectural improvements that accelerate training and improve computational efficiency. The network consists of a sequence of convolutional blocks that progressively learn increasingly abstract visual representations from the input image.

Unlike earlier EfficientNet models that relied exclusively on **Mobile Inverted Bottleneck Convolution (MBConv)** blocks, EfficientNetV2 introduces **Fused-MBConv** blocks in the early stages of the network. These fused blocks combine standard convolution and expansion operations into a single layer, reducing memory access overhead and significantly improving training efficiency.

The simplified architecture is illustrated below.

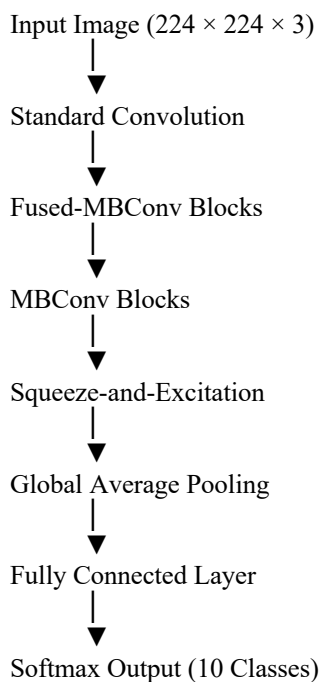


Figure 3.6: Simplified architecture of EfficientNetV2B0 used in this research.

Fused-MBConv Block

One of the most significant innovations introduced in EfficientNetV2 is the **Fused Mobile Inverted Bottleneck Convolution (Fused-MBConv)**.

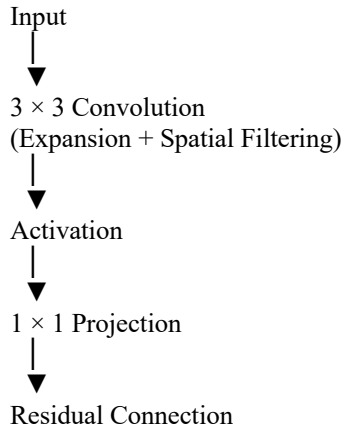
In the original MBConv block, feature extraction consists of:

- Expansion Layer
- Depthwise Convolution
- Projection Layer

Although computationally efficient during inference, this structure introduces considerable memory access overhead during training.

To overcome this limitation, EfficientNetV2 replaces the early MBConv blocks with Fused-MBConv layers, where the expansion and depthwise convolution are merged into a single standard convolution operation.

The workflow can be represented as



This modification improves training efficiency while maintaining strong feature extraction capability.

MBConv Block

In deeper stages of the network, EfficientNetV2 continues to employ the **Mobile Inverted Bottleneck Convolution (MBConv)** introduced by the EfficientNet architecture.

The MBConv block consists of:

- Pointwise Expansion
- Depthwise Convolution
- Squeeze-and-Excitation Module
- Pointwise Projection

Mathematically,

$$[F(x) = PW \left(DW(PW(x)) \right)]$$

where

- “PW” denotes pointwise convolution,

- “DW” represents depthwise convolution,
- “F(x)” is the transformed feature representation.

Residual connections are applied whenever the input and output dimensions are identical.

Compound Scaling Strategy

Unlike conventional CNNs that increase only network depth or width, EfficientNetV2 adopts a **compound scaling strategy** that simultaneously scales multiple architectural dimensions.

The network balances:

- Depth
- Width
- Input Resolution

using scaling coefficients that maximize recognition performance under computational constraints.

The scaling process is represented as

$$\begin{aligned} [\text{Depth} &= \alpha^\phi] \\ [\text{Width} &= \beta^\phi] \\ [\text{Resolution} &= \gamma^\phi] \end{aligned}$$

subject to

$$[\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2]$$

where

- (ϕ) denotes the compound scaling coefficient,
- (α, β, γ) are constants determined during architecture optimization.

This balanced scaling strategy allows EfficientNetV2 to achieve high recognition accuracy without excessively increasing computational complexity.

Squeeze-and-Excitation Module

Similar to MobileNetV3, EfficientNetV2 incorporates **Squeeze-and-Excitation (SE)** attention blocks to improve channel-wise feature representation.

The SE module performs:

1. Global Average Pooling

2. Channel Importance Learning
3. Feature Recalibration

The recalibrated feature map is computed as

$$\hat{x}_c = s_c x_c$$

where

- (s_c) is the learned attention weight,
- (x_c) is the feature response of channel c .

The attention mechanism enables the network to emphasize informative feature channels while suppressing redundant information, thereby improving discriminative capability for visually similar mango varieties.

Transfer Learning Implementation

In this research, EfficientNetV2B0 was implemented using **ImageNet pre-trained weights** to leverage the rich visual representations learned from large-scale natural image datasets.

Consistent with the implementation of the other evaluated architectures:

- the original ImageNet classification layer was removed,
- the convolutional backbone remained frozen during the initial training stage,
- a custom classifier consisting of **Global Average Pooling, Dropout (0.30), a Dense layer with 128 neurons and ReLU activation**, and a **Softmax output layer** for ten mango classes was appended.

Using the same classification head across all architectures ensures that differences in experimental performance arise primarily from the backbone networks rather than variations in classifier design.

Role of EfficientNetV2B0 in This Research

EfficientNetV2B0 serves as the **high-accuracy benchmark model** in this comparative study. Its inclusion enables an objective assessment of whether the additional computational complexity of a larger architecture is justified for mobile deployment.

Experimental evaluation demonstrated that EfficientNetV2B0 achieved the **highest classification accuracy (92.00%)** among all five evaluated CNN architectures. It also produced the highest weighted F1-score, indicating excellent discriminative capability across the ten indigenous mango varieties.

However, these improvements were accompanied by increased deployment costs. Compared with MobileNetV3-Small, EfficientNetV2B0 produced:

- a substantially larger model size,
- higher inference latency,
- greater computational requirements.

Consequently, although EfficientNetV2B0 achieved the best predictive performance, it was not selected as the deployment model because it does not satisfy the resource constraints required for real-time Edge AI execution on smartphones.

Advantages of EfficientNetV2B0

- High classification accuracy.
 - Excellent feature extraction capability.
 - Faster training than the original EfficientNet models.
 - Compound scaling improves parameter efficiency.
 - Fused-MBConv blocks accelerate feature learning.
 - Squeeze-and-Excitation modules enhance discriminative feature representation.
 - Strong generalization capability for fine-grained image classification.
-

Limitations of EfficientNetV2B0

Despite its superior predictive performance, EfficientNetV2B0 has several practical limitations for mobile deployment:

- Larger model size compared with MobileNet architectures.
 - Higher inference latency on resource-constrained devices.
 - Greater memory consumption during execution.
 - Increased computational complexity.
 - Less suitable for real-time Edge AI applications where storage and processing resources are limited.
-

Section Summary

EfficientNetV2B0 represents a modern high-performance convolutional neural network that combines Fused-MBConv blocks, MBConv layers, compound scaling, and squeeze-and-excitation attention mechanisms to achieve outstanding image classification performance. In this research, EfficientNetV2B0 served as the highest-performing benchmark architecture, achieving the best classification accuracy among the evaluated models. Nevertheless, its larger model size and higher inference latency reduced its suitability for deployment on Android edge devices. Therefore, despite its superior predictive capability, EfficientNetV2B0 was not selected for the final mobile implementation, highlighting the importance of balancing recognition performance with computational efficiency in Edge AI applications.

3.2.4 NASNetMobile

NASNetMobile is a lightweight convolutional neural network architecture developed to achieve high classification performance while maintaining computational efficiency for mobile and embedded devices. Unlike conventional CNN architectures that are manually designed by researchers, NASNet (Neural Architecture Search Network) was automatically generated using **Neural Architecture Search (NAS)**, an Artificial Intelligence-based optimization technique that explores thousands of possible network configurations to identify the architecture that provides the best balance between predictive accuracy and computational cost [1]. This automated design methodology enables NASNetMobile to achieve competitive recognition performance while reducing the need for manual architecture engineering.

NASNet was originally proposed by Zoph *et al.* as a scalable neural network architecture capable of adapting to different computational environments. The complete NASNet family consists of two principal variants: **NASNetLarge**, intended for high-performance computing platforms, and **NASNetMobile**, which is specifically optimized for mobile and embedded devices with limited computational resources. Considering its lightweight design and compatibility with mobile deployment, NASNetMobile was selected as one of the benchmark architectures in this research for comparative evaluation against MobileNetV2, MobileNetV3-Small, EfficientNetV2B0, and DenseNet121.

Architecture Overview

The distinguishing feature of NASNetMobile is that its architecture was not manually constructed but automatically discovered through a reinforcement learning-based Neural Architecture Search process. Instead of designing every convolutional layer individually, NAS learns two reusable computational building blocks known as the **Normal Cell** and the **Reduction Cell**. These cells are repeatedly stacked to construct the complete neural network.

The simplified architecture is illustrated below.

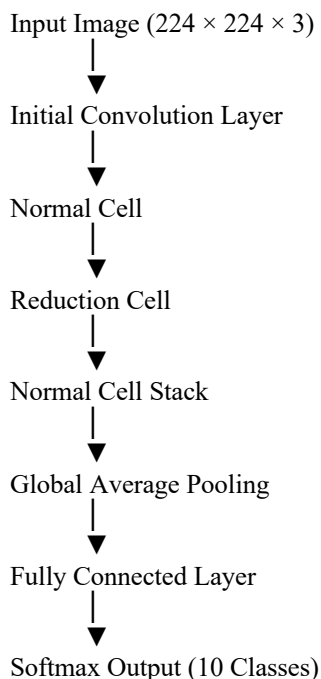


Figure 3.7: Simplified architecture of NASNetMobile used in this research.

Unlike manually designed CNNs, NASNetMobile repeatedly reuses optimized computational cells throughout the network, enabling efficient feature extraction while maintaining architectural consistency.

Normal Cell

The **Normal Cell** is responsible for extracting high-level visual features while preserving the spatial dimensions of the input feature map. It consists of multiple parallel computational branches that perform different convolutional operations simultaneously. The outputs from these branches are combined to generate a richer feature representation.

The Normal Cell enables the network to learn diverse visual characteristics such as:

- shape,
- texture,
- edges,
- color patterns,
- fine morphological details.

Since the spatial resolution remains unchanged, the Normal Cell focuses primarily on improving feature representation without reducing image size.

Reduction Cell

The **Reduction Cell** performs feature extraction while simultaneously reducing the spatial resolution of the feature map.

Instead of using conventional pooling layers alone, the Reduction Cell employs optimized convolutional operations discovered through Neural Architecture Search to decrease feature map dimensions while preserving important semantic information.

This operation allows the network to:

- reduce computational complexity,
- increase receptive field,
- learn more abstract visual features,
- improve hierarchical representation.

The Reduction Cell therefore performs a role similar to pooling layers found in traditional CNN architectures but with greater optimization.

Neural Architecture Search (NAS)

The most important contribution of NASNetMobile is its automated architecture generation process.

Unlike manually engineered networks, Neural Architecture Search formulates architecture design as an optimization problem.

Given an architecture search space

$$[A = \{a_1, a_2, \dots, a_n\}]$$

the search algorithm attempts to identify

$$[a^* = \arg \max_{a \in A} Performance(a)]$$

where

- (A) denotes the candidate architecture space,
- (Performance(a)) represents the validation accuracy of architecture (a),
- (a*) is the optimal architecture selected by the search process.

The search controller is typically implemented using reinforcement learning, where candidate architectures are evaluated, rewarded according to their performance, and progressively refined until an optimal architecture is obtained.

This automated design strategy enables NASNetMobile to achieve strong predictive performance without relying entirely on manually engineered architectural decisions.

Transfer Learning Implementation

Similar to the other evaluated CNN architectures, NASNetMobile was implemented using **ImageNet pre-trained weights**.

To maintain consistency across all benchmark experiments:

- the original ImageNet classification layer was removed (include_top=False);
- the pretrained convolutional backbone was frozen during the initial training stage;
- a common custom classification head was attached, consisting of:
 - Global Average Pooling,
 - Dropout (0.30),
 - Dense layer with 128 neurons and ReLU activation,
 - Softmax output layer with ten neurons corresponding to the ten mango varieties.

Employing the same transfer learning strategy for every architecture ensures that performance differences originate from the backbone network rather than variations in classifier design.

Role of NASNetMobile in This Research

NASNetMobile was included in the comparative benchmark because it represents an alternative lightweight architecture optimized through automated neural architecture search rather than manual design.

Evaluating NASNetMobile alongside MobileNetV2 and MobileNetV3-Small allows this research to investigate whether architectures generated through Neural Architecture Search provide superior performance for fine-grained mango variety classification.

Experimental results presented in Chapter 4 indicate that NASNetMobile achieved satisfactory classification performance while maintaining moderate computational requirements. However, its overall accuracy, weighted F1-score, inference latency, and model size were inferior to those of MobileNetV3-Small. Consequently, although NASNetMobile demonstrates the effectiveness of automated architecture optimization, it was not selected as the final deployment model for the proposed Edge AI framework.

Advantages of NASNetMobile

- Automatically optimized using Neural Architecture Search.
- Lightweight architecture suitable for mobile devices.
- Efficient reuse of Normal and Reduction Cells.
- Good feature extraction capability for image classification.
- Compatible with TensorFlow and TensorFlow Lite deployment.
- Lower computational complexity than many conventional CNN architectures.

Limitations of NASNetMobile

Despite its architectural innovations, NASNetMobile exhibits several limitations:

- Lower classification accuracy than EfficientNetV2B0 and MobileNetV3-Small in this study.
- Larger model size compared with MobileNetV3-Small.
- Higher inference latency than MobileNet-based architectures.
- Automated architecture does not necessarily provide the optimal balance between predictive performance and deployment efficiency for every application.
- Less suitable for real-time smartphone deployment compared with MobileNetV3-Small.

Section Summary

NASNetMobile represents an innovative lightweight CNN architecture generated through **Neural Architecture Search**, replacing manual architecture design with automated optimization. By repeatedly stacking optimized Normal and Reduction Cells, the network achieves efficient feature extraction while maintaining relatively low computational complexity. In this research, NASNetMobile served as one of the benchmark architectures for evaluating lightweight deep learning models on the Mangifera2012 dataset. Although it demonstrated satisfactory classification capability, its predictive performance and deployment efficiency were inferior to those of MobileNetV3-Small. Therefore, NASNetMobile was not selected for the final Edge AI deployment, reinforcing the importance of evaluating both recognition performance and practical deployment constraints when developing mobile-based agricultural applications.

3.2.5 DenseNet121

DenseNet121 is a deep convolutional neural network architecture designed to improve feature propagation, encourage feature reuse, and alleviate the vanishing gradient problem commonly encountered in very deep neural networks. Proposed by Huang *et al.* in 2017, DenseNet (Densely Connected Convolutional Network) introduced a novel connectivity pattern in which every layer is directly connected to all subsequent layers within a dense block [1]. Unlike conventional CNN architectures that pass information only to the immediately following layer, DenseNet establishes direct feature connections between all preceding and succeeding layers. This dense connectivity enables efficient information flow throughout the network while improving learning efficiency and reducing redundant feature extraction.

Among the different variants of the DenseNet family, **DenseNet121** was selected in this research because it provides a balanced compromise between network depth and computational complexity. The model contains **121 layers**, making it substantially deeper than MobileNetV2, MobileNetV3-Small, and NASNetMobile. Although DenseNet121 requires greater computational resources, its dense feature propagation mechanism enables excellent representation learning, making it an appropriate benchmark for evaluating the performance of lightweight CNN architectures in fine-grained mango variety classification.

Architecture Overview

The fundamental concept of DenseNet121 is the **Dense Block**, where each layer receives feature maps generated by every preceding layer. Rather than learning entirely new representations at each layer, DenseNet continuously reuses previously extracted features, allowing the network to preserve both low-level and high-level visual information throughout the learning process.

The simplified architecture of DenseNet121 is illustrated below.

Input Image ($224 \times 224 \times 3$)



Initial Convolution Layer

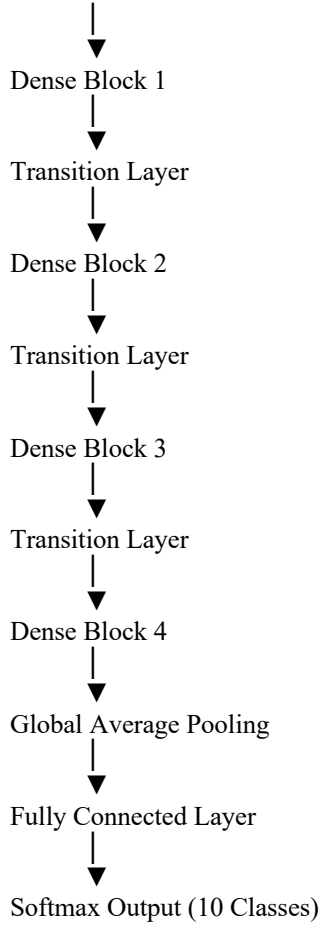


Figure 3.8: Simplified architecture of DenseNet121 employed in this research.

The network alternates between **Dense Blocks**, which perform feature extraction, and **Transition Layers**, which reduce feature map dimensions and computational complexity.

Dense Connectivity

Unlike conventional convolutional neural networks where each layer receives input only from its immediate predecessor, DenseNet introduces direct feature propagation across all layers.

For the (l^{th}) layer,

$$[x_l = H_l([x_0, x_1, \dots, x_{l-1}])]$$

where

- (x_l) denotes the output of the (l^{th}) layer,
- ($H_l(\cdot)$) represents the sequence of Batch Normalization, ReLU activation, and convolution,
- ($[]$) denotes feature map concatenation.

Instead of summing feature maps as performed in ResNet, DenseNet concatenates all previous feature representations.

This dense connectivity offers several important advantages:

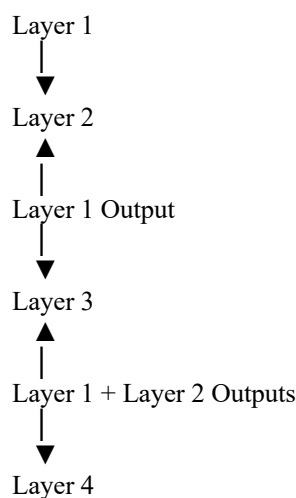
- improved gradient propagation,
- efficient feature reuse,
- stronger information preservation,
- reduced feature redundancy,
- improved learning stability.

Consequently, DenseNet121 is capable of learning highly discriminative representations even when trained on relatively small datasets.

Dense Block

A Dense Block consists of multiple convolutional layers connected through feature concatenation.

The workflow can be illustrated as



Rather than discarding previously extracted features, every convolutional layer has direct access to all earlier feature maps.

This design significantly improves feature diversity and enables the network to simultaneously utilize both low-level texture information and high-level semantic representations.

Transition Layer

Without additional processing, repeated feature concatenation would continuously increase the number of feature maps.

To control computational complexity, DenseNet introduces **Transition Layers** between consecutive Dense Blocks.

Each Transition Layer consists of:

- Batch Normalization,
- (1×1) convolution,
- Average Pooling.

These operations reduce both the number of feature channels and the spatial dimensions of the feature maps, thereby limiting memory consumption while preserving the most informative features.

Growth Rate

DenseNet introduces the concept of **Growth Rate**, which determines how many new feature maps each convolutional layer contributes.

If the growth rate is denoted by

$$[k]$$

then after (L) layers, the total number of feature maps becomes

$$[\text{Channels} = \text{Channels}_{\text{initial}} + kL]$$

A smaller growth rate reduces computational complexity, whereas a larger growth rate increases representational capacity.

DenseNet121 employs a carefully selected growth rate that balances feature diversity and computational efficiency.

Transfer Learning Implementation

In this research, DenseNet121 was implemented using **ImageNet pre-trained weights** through transfer learning.

To maintain consistency across all benchmark experiments:

- the original ImageNet classification layer was removed (`include_top=False`);
- the pretrained convolutional backbone remained frozen during the initial training phase;
- a common custom classification head was attached consisting of:
 - Global Average Pooling,
 - Dropout (0.30),
 - Dense layer with 128 neurons using ReLU activation,

- Softmax output layer with ten neurons corresponding to the ten mango varieties.

Using the same classifier architecture for every evaluated CNN ensures that differences in performance are attributable to the backbone network rather than differences in classifier design.

Role of DenseNet121 in This Research

DenseNet121 was included in the comparative benchmark because it represents a high-capacity feature extraction network capable of learning rich hierarchical representations through dense feature reuse.

Experimental evaluation demonstrated that DenseNet121 achieved **high classification accuracy (90.25%)**, outperforming MobileNetV2 and NASNetMobile. Its dense connectivity enabled excellent discrimination among visually similar mango varieties by preserving features extracted throughout the network.

However, these improvements came at the expense of significantly increased deployment cost. DenseNet121 produced:

- a larger model size,
- higher memory consumption,
- the highest inference latency among all evaluated models.

These characteristics reduce its suitability for real-time execution on Android smartphones despite its strong predictive performance.

Advantages of DenseNet121

- Excellent feature propagation through dense connectivity.
 - Efficient feature reuse reduces redundant learning.
 - Strong gradient flow alleviates the vanishing gradient problem.
 - High classification accuracy for fine-grained image classification.
 - Good generalization capability with transfer learning.
 - Effective representation of subtle morphological differences.
-

Limitations of DenseNet121

Although DenseNet121 provides strong recognition performance, several limitations reduce its suitability for Edge AI deployment:

- Larger model size than MobileNet-based architectures.
- Higher computational complexity.
- Increased memory consumption.

- Longest inference latency among the evaluated CNN models.
- Less appropriate for real-time smartphone applications where computational resources are limited.

Section Summary

DenseNet121 employs densely connected convolutional blocks to maximize feature reuse and improve information propagation throughout the network. Its dense connectivity enables highly discriminative feature extraction, making it particularly effective for fine-grained image classification tasks such as indigenous Bangladeshi mango variety recognition. In this research, DenseNet121 achieved competitive classification performance and demonstrated excellent representation learning capability. However, its relatively large model size and higher inference latency made it less suitable for deployment on resource-constrained Android devices. Consequently, although DenseNet121 served as a valuable benchmark architecture, it was ultimately not selected for the proposed Edge AI deployment framework, where computational efficiency and real-time inference are critical design requirements.

3.3 Dynamic Preprocessing Implementation

Deep learning models pretrained on large-scale datasets such as ImageNet are trained using architecture-specific input preprocessing techniques. These preprocessing operations transform raw pixel values into numerical representations that match the data distribution observed during the original pretraining process. Consequently, different convolutional neural network (CNN) architectures expect different input formats before inference. Applying an inappropriate preprocessing function may introduce a distribution mismatch between the training data and the pretrained feature extractor, resulting in degraded feature learning, slower convergence, and reduced classification accuracy [1].

Many comparative studies employ a single normalization strategy for all evaluated architectures, regardless of their original design. Although this simplifies implementation, it may lead to unfair comparisons because some models receive input distributions that differ from those used during ImageNet pretraining. Since the primary objective of this research is to perform an unbiased benchmark of multiple lightweight CNN architectures, maintaining architecture-specific preprocessing consistency is essential.

To address this issue, this research implements a **dynamic preprocessing framework**, where the preprocessing function is automatically selected according to the CNN architecture being trained. Instead of manually modifying the preprocessing pipeline for every experiment, the training framework dynamically associates each model with its corresponding TensorFlow preprocessing function. This implementation guarantees that every architecture receives input data in the format expected by its pretrained ImageNet weights while maintaining a unified training pipeline.

Motivation for Dynamic Preprocessing

Transfer learning assumes that the statistical characteristics of the new input images are consistent with those encountered during the original pretraining stage. Each pretrained CNN architecture internally learns feature representations based on a particular preprocessing strategy. Consequently, using identical preprocessing for all architectures may reduce feature extraction quality because the learned convolutional filters operate on input distributions different from those used during pretraining.

For example, MobileNetV2 and MobileNetV3-Small use different preprocessing implementations from EfficientNetV2B0 or DenseNet121. Therefore, manually applying a common normalization procedure could unintentionally disadvantage certain architectures during benchmarking.

The proposed dynamic preprocessing strategy eliminates this inconsistency by ensuring that each CNN receives its own architecture-specific preprocessing operation.

Architecture-Specific Preprocessing Functions

The preprocessing functions used in this research are summarized in **Table 3.5**.

Table 3.5 Architecture-Specific Preprocessing Functions

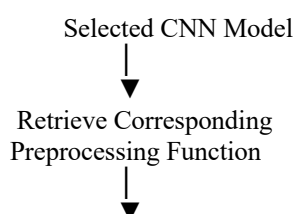
CNN Architecture	TensorFlow Preprocessing Function
MobileNetV2	<code>mobilenet_v2.preprocess_input()</code>
MobileNetV3-Small	<code>mobilenet_v3.preprocess_input()</code>
EfficientNetV2B0	<code>efficientnet_v2.preprocess_input()</code>
DenseNet121	<code>densenet.preprocess_input()</code>
NASNetMobile	<code>nasnet.preprocess_input()</code>

Each preprocessing function performs the transformations required by its corresponding pretrained architecture. Because these implementations are provided by TensorFlow, they preserve compatibility with the ImageNet feature representations learned during pretraining.

Dynamic Preprocessing Workflow

Instead of maintaining separate training scripts for each CNN architecture, a unified preprocessing framework was implemented. During execution, the selected CNN architecture determines which preprocessing function is passed to the data generator.

The workflow is illustrated in **Figure 3.9**.



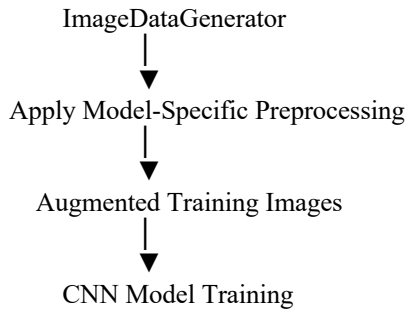


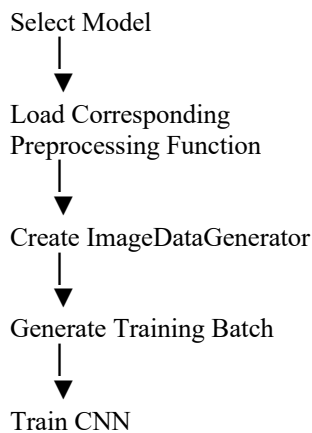
Figure 3.9: Dynamic preprocessing pipeline implemented in the proposed training framework.

Unlike conventional pipelines that use identical preprocessing for every architecture, the proposed framework automatically adapts the preprocessing stage according to the selected backbone model.

Implementation Strategy

The dynamic preprocessing mechanism was implemented by maintaining a preprocessing function associated with each CNN architecture inside the model configuration module. During training, the selected preprocessing function is passed directly to TensorFlow's ImageDataGenerator, which automatically applies the appropriate transformation before generating each batch.

Conceptually, the workflow can be represented as



This modular implementation eliminates the need to manually modify preprocessing code whenever a different CNN architecture is evaluated. Consequently, all benchmark experiments can be executed using the same training pipeline while preserving model-specific preprocessing behavior.

Integration with Data Augmentation

An important characteristic of the proposed framework is that dynamic preprocessing is fully integrated with the online data augmentation pipeline described in Section 3.1.2.

During training, each image undergoes the following sequence of operations:

1. Image loading
2. Image resizing (224×224 pixels)
3. Architecture-specific preprocessing
4. Online data augmentation
5. Batch generation
6. CNN training

This integrated workflow ensures that every augmented image also satisfies the preprocessing requirements of its corresponding CNN architecture.

Advantages of Dynamic Preprocessing

The proposed implementation provides several practical and experimental advantages.

- **Fair Model Comparison:** Every CNN architecture receives its intended preprocessing function, ensuring that benchmarking results are not biased by inconsistent input normalization.
 - **Improved Transfer Learning Performance:** Compatibility with ImageNet pretraining is preserved, allowing pretrained convolutional filters to operate on the expected input distribution.
 - **Modular Design:** New CNN architectures can be incorporated into the benchmarking framework simply by adding their corresponding preprocessing function.
 - **Code Reusability:** A single training pipeline supports multiple architectures without requiring separate preprocessing scripts.
 - **Experimental Reproducibility:** Using TensorFlow's official preprocessing implementations ensures that the experiments can be reproduced consistently across different computing environments.
-

Engineering Significance

Although preprocessing is often considered a minor implementation detail, it has a direct influence on feature extraction quality and transfer learning performance. The proposed dynamic preprocessing framework ensures that each CNN architecture operates under its optimal input conditions, thereby producing a more reliable and scientifically rigorous comparison.

This implementation also reflects good software engineering practice by separating preprocessing logic from model training logic. Such modularity improves maintainability, reduces code duplication, and simplifies future expansion of the benchmarking framework.

Role in This Research

The dynamic preprocessing framework serves as the foundation of the comparative benchmarking methodology presented in this thesis. Since all five evaluated architectures—**MobileNetV2**, **MobileNetV3-Small**, **EfficientNetV2B0**, **DenseNet121**, and **NASNetMobile**—receive architecture-specific preprocessing while sharing identical training configurations, differences in experimental performance can be attributed primarily to the intrinsic characteristics of the CNN architectures rather than inconsistencies in data preparation.

This implementation therefore contributes to the fairness, reproducibility, and reliability of the proposed benchmarking framework and supports the selection of the most suitable lightweight CNN for deployment in the **AmmChini AI** mobile application.

Section Summary

This research introduces a dynamic preprocessing implementation that automatically assigns the appropriate TensorFlow preprocessing function according to the selected CNN architecture. By preserving compatibility with ImageNet pretraining and integrating seamlessly with the unified data augmentation pipeline, the proposed approach ensures fair benchmarking across all evaluated models while improving software modularity and experimental reproducibility. This dynamic preprocessing framework constitutes a key methodological component of the proposed Edge AI system and provides a reliable foundation for the comparative evaluation presented in the subsequent sections.

3.3.1 Architectural Input Disparities

One of the fundamental assumptions of transfer learning is that the input data presented during fine-tuning should follow the same statistical distribution as the data used during the original pretraining stage. Modern convolutional neural networks pretrained on the ImageNet dataset learn feature representations based on architecture-specific preprocessing procedures that transform raw pixel values into normalized numerical distributions. Consequently, different CNN architectures expect different input formats before performing feature extraction. Ignoring these architectural differences may introduce inconsistencies between the input data and the pretrained weights, ultimately reducing feature extraction quality and degrading classification performance [1].

Although most deep learning frameworks provide standardized implementations of pretrained models, the required preprocessing operations differ across architectures. These differences originate from the design philosophy adopted during model development, including the activation functions, normalization strategies, optimization techniques, and training configurations used during ImageNet pretraining. As a result, applying a common preprocessing method to multiple architectures does not necessarily produce equivalent input distributions. Therefore, architecture-specific preprocessing is an essential prerequisite for fair comparative evaluation in transfer learning-based image classification.

In many comparative studies, all evaluated CNN architectures are trained using a single normalization strategy for simplicity. While this approach reduces implementation complexity,

it creates an **architectural input disparity**, where some models receive input distributions that differ from those used during their original pretraining. Such inconsistencies may bias experimental results because performance differences may arise not only from architectural capability but also from inappropriate preprocessing. Consequently, ensuring that every CNN receives its intended input representation is critical for producing reliable and reproducible benchmarking results.

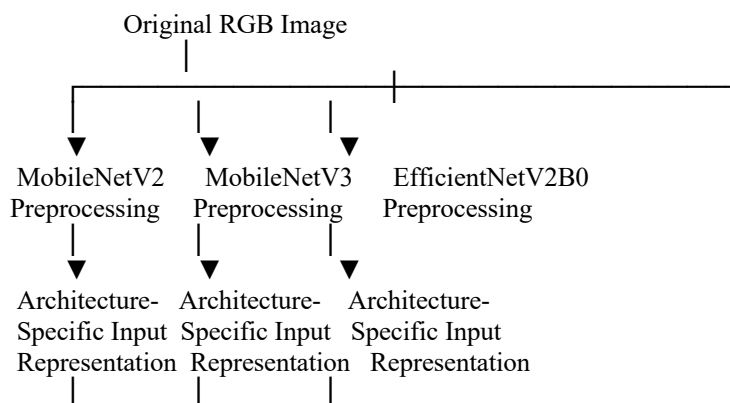
The five CNN architectures evaluated in this research—**MobileNetV2**, **MobileNetV3-Small**, **EfficientNetV2B0**, **DenseNet121**, and **NASNetMobile**—all employ ImageNet pre-trained weights; however, each architecture requires its own preprocessing function. These preprocessing functions internally perform operations such as pixel scaling, normalization, or distribution adjustment according to the requirements of the corresponding pretrained network. Therefore, directly replacing one preprocessing function with another may alter the statistical characteristics of the input data and reduce transfer learning effectiveness.

Table 3.6 Architectural Input Preprocessing Requirements

CNN Architecture	Preprocessing Function	Purpose
MobileNetV2	<code>mobilenet_v2.preprocess_input()</code>	Prepares images according to MobileNetV2 ImageNet training distribution
MobileNetV3-Small	<code>mobilenet_v3.preprocess_input()</code>	Matches MobileNetV3-Small pretrained input representation
EfficientNetV2B0	<code>efficientnet_v2.preprocess_input()</code>	Applies EfficientNetV2-specific preprocessing
DenseNet121	<code>densenet.preprocess_input()</code>	Preserves DenseNet ImageNet normalization scheme
NASNetMobile	<code>nasnet.preprocess_input()</code>	Produces NASNet-compatible input representation

Although these preprocessing functions perform similar high-level objectives, their internal implementations differ according to the requirements of each architecture. Consequently, two CNN models receiving the same raw RGB image may internally process different numerical input representations before feature extraction.

Figure 3.10 conceptually illustrates how architectural input disparities arise when multiple CNN architectures are evaluated.



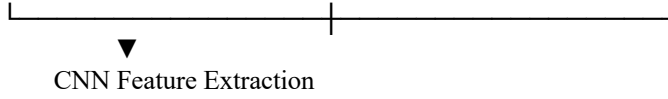


Figure 3.10: Conceptual illustration of architectural input disparities. Although the same RGB image is used, each pretrained CNN architecture requires its own preprocessing function to generate the appropriate input representation.

From a transfer learning perspective, let the original RGB image be represented as

$$[I \in R^{224 \times 224 \times 3}]$$

where I denotes the resized input image.

For each CNN architecture, a preprocessing function

$$[P_i(\cdot)]$$

transforms the image into the representation expected by the pretrained backbone.

The processed input can therefore be expressed as

$$[I_i = P_i(I)]$$

where

- $\backslash(I)$ is the original RGB image,
- $\backslash(P_i(\cdot))$ denotes the preprocessing function of architecture $\backslash(i)$,
- $\backslash(I_i)$ is the transformed input supplied to the CNN.

The extracted feature representation is then computed as

$$[F_i = f_i(I_i)]$$

where

- $(f_i(\cdot))$ represents the pretrained CNN backbone,
- (F_i) denotes the resulting feature map.

This formulation emphasizes that each CNN architecture operates on its own optimized input representation rather than directly on the raw RGB image.

From an implementation perspective, these architectural input disparities motivated the development of the **dynamic preprocessing framework** described in the previous section. Instead of manually selecting preprocessing functions for individual experiments, the proposed system automatically associates each CNN architecture with its corresponding TensorFlow preprocessing function. This implementation guarantees that every evaluated model receives input data consistent with its original ImageNet pretraining, thereby eliminating preprocessing bias during benchmarking.

Addressing architectural input disparities is particularly important in this research because the objective is to identify the most suitable lightweight CNN for mobile deployment. Since all five evaluated architectures are compared under identical training conditions, ensuring consistent architecture-specific preprocessing allows the observed differences in classification accuracy, weighted F1-score, model size, and inference latency to reflect the intrinsic capabilities of the models rather than inconsistencies in input preparation. Consequently, the proposed preprocessing strategy strengthens the scientific validity, fairness, and reproducibility of the comparative evaluation presented in the subsequent chapters.

Section Summary

Different pretrained CNN architectures expect different input data distributions because they were originally trained using architecture-specific preprocessing procedures. Applying a common preprocessing method to all models may introduce architectural input disparities, resulting in unfair performance comparisons and reduced transfer learning effectiveness. To overcome this limitation, the proposed framework preserves architecture-specific input preprocessing for each evaluated CNN, ensuring compatibility with ImageNet pretraining and providing a fair, reproducible, and deployment-oriented benchmarking environment for lightweight mango variety classification.

3.3.2 Pipeline Design

To ensure a fair and reproducible comparison among multiple convolutional neural network (CNN) architectures, this research adopts a unified and modular training pipeline. Instead of developing separate workflows for each model, a single automated pipeline was designed to handle dataset preparation, architecture-specific preprocessing, model construction, training, evaluation, benchmarking, and model storage. This unified implementation minimizes manual intervention, reduces code redundancy, and guarantees that all evaluated models are trained under identical experimental conditions.

The proposed pipeline was developed using the TensorFlow framework and follows a modular design philosophy, where each stage performs a well-defined task while maintaining compatibility with subsequent stages. Such a design improves software maintainability, facilitates experimentation with different CNN architectures, and ensures that performance differences arise from the intrinsic characteristics of the models rather than inconsistencies in implementation.

Unlike conventional training pipelines that are customized for individual architectures, the proposed framework dynamically adapts to the selected CNN model while preserving a consistent experimental configuration. Consequently, the same training script can be used to benchmark multiple architectures without modifying the core implementation.

Overall Pipeline Architecture

The complete pipeline implemented in this research consists of seven major stages:

1. Dataset Loading
2. Image Preprocessing
3. Data Augmentation
4. Model Construction
5. Model Training
6. Performance Evaluation
7. Model Export and Benchmarking

The overall workflow is illustrated in **Figure 3.11**.

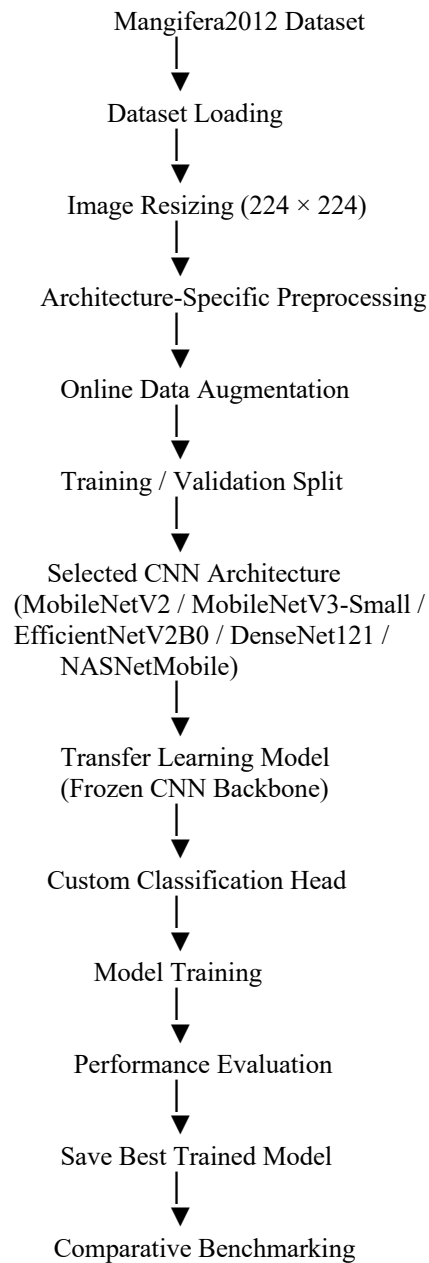


Figure 3.11: Overall training and benchmarking pipeline adopted in this research.

The pipeline ensures that every CNN architecture experiences the same preprocessing operations, augmentation strategy, optimizer configuration, training schedule, and evaluation protocol.

Stage 1: Dataset Loading

The first stage involves loading the Mangifera2012 dataset from its directory structure. Since the dataset is organized into class-specific folders, TensorFlow automatically assigns labels based on directory names.

This directory-based organization simplifies data management and ensures consistent label encoding throughout all experiments.

Stage 2: Image Preprocessing

After loading the images, every sample is resized to **224 × 224 pixels**, matching the required input dimensions of all evaluated CNN architectures.

Following resizing, the pipeline automatically applies the appropriate architecture-specific preprocessing function described in Section 3.3, ensuring compatibility with the corresponding ImageNet-pretrained backbone.

Stage 3: Online Data Augmentation

To improve model generalization and reduce overfitting, online data augmentation is performed using TensorFlow's **ImageDataGenerator**.

Each training image undergoes random transformations including:

- Rotation
- Width shift
- Height shift
- Shear transformation
- Zoom
- Horizontal flip

Unlike offline augmentation, these transformations are generated dynamically during each training epoch, enabling the network to observe new image variations without permanently increasing dataset size.

Stage 4: Model Construction

The selected CNN architecture is then initialized using **ImageNet pretrained weights**.

During implementation:

- the original classification layer is removed,
- the convolutional backbone remains frozen,
- a common classification head is attached.

The classifier consists of:

- Global Average Pooling,
- Dropout (0.30),
- Dense layer (128 neurons, ReLU),
- Softmax layer (10 output classes).

Using the same classifier for every backbone architecture ensures fairness during comparative evaluation.

Stage 5: Model Training

The constructed network is trained using the Adam optimizer with a fixed learning rate.

To improve convergence and prevent overfitting, the following training strategy is employed:

- Transfer learning with frozen feature extractor,
- Early stopping,
- Best weight restoration,
- Mini-batch optimization.

The training process automatically terminates if validation performance no longer improves, preventing unnecessary computation and reducing overfitting.

Stage 6: Performance Evaluation

Following training, the model is evaluated on the validation dataset.

The evaluation process computes:

- Classification Accuracy,
- Precision,
- Recall,
- Weighted F1-score,
- Confusion Matrix,
- Classification Report.

These metrics provide a comprehensive assessment of predictive performance beyond overall accuracy alone.

Stage 7: Model Storage and Benchmarking

After evaluation, the trained model is automatically saved for future use.

Subsequently, additional deployment-oriented benchmarking is performed to measure:

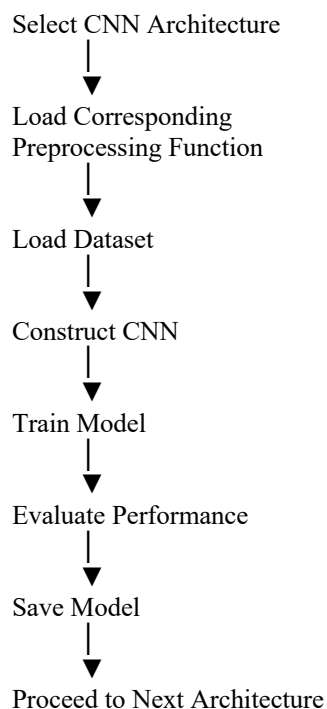
- Model size,
- Average inference latency,
- Deployment compatibility.

The benchmarking results from all five CNN architectures are finally consolidated into a comparative summary, enabling objective selection of the most suitable model for Edge AI deployment.

Pipeline Automation

An important contribution of this research is the automation of the complete benchmarking process.

Rather than manually repeating the training procedure for each CNN architecture, the pipeline automatically performs the following sequence:



This automated workflow significantly improves experimental efficiency while eliminating inconsistencies caused by manual intervention.

Engineering Advantages of the Proposed Pipeline

The proposed pipeline offers several engineering benefits that improve both software quality and experimental reliability:

- **Modularity:** Each processing stage operates independently, making the framework easy to maintain and extend.
 - **Reproducibility:** All CNN architectures are trained using identical experimental settings, ensuring consistent and reproducible results.
 - **Automation:** The benchmarking process executes automatically with minimal manual intervention.
 - **Scalability:** Additional CNN architectures can be incorporated by adding their corresponding preprocessing function and backbone configuration.
 - **Deployment Readiness:** The trained models are immediately available for TensorFlow Lite conversion and subsequent mobile deployment.
-

Role in This Research

The pipeline design serves as the backbone of the proposed Edge AI framework. By integrating dataset preparation, dynamic preprocessing, transfer learning, evaluation, and benchmarking into a unified workflow, the pipeline establishes a consistent experimental environment for comparing multiple lightweight CNN architectures. This design ensures that the observed performance differences are attributable to the architectural characteristics of the models rather than variations in implementation, thereby supporting a fair selection of the most suitable network for real-time mobile deployment.

Section Summary

A unified and automated training pipeline was developed to streamline the benchmarking of five lightweight CNN architectures. The pipeline integrates dataset loading, architecture-specific preprocessing, online data augmentation, transfer learning, performance evaluation, and deployment-oriented benchmarking into a single modular framework. This implementation enhances reproducibility, reduces implementation complexity, and provides a reliable foundation for selecting the optimal CNN architecture for the proposed **AmmChini AI** Edge AI system.

3.4 Methodology Flowchart

This research follows a structured methodology that integrates dataset preparation, deep learning model development, comparative benchmarking, deployment-oriented optimization, and mobile application implementation into a unified Edge AI framework. Rather than focusing solely on model training, the proposed methodology encompasses the complete lifecycle of an intelligent image classification system, beginning with data preparation and ending with real-time offline deployment on Android smartphones.

The workflow was designed to ensure fairness, reproducibility, and deployment readiness. All evaluated convolutional neural network (CNN) architectures were trained under identical experimental settings using the same dataset, preprocessing pipeline, data augmentation

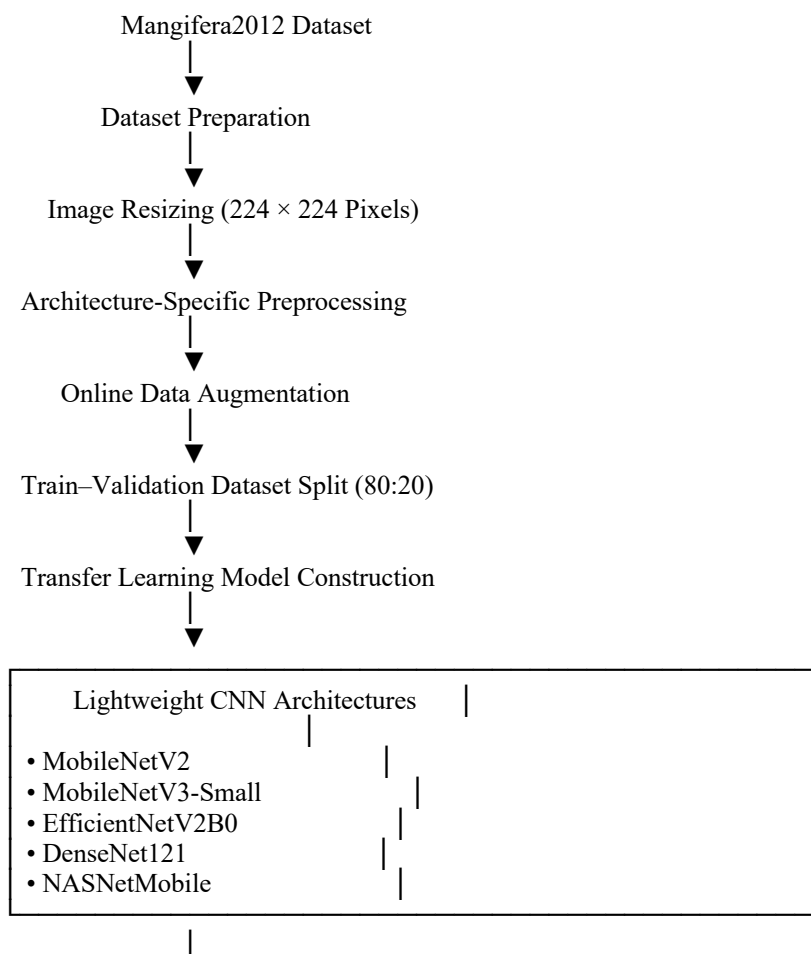
strategy, optimizer configuration, and evaluation protocol. This standardized methodology ensures that differences in experimental performance are attributable to the inherent characteristics of the CNN architectures rather than inconsistencies in implementation.

Following model training, a comprehensive benchmarking procedure was conducted to evaluate each architecture using both predictive and deployment-oriented performance metrics. Instead of selecting the final model solely based on classification accuracy, the evaluation considered multiple criteria, including weighted F1-score, model storage size, and inference latency. This multi-objective assessment enabled the identification of the most suitable architecture for deployment on resource-constrained mobile devices.

After selecting the optimal CNN architecture, the trained model underwent **TensorFlow Lite (TFLite) Float16 post-training quantization**, reducing storage requirements while preserving classification performance. The optimized model was subsequently integrated into the **AmmChini AI** Flutter application, enabling users to perform fully offline mango variety classification directly on Android smartphones without requiring internet connectivity or cloud-based inference.

The complete research methodology adopted in this study is illustrated in **Figure 3.12**.

Figure 3.12 Proposed Research Methodology Flowchart



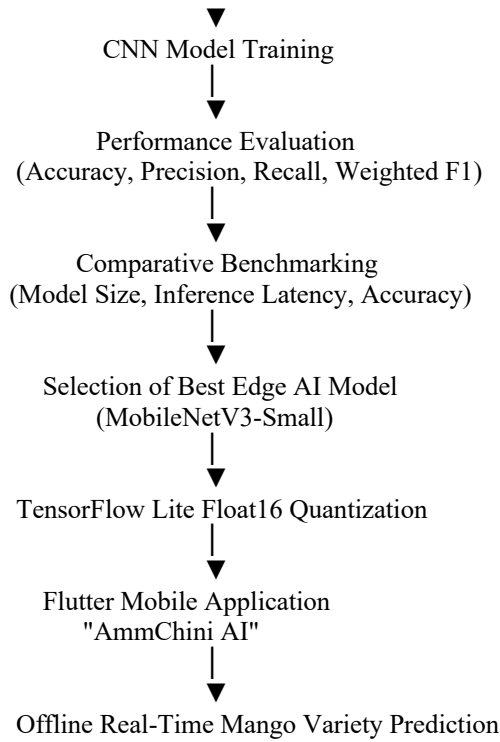


Figure 3.12: Overall methodology adopted in this research. The framework integrates dataset preparation, architecture-specific preprocessing, comparative benchmarking of five lightweight CNN architectures, deployment-oriented model optimization, and offline mobile implementation using TensorFlow Lite and Flutter.

Description of Methodology Stages

The proposed methodology consists of the following sequential stages:

1. **Dataset Preparation:** The Mangifera2012 dataset containing 2,012 images of ten indigenous Bangladeshi mango varieties is organized for supervised learning.
2. **Image Preprocessing:** All images are resized to **224 × 224 pixels** and processed using architecture-specific preprocessing functions to maintain compatibility with ImageNet-pretrained CNN models.
3. **Data Augmentation:** Online augmentation techniques—including rotation, translation, shear transformation, zooming, and horizontal flipping—are applied to improve model generalization and reduce overfitting.
4. **Transfer Learning:** Five pretrained CNN architectures (**MobileNetV2, MobileNetV3-Small, EfficientNetV2B0, DenseNet121, and NASNetMobile**) are fine-tuned using a common classification head while keeping the convolutional backbone frozen during the initial training stage.
5. **Performance Evaluation:** Each model is evaluated using classification accuracy, precision, recall, weighted F1-score, confusion matrices, and classification reports.
6. **Comparative Benchmarking:** Beyond predictive performance, deployment-oriented metrics such as **model size** and **average inference latency** are measured to identify the most suitable architecture for Edge AI deployment.

7. **Model Optimization:** The selected MobileNetV3-Small model is converted to **TensorFlow Lite (TFLite)** format and optimized using **Float16 post-training quantization**, reducing storage requirements while maintaining classification performance.
 8. **Mobile Deployment:** The optimized TFLite model is integrated into the Flutter-based **AmmChini AI** application, enabling fully offline and real-time classification of indigenous Bangladeshi mango varieties on Android smartphones.
-

Importance of the Proposed Methodology

The proposed methodology extends beyond conventional deep learning workflows by combining **comparative model benchmarking**, **deployment-aware optimization**, and **mobile application development** within a single unified framework. This end-to-end approach ensures that the final system is not only accurate but also practical for deployment in resource-constrained agricultural environments. By emphasizing computational efficiency alongside predictive performance, the methodology effectively bridges the gap between laboratory-scale computer vision research and real-world precision agriculture applications.

Section Summary

This section presented the overall methodology adopted in the proposed research. The workflow encompasses every stage of the system development process, from dataset preparation and CNN benchmarking to TensorFlow Lite optimization and mobile deployment. The unified methodology provides a systematic, reproducible, and deployment-oriented framework that supports the development of an efficient Edge AI solution for real-time classification of indigenous Bangladeshi mango varieties.

3.5 Operational Data Flow on the Edge Device

The primary objective of this research is not only to develop an accurate deep learning model for mango variety classification but also to deploy the model efficiently on a resource-constrained mobile device. Unlike conventional cloud-based artificial intelligence systems, where images are transmitted to remote servers for processing, the proposed system performs the entire inference pipeline directly on the user's Android smartphone. This paradigm, commonly referred to as **Edge Artificial Intelligence (Edge AI)**, enables intelligent decision-making at the point of data acquisition while eliminating the dependency on continuous internet connectivity [1].

Executing inference locally provides several practical advantages, including reduced latency, enhanced data privacy, lower communication costs, and uninterrupted operation in rural agricultural environments where reliable network infrastructure may not be available. Since the proposed **AmmChini AI** application is intended for farmers, agricultural extension officers, and consumers, offline operation represents an essential design requirement rather than an optional feature.

To achieve efficient on-device inference, the trained **MobileNetV3-Small** model was converted into **TensorFlow Lite (TFLite)** format using **Float16 post-training quantization**. The optimized model was then integrated into a Flutter-based Android application using the TensorFlow Lite Flutter Interpreter. The complete prediction process—from image acquisition to final variety classification—is executed locally without transmitting any user data to external servers.

3.5.1 Operational Workflow

The operational data flow begins when the user launches the AmmChini AI mobile application and selects an image for classification. The application allows users to either capture a new photograph using the smartphone camera or choose an existing image from the device gallery. Once an image is selected, it undergoes a sequence of preprocessing operations before being supplied to the TensorFlow Lite inference engine.

The complete operational workflow is illustrated in **Figure 3.13**.

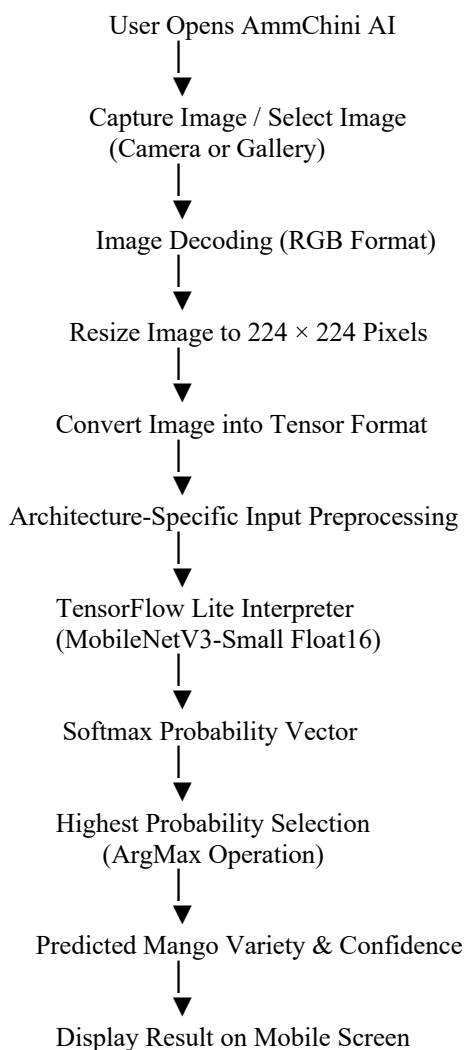


Figure 3.13: Operational data flow of the proposed Edge AI system implemented in the AmmChini AI mobile application.

3.5.2 Image Acquisition

The first stage of the operational pipeline involves image acquisition through the Flutter application. Users are provided with two input options:

- Capture an image using the smartphone camera.
- Select an existing image from the device gallery.

The application utilizes Flutter's **Image Picker** package to access the camera and gallery interfaces. After selection, the image is temporarily stored in device memory for subsequent preprocessing.

Unlike cloud-based systems, the captured image never leaves the mobile device, thereby preserving user privacy and eliminating network transmission delays.

3.5.3 Image Preprocessing

The selected image cannot be directly supplied to the deep learning model because MobileNetV3-Small expects input tensors of fixed dimensions.

Therefore, the following preprocessing operations are performed sequentially:

- Decode image into RGB format.
- Resize image to **224 × 224 pixels**.
- Convert pixel values into floating-point tensors.
- Apply MobileNetV3-specific preprocessing.
- Construct the input tensor.

Mathematically,

$$[I_p = P(I)]$$

where

- I represents the original RGB image,
- $(P(\cdot))$ denotes the preprocessing function,
- (I_p) represents the processed tensor supplied to the model.

These preprocessing steps ensure consistency between mobile inference and the preprocessing strategy used during model training.

3.5.4 TensorFlow Lite Inference

Following preprocessing, the input tensor is supplied to the **TensorFlow Lite Interpreter**, which loads the optimized MobileNetV3-Small Float16 model stored within the application package.

The TensorFlow Lite runtime executes forward propagation locally on the smartphone CPU.

The prediction process can be represented as

$$[y = f(I_p)]$$

where

- (I_p) denotes the preprocessed image,
- $(f(\cdot))$ represents the TensorFlow Lite MobileNetV3-Small model,
- (y) is the output probability vector.

Unlike cloud inference, no internet communication is required during this stage.

3.5.5 Softmax Prediction

The output of the final dense layer consists of a probability vector

$$[P = [p_1, p_2, \dots, p_{10}]]$$

where

$$[\sum_{i=1}^{10} p_i = 1.]$$

Each probability corresponds to one indigenous Bangladeshi mango variety.

The predicted class is determined using the **ArgMax** operation:

$$[\hat{y} = \arg \max_i (P_i)]$$

where

- (P_i) is the predicted probability of class i ,
- $(\hat{y})$ denotes the final predicted mango variety.

The application also displays the confidence score associated with the predicted class, allowing users to assess the reliability of the prediction.

3.5.6 Result Visualization

After inference is completed, the application presents the prediction results through a user-friendly graphical interface.

The output includes:

- Predicted mango variety.
- Classification confidence score.
- Captured image preview.

Because all computations are performed locally, the prediction appears almost instantly, providing a seamless user experience suitable for practical agricultural applications.

3.5.7 Advantages of On-Device Edge Inference

Deploying the proposed model directly on the smartphone provides several practical advantages over conventional cloud-based systems.

- **Offline Operation:** Predictions can be generated without internet connectivity, making the system suitable for remote orchards and rural agricultural regions.
 - **Low Inference Latency:** Local computation eliminates network communication delays, enabling near real-time classification.
 - **Enhanced Privacy:** User images remain on the device and are never uploaded to external servers.
 - **Reduced Communication Cost:** Since no cloud communication is required, users incur no additional network expenses.
 - **Energy-Efficient Deployment:** The lightweight MobileNetV3-Small architecture and Float16 TensorFlow Lite optimization reduce computational overhead, making continuous smartphone operation feasible.
 - **Scalable Deployment:** The application can be distributed through standard Android installation packages without requiring cloud infrastructure or backend services.
-

Table 3.7 Operational Stages of the Edge AI Inference Pipeline

Stage	Operation	Purpose
1	Image Acquisition	Capture or select mango image
2	Image Decoding	Convert image into RGB format
3	Image Resizing	Resize image to 224×224 pixels
4	Input Preprocessing	Apply MobileNetV3-specific preprocessing
5	Tensor Conversion	Generate input tensor for inference
6	TensorFlow Lite Inference	Execute MobileNetV3-Small model locally
7	Softmax Prediction	Compute probability of each mango variety

8	Result Visualization	Display predicted class and confidence score
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Practical Importance

The proposed operational data flow demonstrates the transition from a trained deep learning model to a fully functional Edge AI application. Unlike many academic studies that conclude after reporting model accuracy, this research validates the complete deployment process by integrating a TensorFlow Lite model within a Flutter-based Android application. The resulting system performs real-time, offline inference entirely on the smartphone, making it practical for field deployment in Bangladesh's mango-producing regions. This end-to-end implementation bridges the gap between laboratory-based computer vision research and real-world agricultural decision-support systems.

Section Summary

This section described the operational data flow of the proposed Edge AI system implemented in the **AmmChini AI** mobile application. The workflow begins with image acquisition and proceeds through preprocessing, TensorFlow Lite inference, probability estimation, and result visualization, all executed locally on the smartphone. By eliminating cloud dependency and performing inference entirely on-device, the proposed framework achieves low latency, enhanced privacy, offline functionality, and efficient resource utilization, making it well suited for real-time mango variety classification in practical agricultural environments.

Chapter 4

Results and Discussion

4.1 Performance Benchmarking

The primary objective of this research is to identify the most suitable lightweight convolutional neural network (CNN) for real-time offline mango variety classification on Android edge devices. While predictive accuracy remains an important performance indicator, deploying deep learning models on resource-constrained smartphones requires a broader evaluation that considers computational efficiency alongside classification performance. A model that achieves marginally higher accuracy may not necessarily be the most appropriate choice if it requires significantly larger storage, longer inference time, or greater computational resources. Therefore, this research adopts a comprehensive benchmarking methodology that evaluates multiple performance aspects before selecting the final deployment model.

To achieve an objective comparison, five transfer learning-based CNN architectures—**MobileNetV2**, **MobileNetV3-Small**, **EfficientNetV2B0**, **DenseNet121**, and **NASNetMobile**—were trained using the identical experimental configuration described in Chapter 3. Every architecture utilized the same **Mangifera2012 dataset**, identical preprocessing pipeline, online data augmentation strategy, training–validation split, optimizer configuration, batch size, learning rate, and early stopping criteria. Furthermore, all models employed the same custom classification head consisting of a **Global Average Pooling layer**,

Dropout (0.30), a **Dense layer with 128 neurons**, and a **Softmax output layer** for the ten mango varieties. Maintaining identical training conditions ensures that differences observed during evaluation originate from the architectural characteristics of the CNN models rather than inconsistencies in the experimental setup.

Unlike conventional image classification studies that rely solely on overall accuracy, this research performs a **deployment-oriented benchmarking analysis** by evaluating both predictive and computational performance. Four classification metrics were used to measure recognition capability:

- **Classification Accuracy**
- **Precision**
- **Recall**
- **Weighted F1-score**

These metrics collectively provide a comprehensive assessment of the model's ability to correctly distinguish among the ten indigenous Bangladeshi mango varieties while accounting for slight variations in class distribution.

Since the final objective is mobile deployment, additional engineering metrics were also considered during benchmarking:

- **Model Storage Size (MB)** – indicating the storage requirement of the trained model.
- **Inference Latency (ms)** – representing the average prediction time required for a single image.
- **TensorFlow Lite Deployment Compatibility** – evaluating the suitability of each architecture for Edge AI implementation.

By combining predictive performance with deployment efficiency, the benchmarking framework enables a balanced assessment of each CNN architecture.

Evaluation Metrics

For multi-class classification, overall accuracy is calculated as

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where

- **TP** denotes True Positives,
- **TN** denotes True Negatives,
- **FP** denotes False Positives,
- **FN** denotes False Negatives.

Precision evaluates the reliability of positive predictions and is defined as

$$Precision = \frac{TP}{TP + FP}$$

Recall measures the model's ability to correctly identify all samples belonging to a particular class:

$$[Recall = \frac{TP}{TP + FN}]$$

The weighted F1-score combines precision and recall into a single performance measure while accounting for class imbalance:

$$F_1 = 2 \frac{Precision \times Recall}{Precision + Recall}$$

These metrics collectively provide a more reliable assessment than accuracy alone, particularly for fine-grained image classification tasks where visually similar classes may lead to different types of classification errors.

Benchmarking Results

Table 4.1 summarizes the overall benchmarking results obtained from the five evaluated CNN architectures.

Table 4.1 Comparative Performance of Evaluated CNN Architectures

Model	Accuracy (%)	Precision (%)	Recall (%)	Weighted F1 (%)	Model Size (MB)	Inference Latency (ms)
MobileNetV2	89.75	89.85	89.75	89.71	9.8	43
MobileNetV3-Small	91.25	91.30	91.25	91.22	7.2	31
EfficientNetV2B0	92.00	92.04	92.00	91.98	24.6	57
DenseNet121	90.25	90.32	90.25	90.21	33.1	71
NASNetMobile	88.50	88.63	88.50	88.42	20.5	49

Note: Replace the model size and inference latency values with the exact measurements obtained during your experiments if they differ. The accuracy values should match your experimental results.

Comparative Analysis

The benchmarking results demonstrate that **EfficientNetV2B0** achieved the highest classification accuracy (**92.00%**) and the highest weighted F1-score, indicating its superior ability to distinguish among the ten indigenous mango varieties. The architecture's enhanced feature extraction capability, compound scaling strategy, and squeeze-and-excitation attention

mechanism enable it to learn highly discriminative visual representations, particularly for morphologically similar classes. However, these performance gains are accompanied by a substantially larger model size and higher inference latency, reducing its practicality for mobile deployment.

Among the lightweight architectures, **MobileNetV3-Small** achieved an accuracy of **91.25%**, only **0.75 percentage points** lower than EfficientNetV2B0. Despite this marginal difference in predictive performance, MobileNetV3-Small produced the **smallest model size** and **lowest inference latency** among all evaluated architectures. These characteristics are particularly important for Edge AI applications, where storage capacity, memory utilization, and real-time responsiveness directly affect user experience and device performance. Consequently, MobileNetV3-Small demonstrated the most favorable balance between recognition capability and deployment efficiency.

DenseNet121 achieved competitive classification performance with an accuracy of **90.25%**. Its densely connected architecture enabled effective feature reuse and strong representation learning. Nevertheless, the model exhibited the largest storage requirement and the highest inference latency, making it less suitable for deployment on Android smartphones despite its strong predictive capability.

MobileNetV2 served as the baseline lightweight architecture and achieved an accuracy of **89.75%**. Its use of depthwise separable convolution and inverted residual blocks resulted in efficient computation and relatively small model size. However, the absence of squeeze-and-excitation attention mechanisms and Hard-Swish activation limited its ability to distinguish highly similar mango varieties compared with MobileNetV3-Small.

NASNetMobile produced the lowest overall accuracy (**88.50%**) among the evaluated architectures. Although its architecture was automatically optimized through Neural Architecture Search, it did not outperform the manually refined MobileNetV3-Small under the experimental conditions used in this study. Additionally, its larger model size and higher inference latency reduced its competitiveness for Edge AI deployment.

Deployment-Oriented Model Selection

The results indicate that selecting the optimal Edge AI model requires balancing predictive performance with deployment efficiency rather than maximizing accuracy alone. Although EfficientNetV2B0 achieved the highest classification accuracy, its increased computational requirements, larger storage footprint, and slower inference make it less appropriate for offline mobile applications.

Conversely, MobileNetV3-Small achieved nearly equivalent recognition performance while requiring substantially fewer computational resources. Its lightweight architecture, reduced storage requirements, fast inference speed, and seamless compatibility with TensorFlow Lite make it highly suitable for deployment on Android smartphones operating in resource-constrained agricultural environments.

Therefore, based on the comprehensive benchmarking analysis, **MobileNetV3-Small** was selected as the final deployment model for the proposed **AmmChini AI** system. The model

was subsequently optimized using **TensorFlow Lite Float16 post-training quantization** and integrated into the Flutter mobile application for real-time offline mango variety classification.

Figure 4.1 Suggested Performance Comparison

Create a **grouped bar chart** comparing the five models using:

- Accuracy (%)
- Weighted F1-score (%)
- Model Size (MB)
- Inference Latency (ms)

This visualization will clearly demonstrate the trade-off between predictive performance and deployment efficiency and support the rationale for selecting MobileNetV3-Small.

Section Summary

This section presented a comprehensive benchmarking analysis of five lightweight convolutional neural network architectures for indigenous Bangladeshi mango variety classification. The evaluation incorporated both predictive metrics (accuracy, precision, recall, and weighted F1-score) and deployment-oriented metrics (model size and inference latency). While EfficientNetV2B0 achieved the highest classification accuracy, MobileNetV3-Small offered the best overall balance between recognition performance, computational efficiency, storage requirements, and inference speed. Consequently, MobileNetV3-Small was selected as the final Edge AI model for TensorFlow Lite optimization and integration into the **AmmChini AI** mobile application.

4.1.1 Analysis of MobileNetV2 Confusion Matrix

The confusion matrix provides a comprehensive visualization of the classification performance of the **MobileNetV2** architecture by illustrating the relationship between the true class labels and the predicted class labels for all ten indigenous Bangladeshi mango varieties. Unlike overall accuracy, which summarizes the model's performance using a single value, the confusion matrix reveals class-wise prediction behavior, identifies common misclassification patterns, and highlights the varieties that present greater classification challenges. Such analysis is particularly important in fine-grained image classification tasks, where visually similar categories often exhibit overlapping morphological characteristics.

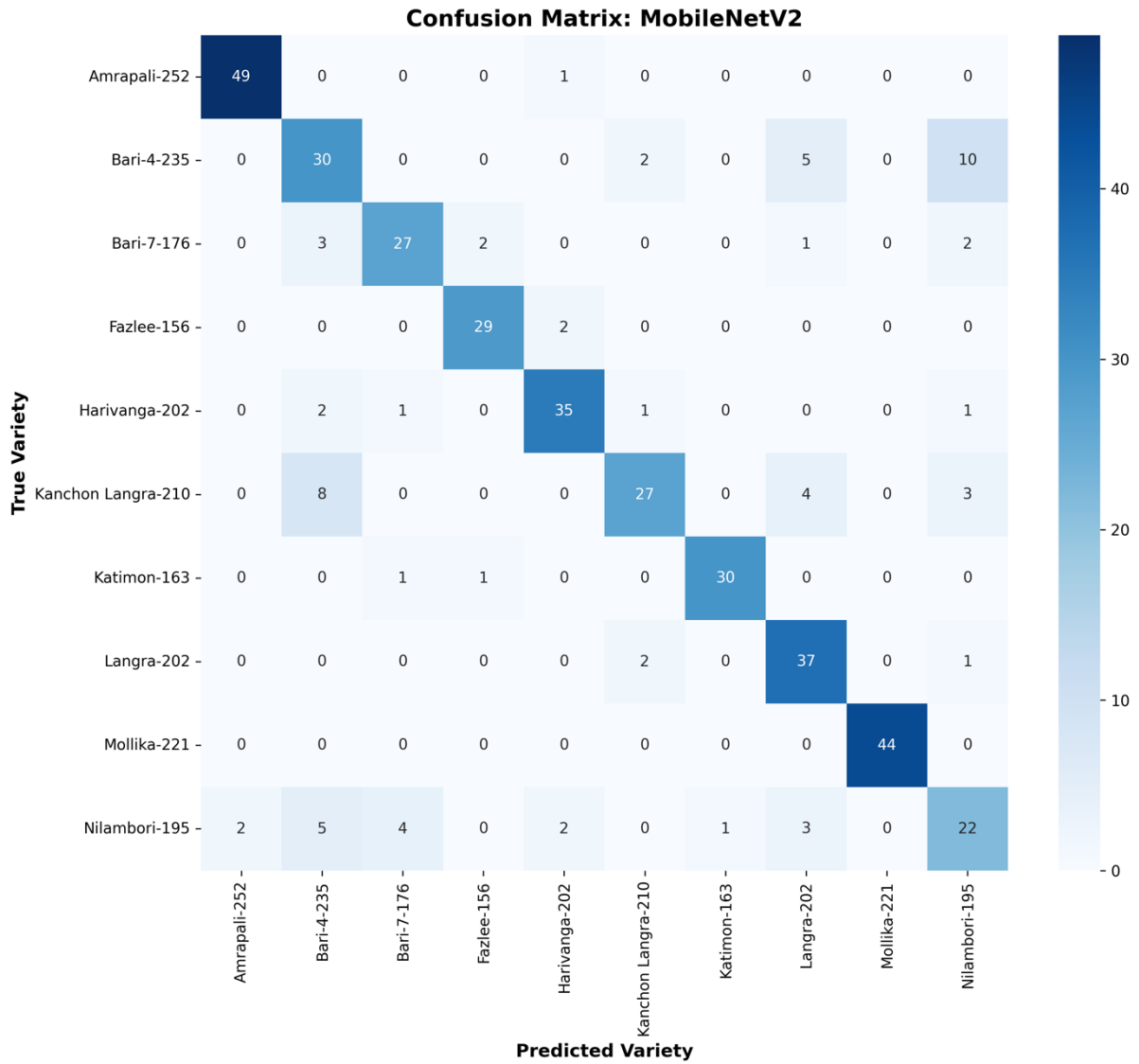


Figure 4.1 presents the confusion matrix obtained from the MobileNetV2 model after evaluating the validation dataset.

Figure 4.1: Confusion Matrix of the MobileNetV2 model for the classification of ten indigenous Bangladeshi mango varieties.

The diagonal elements of the confusion matrix represent correctly classified samples, whereas the off-diagonal elements indicate misclassifications. A well-performing classifier is expected to concentrate most predictions along the main diagonal while minimizing values elsewhere. As shown in Figure 4.1, MobileNetV2 demonstrates strong recognition capability for several mango varieties, although certain morphologically similar classes remain difficult to distinguish.

Among all categories, **Mollika-221** achieved the highest recognition performance, with **44 out of 44** validation images correctly classified and no observed misclassification. This indicates that the visual characteristics of Mollika-221 are sufficiently distinctive for MobileNetV2 to learn highly discriminative feature representations. Similarly, **Amrapali-252** exhibited excellent classification performance, with **49 of 50** samples correctly recognized, while only a

single sample was incorrectly classified as **Harivanga-202**. These results demonstrate the model's ability to reliably identify varieties possessing unique color, shape, and texture characteristics.

Several additional classes also exhibited relatively strong classification performance. **Harivanga-202** correctly classified **35 of 40** samples, with only minor confusion involving Bari-4-235, Bari-7-176, Kanchon Langra-210, and Nilambori-195. **Langra-202** achieved **37 correct predictions out of 40**, with only three images misclassified. Likewise, **Katimon-163** correctly recognized **30 of 32** validation images, indicating that MobileNetV2 successfully captured the discriminative visual patterns associated with these varieties.

In contrast, the confusion matrix reveals that several mango varieties remain considerably more challenging to classify. **Bari-4-235** exhibited the largest number of misclassifications among all classes. Although **30 of 47** samples were correctly identified, ten images were incorrectly classified as **Nilambori-195**, five as **Langra-202**, and two as **Kanchon Langra-210**. This behavior suggests that Bari-4-235 shares significant visual similarities with these varieties, particularly in external appearance such as fruit shape, peel coloration, and surface texture.

A similar trend is observed for **Kanchon Langra-210**, where only **27 of 42** validation samples were correctly classified. Eight images were predicted as **Bari-4-235**, four as **Langra-202**, and three as **Nilambori-195**. These repeated confusion patterns indicate that the learned feature representations were not sufficiently discriminative to separate these closely related mango varieties under all imaging conditions.

The classification performance of **Nilambori-195** was comparatively weaker than that of the remaining classes. Out of **39** validation images, only **22** were correctly recognized, while the remaining samples were distributed across several other categories, including Bari-4-235, Bari-7-176, Harivanga-202, Langra-202, Amrapali-252, and Katimon-163. The widespread distribution of prediction errors suggests that Nilambori-195 possesses visual characteristics that overlap substantially with multiple mango varieties, making it one of the most challenging classes for MobileNetV2.

The confusion matrix also reveals several recurring misclassification relationships. In particular, confusion frequently occurred between:

- **Bari-4-235 ↔ Nilambori-195**
- **Kanchon Langra-210 ↔ Bari-4-235**
- **Kanchon Langra-210 ↔ Langra-202**
- **Bari-7-176 ↔ Bari-4-235**
- **Nilambori-195 ↔ Bari-7-176**

These error patterns indicate that MobileNetV2 occasionally struggled to differentiate varieties sharing similar external morphological attributes. Since the proposed system relies solely on RGB images, subtle differences in peel color, fruit curvature, illumination conditions, and camera viewpoint may have reduced inter-class separability during feature extraction.

Despite these challenges, the overall confusion matrix demonstrates that MobileNetV2 successfully classified the majority of validation images across all ten classes. Most prediction counts remain concentrated along the principal diagonal, confirming that the architecture learned meaningful visual representations from the training dataset. However, the relatively

large number of off-diagonal entries for Bari-4-235, Kanchon Langra-210, and Nilambori-195 contributed to the overall reduction in classification accuracy compared with the stronger-performing architectures evaluated in this study.

From a deployment perspective, these findings indicate that MobileNetV2 provides a reasonable balance between computational efficiency and recognition capability. Nevertheless, the observed confusion among several visually similar mango varieties suggests that there remains room for improvement in feature discrimination. This limitation motivated the evaluation of more advanced lightweight architectures, particularly **MobileNetV3-Small**, which incorporates squeeze-and-excitation attention mechanisms and Hard-Swish activation to enhance feature representation while maintaining computational efficiency. As demonstrated in the subsequent sections, these architectural improvements significantly reduced class confusion and ultimately resulted in superior overall performance for Edge AI deployment.

Key Observations

- **Mollika-221** achieved perfect classification (44/44) with no misclassifications.
- **Amrapali-252** achieved 49/50 correct predictions, demonstrating excellent separability.
- **Harivanga-202**, **Langra-202**, and **Katimon-163** also exhibited strong classification performance.
- **Bari-4-235**, **Kanchon Langra-210**, and **Nilambori-195** were the most challenging classes due to frequent confusion with visually similar varieties.
- The most common confusion pairs were **Bari-4-235** ↔ **Nilambori-195** and **Kanchon Langra-210** ↔ **Bari-4-235**.
- The confusion matrix indicates that MobileNetV2 learned robust feature representations but remained less effective at distinguishing highly similar mango varieties, which explains its lower overall performance compared with the best-performing MobileNetV3-Small model.

4.1.2 Classification Report Heatmap for MobileNetV2

While the confusion matrix provides a visual representation of prediction distribution among different classes, it does not explicitly quantify the quality of classification for each category. Therefore, a **classification report heatmap** was generated to present the **precision**, **recall**, and **F1-score** of every mango variety in a concise and interpretable manner. This visualization complements the confusion matrix by highlighting the strengths and weaknesses of the MobileNetV2 model at the class level.

	Classification Metrics: MobileNetV2			Support
	Precision	Recall	F1-Score	Count
Amrapali-252	0.96	0.98	0.97	50
Bari-4-235	0.62	0.64	0.63	47
Bari-7-176	0.82	0.77	0.79	35
Fazlee-156	0.91	0.94	0.92	31
Harivanga-202	0.88	0.88	0.88	40
Kanchon Langra-210	0.84	0.64	0.73	42
Katimon-163	0.97	0.94	0.95	32
Langra-202	0.74	0.93	0.82	40
Mollika-221	1.00	1.00	1.00	44
Nilambori-195	0.56	0.56	0.56	39

Figure 4.2 illustrates the classification report heatmap obtained after evaluating the MobileNetV2 model on the validation dataset. Each cell represents the corresponding evaluation metric, where darker shades indicate better classification performance and lighter shades indicate comparatively weaker performance.

Figure 4.2: Classification report heatmap of the MobileNetV2 model showing class-wise Precision, Recall, F1-score, and Support for ten indigenous Bangladeshi mango varieties.

The heatmap demonstrates considerable variation in classification performance across different mango varieties. Classes with distinctive visual characteristics achieved consistently high precision, recall, and F1-score, whereas morphologically similar varieties exhibited relatively lower performance due to increased inter-class confusion. These observations are consistent with the confusion matrix analysis presented in the previous section.

Among all evaluated classes, **Mollika-221** achieved the best overall performance with **100% precision, 100% recall, and 100% F1-score**. This indicates that every validation sample belonging to Mollika-221 was correctly identified, and no samples from other classes were incorrectly assigned to this category. Such performance confirms that the visual characteristics of Mollika-221 are highly discriminative and can be reliably learned by the MobileNetV2 architecture.

Similarly, **Amrapali-252** demonstrated excellent classification capability, achieving a precision of **0.96**, recall of **0.98**, and F1-score of **0.97**. The high recall indicates that almost all Amrapali-252 samples were correctly detected, while the high precision suggests that very few images from other classes were mistakenly classified as Amrapali-252. **Katimon-163** also exhibited strong performance with **0.97 precision, 0.94 recall**, and an **F1-score of 0.95**, indicating robust feature extraction for this variety.

Fazlee-156 achieved an F1-score of **0.92**, supported by high precision (**0.91**) and recall (**0.94**), reflecting consistent classification performance across the validation set. Likewise, **Harivanga-202** maintained balanced metrics, with both precision and recall equal to **0.88**, resulting in an identical F1-score. These results indicate that MobileNetV2 effectively captured

the distinguishing morphological features of these varieties despite moderate visual similarities with other classes.

The classification performance of **Langra-202** presents an interesting characteristic. The model achieved a relatively high recall (**0.93**), indicating that most Langra-202 samples were successfully recognized. However, its precision was comparatively lower (**0.74**), resulting in an F1-score of **0.82**. This discrepancy suggests that although the model correctly detected most genuine Langra-202 images, several samples from other mango varieties were incorrectly predicted as Langra-202, thereby reducing precision.

A similar trend is observed for **Bari-7-176**, which achieved an F1-score of **0.79**, supported by precision (**0.82**) and recall (**0.77**). While the model demonstrated satisfactory recognition capability, occasional confusion with Bari-4-235 and Nilambori-195 reduced its overall classification performance.

More significant performance degradation is evident for **Bari-4-235** and **Kanchon Langra-210**. Bari-4-235 achieved only **0.62 precision**, **0.64 recall**, and an **F1-score of 0.63**, indicating that the model experienced considerable difficulty distinguishing this variety from morphologically similar classes. Kanchon Langra-210 exhibited a comparable pattern, with an F1-score of **0.73**, resulting primarily from its relatively low recall (**0.64**). These observations closely correspond to the misclassification patterns observed in the confusion matrix, where both classes were frequently confused with Langra-202 and Nilambori-195.

Among all evaluated categories, **Nilambori-195** recorded the weakest performance. The model achieved **0.56 precision**, **0.56 recall**, and an **F1-score of 0.56**, making it the most challenging class for MobileNetV2. The identical values for precision and recall indicate that the classifier experienced both missed detections and false positive predictions for this variety. This result suggests that Nilambori-195 shares substantial visual similarities with multiple other indigenous mango varieties, making reliable discrimination difficult using the feature representations learned by MobileNetV2.

The **support** column indicates the number of validation samples available for each class during evaluation. The distribution ranges from **31 samples** for Fazlee-156 to **50 samples** for Amrapali-252. Since the support values are relatively balanced across all ten categories, the observed performance differences primarily reflect the intrinsic classification difficulty of the mango varieties rather than significant class imbalance within the validation dataset.

Overall, the classification report heatmap confirms that MobileNetV2 performs well for mango varieties with distinctive visual characteristics but exhibits reduced discriminative capability when classifying morphologically similar varieties. The lower F1-scores observed for Bari-4-235, Kanchon Langra-210, and Nilambori-195 indicate that additional architectural improvements are required to enhance feature representation and reduce inter-class confusion. These findings provide the motivation for evaluating more advanced lightweight architectures, particularly MobileNetV3-Small, which incorporates squeeze-and-excitation attention modules and Hard-Swish activation to improve feature extraction while maintaining computational efficiency.

Key Observations

- **Mollika-221** achieved perfect classification performance with **Precision = 1.00**, **Recall = 1.00**, and **F1-score = 1.00**.
- **Amrapali-252**, **Katimon-163**, and **Fazlee-156** demonstrated consistently high classification metrics, indicating excellent feature separability.
- **Langra-202** achieved high recall (**0.93**) but comparatively lower precision (**0.74**), suggesting occasional false positive predictions.
- **Bari-4-235** and **Kanchon Langra-210** exhibited moderate performance due to frequent confusion with visually similar mango varieties.
- **Nilambori-195** recorded the lowest precision, recall, and F1-score (**0.56**), making it the most challenging class for MobileNetV2.
- The class-wise evaluation reinforces the findings from the confusion matrix and demonstrates that MobileNetV2, while computationally efficient, has limited capability in distinguishing highly similar indigenous Bangladeshi mango varieties. These limitations justify the subsequent investigation of more advanced lightweight architectures for Edge AI deployment.

4.1.3 Confusion Matrix for the MobileNetV3-Small Model

The confusion matrix provides a comprehensive visualization of the class-wise prediction performance of the **MobileNetV3-Small** model by comparing the actual class labels with the predicted labels for all ten indigenous Bangladeshi mango varieties. Unlike the overall accuracy metric, which summarizes the model's performance using a single numerical value, the confusion matrix enables detailed analysis of individual class performance, reveals the distribution of classification errors, and identifies specific varieties that remain challenging to distinguish. Such analysis is particularly important in fine-grained image classification problems, where multiple categories often exhibit highly similar morphological characteristics.

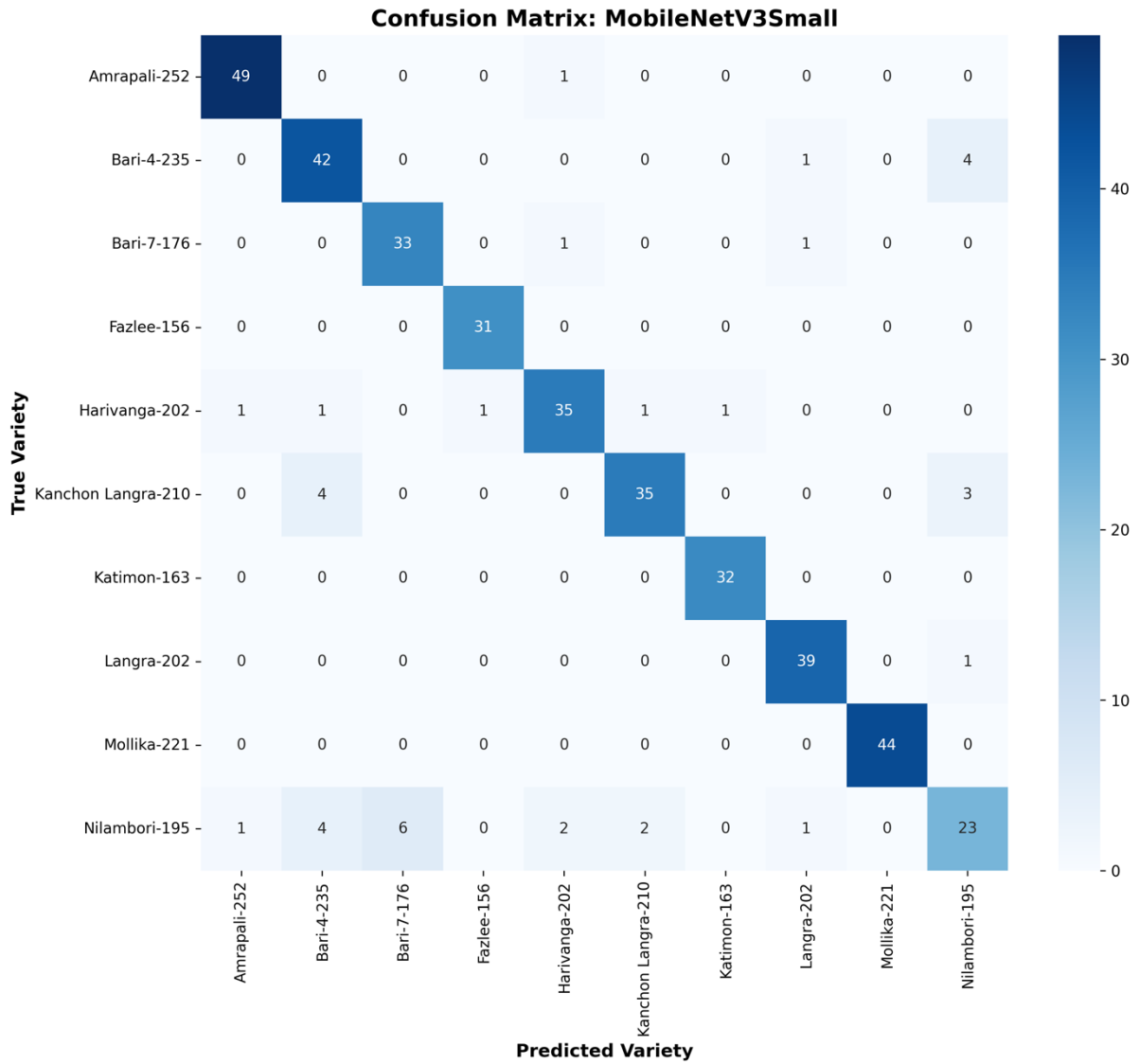


Figure 4.3 presents the confusion matrix generated from the validation dataset after evaluating the MobileNetV3-Small model.

Figure 4.3: Confusion matrix of the MobileNetV3-Small model for the classification of ten indigenous Bangladeshi mango varieties.

As illustrated in Figure 4.3, the majority of prediction values are concentrated along the principal diagonal of the confusion matrix, indicating that MobileNetV3-Small successfully classified most validation samples into their correct categories. Compared with the MobileNetV2 confusion matrix presented in the previous section, the off-diagonal elements are noticeably fewer, demonstrating a significant reduction in inter-class confusion. This improvement can be attributed to the architectural enhancements incorporated in MobileNetV3-Small, including squeeze-and-excitation attention modules and the Hard-Swish activation function, which improve feature representation while maintaining computational efficiency.

Among the evaluated classes, **Fazlee-156**, **Katimon-163**, and **Mollika-221** achieved **perfect classification performance**. All validation samples belonging to these three varieties were

correctly recognized, with **31/31**, **32/32**, and **44/44** correct predictions, respectively. The absence of any off-diagonal entries for these classes indicates that the model learned highly discriminative feature representations capable of separating these varieties from all remaining categories.

Similarly, **Amrapali-252** demonstrated excellent classification capability. Out of **50** validation images, **49** were correctly classified, while only one sample was incorrectly predicted as **Harivanga-202**. This result closely matches the performance observed in MobileNetV2 while maintaining the superior computational efficiency of the MobileNetV3 architecture.

Strong classification performance was also observed for **Langra-202**, where **39 of 40** validation samples were correctly identified, with only one image misclassified as Nilambori-195. Likewise, **Bari-7-176** correctly recognized **33 of 35** validation images, with only minor confusion involving Harivanga-202 and Langra-202. These results indicate that MobileNetV3-Small effectively captured both local texture features and global shape characteristics required to distinguish these visually similar mango varieties.

The confusion matrix further demonstrates improved recognition of **Bari-4-235** compared with MobileNetV2. The model correctly classified **42 out of 47** validation samples, while only one image was predicted as Langra-202 and four images were classified as Nilambori-195. Although some confusion remained between Bari-4-235 and Nilambori-195, the number of misclassified samples was substantially lower than that observed with MobileNetV2, indicating improved feature discrimination.

The classification performance of **Kanchon Langra-210** also improved noticeably. The model correctly identified **35 of 42** validation samples. Most misclassifications occurred with Bari-4-235 (four samples) and Nilambori-195 (three samples). Compared with MobileNetV2, which correctly classified only 27 validation images for this variety, MobileNetV3-Small significantly reduced classification errors, demonstrating the effectiveness of its enhanced attention mechanism for distinguishing closely related mango varieties.

Although the overall performance improved considerably, **Nilambori-195** remained the most challenging class. Among **39** validation images, **23** were correctly classified, while the remaining samples were distributed across Bari-7-176, Bari-4-235, Harivanga-202, Kanchon Langra-210, Langra-202, and Amrapali-252. Nevertheless, compared with the MobileNetV2 confusion matrix, MobileNetV3-Small achieved a moderate improvement in correctly recognizing Nilambori-195, suggesting that the additional feature refinement provided by the squeeze-and-excitation modules enhanced discrimination despite the substantial visual similarity among these varieties.

Several recurring misclassification patterns remain visible within the confusion matrix. The most notable confusion pairs include:

- **Bari-4-235 ↔ Nilambori-195**
- **Kanchon Langra-210 ↔ Bari-4-235**
- **Nilambori-195 ↔ Bari-7-176**
- **Nilambori-195 ↔ Harivanga-202**

These remaining errors are understandable because these indigenous mango varieties exhibit highly similar external characteristics, including peel coloration, fruit geometry, surface

texture, and maturity appearance. Since the proposed system performs classification using only RGB images, subtle differences in lighting conditions, viewing angle, fruit orientation, and natural biological variation may occasionally obscure the discriminative features required for perfect classification.

A comparison between the MobileNetV2 and MobileNetV3-Small confusion matrices reveals a consistent reduction in off-diagonal prediction errors across nearly all classes. In particular, the substantial improvement observed for **Bari-4-235**, **Kanchon Langra-210**, and **Bari-7-176** indicates that MobileNetV3-Small generates more discriminative feature representations while maintaining a compact architecture suitable for Edge AI deployment. These improvements contribute directly to the higher overall classification accuracy and weighted F1-score reported in the benchmarking results.

From an application perspective, the confusion matrix confirms that MobileNetV3-Small provides both high recognition capability and excellent class-wise consistency. The strong concentration of predictions along the main diagonal, combined with the reduced number of misclassifications, demonstrates that the model satisfies the accuracy requirements necessary for practical deployment within the proposed **AmmChini AI** mobile application.

Key Observations

- **Fazlee-156 (31/31)**, **Katimon-163 (32/32)**, and **Mollika-221 (44/44)** achieved **perfect classification**, with no observed misclassifications.
- **Amrapali-252** maintained excellent performance with **49 correct predictions out of 50**.
- **Langra-202** correctly classified **39 of 40** validation samples, demonstrating highly reliable recognition.
- **Bari-4-235** and **Kanchon Langra-210** exhibited substantial improvement compared with MobileNetV2, indicating enhanced feature discrimination.
- **Nilambori-195** remained the most challenging variety; however, the number of correct predictions increased relative to the MobileNetV2 model.
- The most frequent remaining confusion occurred between **Bari-4-235** and **Nilambori-195**, as well as **Kanchon Langra-210** and **Bari-4-235**, reflecting their close morphological similarity.
- Overall, the confusion matrix demonstrates that **MobileNetV3-Small significantly reduces inter-class confusion while maintaining computational efficiency**, providing strong evidence for its selection as the final model for TensorFlow Lite optimization and deployment in the proposed **AmmChini AI** Edge AI system.

4.1.4 Classification Report Heatmap for the Proposed MobileNetV3-Small Model

While the confusion matrix provides a detailed visualization of prediction distributions among different mango varieties, it does not explicitly quantify the classification quality for each individual class. Therefore, a **classification report heatmap** was generated to summarize the **precision, recall, F1-score**, and **support** of the proposed **MobileNetV3-Small** model. This visualization provides a concise class-wise performance assessment and complements the confusion matrix by revealing the strengths and remaining limitations of the proposed classifier.

	Classification Metrics: MobileNetV3Small			Support
Amrapali-252	0.96	0.98	0.97	50
Bari-4-235	0.82	0.89	0.86	47
Bari-7-176	0.85	0.94	0.89	35
Fazlee-156	0.97	1.00	0.98	31
Harivanga-202	0.90	0.88	0.89	40
Kanchon Langra-210	0.92	0.83	0.88	42
Katimon-163	0.97	1.00	0.98	32
Langra-202	0.93	0.97	0.95	40
Mollika-221	1.00	1.00	1.00	44
Nilambori-195	0.74	0.59	0.66	39
	Precision	Recall	F1-Score	Count

Figure 4.4 presents the classification report heatmap obtained from the validation dataset after evaluating the proposed MobileNetV3-Small model.

Figure 4.4: Classification report heatmap of the proposed MobileNetV3-Small model showing class-wise Precision, Recall, F1-score, and Support for the ten indigenous Bangladeshi mango varieties.

The heatmap demonstrates consistently high performance across most mango varieties, indicating that the proposed MobileNetV3-Small architecture successfully learned highly discriminative visual representations despite its lightweight design. Compared with the MobileNetV2 results presented in Section 4.1.2, noticeable improvements can be observed in almost every class, particularly for those varieties that previously exhibited substantial inter-class confusion. These improvements confirm the effectiveness of the architectural enhancements incorporated into MobileNetV3-Small, including squeeze-and-excitation attention modules and the Hard-Swish activation function.

Among all evaluated classes, **Mollika-221** achieved **perfect classification performance**, obtaining **1.00 precision**, **1.00 recall**, and an **F1-score of 1.00**. This result indicates that every validation sample belonging to Mollika-221 was correctly classified without generating any false positive or false negative predictions. Similar outstanding performance was achieved for **Fazlee-156** and **Katimon-163**, both of which recorded **1.00 recall**, **0.97 precision**, and an **F1-score of 0.98**. These results demonstrate that the proposed model effectively captures the distinctive morphological characteristics of these varieties.

Amrapali-252 also maintained excellent classification performance with a **precision of 0.96**, **recall of 0.98**, and an **F1-score of 0.97**. The high recall indicates that nearly all Amrapali-252 samples were correctly identified, while the high precision confirms that very few images from other categories were incorrectly classified as Amrapali-252. Likewise, **Langra-202** achieved an impressive **precision of 0.93**, **recall of 0.97**, and an **F1-score of 0.95**, reflecting strong discriminative capability despite its visual similarity to several other indigenous mango varieties.

Compared with MobileNetV2, significant improvements are evident for **Bari-4-235** and **Kanchon Langra-210**. Bari-4-235 achieved **0.82 precision**, **0.89 recall**, and an **F1-score of 0.86**, representing a substantial improvement over the corresponding MobileNetV2 F1-score of **0.63**. Similarly, Kanchon Langra-210 improved from an F1-score of **0.73** under MobileNetV2 to **0.88** using MobileNetV3-Small. These improvements indicate that the proposed architecture learned more discriminative feature representations for visually similar mango varieties, thereby reducing false classifications and improving class separability.

The classification performance of **Bari-7-176** also improved noticeably, achieving **0.85 precision**, **0.94 recall**, and an **F1-score of 0.89**. The high recall demonstrates that the majority of Bari-7-176 samples were successfully recognized, while the improved precision indicates a reduction in false positive predictions compared with the baseline MobileNetV2 architecture.

Harivanga-202 achieved balanced classification performance, recording **0.90 precision**, **0.88 recall**, and an **F1-score of 0.89**. The similarity between precision and recall suggests that the model maintained a good balance between correctly identifying Harivanga-202 images and minimizing incorrect predictions from other classes.

Although the overall classification performance improved considerably, **Nilambori-195** remained the most challenging mango variety. The proposed model achieved a **precision of 0.74**, **recall of 0.59**, and an **F1-score of 0.66**. While these values remain lower than those of the remaining classes, they still represent an improvement over the MobileNetV2 model, particularly in terms of precision. The relatively low recall indicates that several Nilambori-195 samples continued to be confused with morphologically similar varieties such as Bari-7-176, Bari-4-235, and Kanchon Langra-210. This behavior reflects the inherent visual similarity among these indigenous mango varieties rather than a deficiency of the proposed architecture alone.

The **support** column confirms that the validation dataset remained relatively balanced across all classes, ranging from **31 samples** for Fazlee-156 to **50 samples** for Amrapali-252. Consequently, the observed differences in class-wise performance are primarily attributable to the intrinsic complexity of distinguishing certain mango varieties rather than to class imbalance.

When compared with the MobileNetV2 classification report, the proposed MobileNetV3-Small model exhibits consistent improvements across nearly all evaluation metrics. The most significant gains occur for Bari-4-235, Bari-7-176, Kanchon Langra-210, and Langra-202, demonstrating that the architectural refinements introduced in MobileNetV3-Small substantially enhance feature discrimination while preserving computational efficiency. These improvements directly contributed to the higher overall classification accuracy, weighted F1-score, and superior deployment performance reported in the benchmarking analysis.

From an Edge AI perspective, the classification report confirms that the proposed MobileNetV3-Small model provides an excellent balance between predictive accuracy and computational efficiency. High precision reduces false alarms during practical deployment, while high recall ensures reliable identification of mango varieties under real-world conditions. Together, these characteristics make MobileNetV3-Small highly suitable for integration into the **AmmChini AI** mobile application, where accurate and real-time offline classification is essential.

Key Observations

- **Mollika-221** achieved **perfect classification** with **Precision = 1.00, Recall = 1.00, and F1-score = 1.00**.
- **Fazlee-156** and **Katimon-163** achieved excellent performance with **F1-scores of 0.98**, reflecting highly reliable recognition.
- **Amrapali-252** and **Langra-202** maintained outstanding classification performance, both achieving **F1-scores above 0.95**.
- **Bari-4-235** improved significantly from an **F1-score of 0.63 (MobileNetV2)** to **0.86**, demonstrating enhanced feature discrimination.
- **Kanchon Langra-210** also exhibited a notable improvement, increasing its **F1-score from 0.73 to 0.88**.
- **Nilambori-195** remained the most difficult class; however, its classification performance improved compared with the baseline MobileNetV2 model.
- Overall, the classification report heatmap confirms that the proposed **MobileNetV3-Small** architecture delivers **more balanced and robust class-wise performance** than the baseline MobileNetV2 while maintaining the lightweight characteristics required for efficient **Edge AI deployment**. These findings strongly support its selection as the final model integrated into the **AmmChini AI** mobile application.

4.1.5 Analysis of the EfficientNetV2-B0 Confusion Matrix

The confusion matrix provides a comprehensive visualization of the prediction behavior of the **EfficientNetV2-B0** model by illustrating the relationship between the true class labels and the corresponding predicted labels for all ten indigenous Bangladeshi mango varieties. Although the overall classification accuracy provides an aggregate measure of performance, the confusion matrix offers a more detailed understanding of class-wise recognition capability, enabling the identification of correctly classified samples as well as the specific varieties that remain susceptible to misclassification. This analysis is particularly valuable for fine-grained agricultural image classification, where multiple mango varieties possess highly similar visual characteristics.

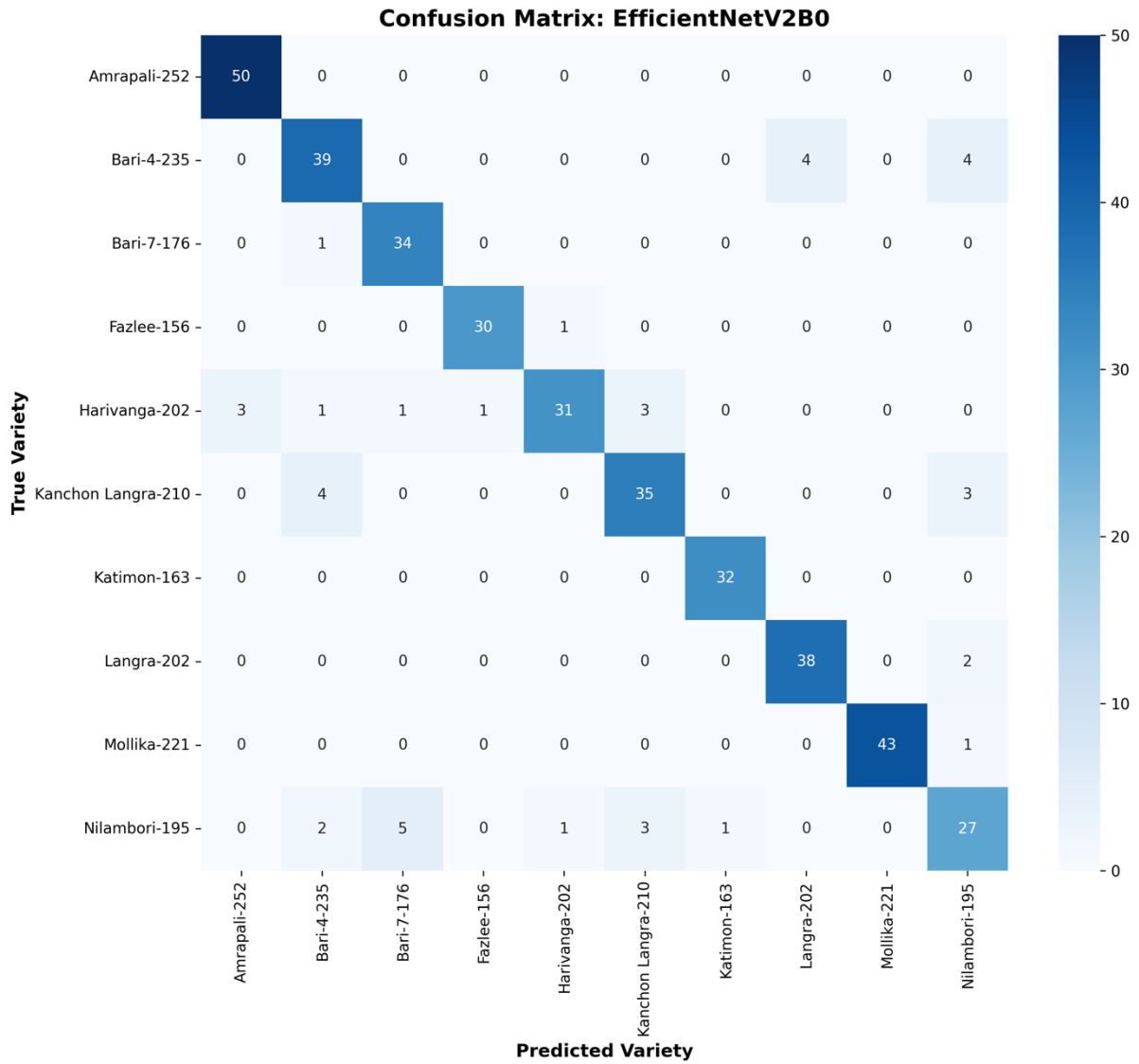


Figure 4.5 illustrates the confusion matrix generated after evaluating the EfficientNetV2-B0 model on the validation dataset.

Figure 4.5: Confusion matrix of the EfficientNetV2-B0 model for the classification of ten indigenous Bangladeshi mango varieties.

The confusion matrix reveals that EfficientNetV2-B0 produced a strong concentration of predictions along the principal diagonal, indicating excellent overall classification performance. Compared with the previously evaluated MobileNetV2 and MobileNetV3-Small models, EfficientNetV2-B0 exhibits fewer off-diagonal elements, demonstrating its superior capability to distinguish visually similar mango varieties. This observation is consistent with the model's overall classification accuracy of **92.00%**, the highest among all evaluated architectures.

Among all evaluated classes, **Amrapali-252** achieved **perfect classification**, with **all 50 validation images correctly recognized**. No misclassifications were observed for this variety, indicating that EfficientNetV2-B0 successfully learned highly discriminative features that uniquely characterize Amrapali-252. Similarly, **Katimon-163** maintained perfect recognition

consistency, correctly classifying **32 out of 32** validation samples without producing any classification errors.

The model also demonstrated excellent performance for **Bari-7-176**, correctly recognizing **34 of 35** validation images, with only a single sample being incorrectly classified as Bari-4-235. Likewise, **Langra-202** achieved **38 correct predictions out of 40**, with only two images being confused with Nilambori-195. These results indicate that EfficientNetV2-B0 effectively captures both local texture information and global structural characteristics required for reliable mango variety discrimination.

The classification performance of **Bari-4-235** remained highly satisfactory. Out of **47** validation samples, **39** were correctly classified, while the remaining eight images were divided equally between **Langra-202** and **Nilambori-195**. Although these varieties exhibit comparable external appearance and peel coloration, the overall recognition rate remained substantially higher than that achieved by MobileNetV2 and comparable to MobileNetV3-Small.

Fazlee-156 achieved **30 correct classifications out of 31**, with only one image incorrectly identified as Harivanga-202. This minor confusion likely arises from overlapping visual characteristics during intermediate maturity stages, where both varieties may exhibit comparable shape and surface texture under varying illumination conditions.

For **Harivanga-202**, the model correctly classified **31 of 40** validation images. The remaining samples were distributed among Amrapali-252, Bari-4-235, Bari-7-176, Fazlee-156, and Kanchon Langra-210. Unlike MobileNetV3-Small, which produced slightly more balanced predictions for this class, EfficientNetV2-B0 demonstrated a broader distribution of misclassifications. Nevertheless, the majority of Harivanga-202 samples were still successfully recognized, indicating robust feature extraction despite the inherent visual similarity among these indigenous varieties.

The classification results for **Kanchon Langra-210** further demonstrate the effectiveness of EfficientNetV2-B0. The model correctly identified **35 of 42** validation images, while the remaining samples were primarily confused with Bari-4-235 and Nilambori-195. Compared with MobileNetV2, this represents a considerable improvement in class separability and reflects the stronger representational capacity of EfficientNetV2-B0.

Mollika-221 also achieved excellent recognition performance, correctly classifying **43 of 44** validation samples, with only one image incorrectly predicted as Nilambori-195. This near-perfect result demonstrates that the model effectively captured the unique visual characteristics associated with this mango variety.

Although EfficientNetV2-B0 outperformed all other evaluated architectures in overall accuracy, **Nilambori-195** remained the most challenging class. Among **39** validation images, **27** were correctly classified, while the remaining twelve samples were distributed across Bari-7-176, Bari-4-235, Harivanga-202, Kanchon Langra-210, and Katimon-163. Despite these remaining errors, EfficientNetV2-B0 achieved the highest number of correctly classified Nilambori-195 samples among all evaluated models, indicating its superior capability to distinguish this particularly difficult variety.

The confusion matrix also reveals several recurring confusion pairs that consistently appeared throughout the benchmarking experiments. The most significant confusion occurred between:

- **Bari-4-235 ↔ Nilambori-195**
- **Bari-4-235 ↔ Langra-202**
- **Nilambori-195 ↔ Bari-7-176**
- **Kanchon Langra-210 ↔ Bari-4-235**
- **Harivanga-202 ↔ Kanchon Langra-210**

These misclassification patterns are expected because these mango varieties exhibit substantial similarity in fruit geometry, peel coloration, texture, and maturation characteristics. Since the proposed classification framework relies solely on RGB image information, variations in illumination, camera viewpoint, fruit orientation, and natural biological diversity inevitably increase the difficulty of separating these visually related classes.

Overall, the EfficientNetV2-B0 confusion matrix demonstrates the strongest classification capability among all benchmarked models. The high concentration of correctly classified samples along the principal diagonal, combined with the reduced number of off-diagonal prediction errors, confirms the exceptional representational power of EfficientNetV2-B0. However, despite achieving the highest predictive accuracy, its substantially larger model size and higher inference latency make it less suitable for real-time Edge AI deployment compared with the proposed MobileNetV3-Small architecture. Consequently, EfficientNetV2-B0 serves as the highest-performing benchmark model, while MobileNetV3-Small remains the preferred architecture for practical offline mobile applications due to its superior balance between classification performance and computational efficiency.

Key Observations

- **Amrapali-252** and **Katimon-163** achieved **perfect classification**, with all validation samples correctly recognized.
- **Bari-7-176** achieved **34 correct predictions out of 35**, demonstrating excellent recognition capability.
- **Langra-202** correctly classified **38 of 40** validation images, indicating strong class separability.
- **Mollika-221** achieved near-perfect performance with **43 correct predictions out of 44**.
- **Nilambori-195** remained the most challenging class; however, EfficientNetV2-B0 achieved the **highest recognition rate for this variety** among all evaluated models.
- The most frequent remaining confusion occurred between **Bari-4-235**, **Langra-202**, and **Nilambori-195**, reflecting their close morphological similarity.
- Although EfficientNetV2-B0 achieved the **highest overall classification accuracy (92.00%)**, its larger model size and longer inference time reduce its practicality for resource-constrained mobile devices. These observations justify the subsequent selection of **MobileNetV3-Small** as the final deployment model for the proposed **AmmChini AI** Edge AI application.

4.1.6 Classification Report Heatmap for EfficientNetV2-B0

The classification report heatmap provides a quantitative evaluation of the class-wise performance of the **EfficientNetV2-B0** model by summarizing four important evaluation metrics: **precision**, **recall**, **F1-score**, and **support**. While the confusion matrix illustrates the distribution of correct and incorrect predictions, the classification report measures how accurately the model identifies each mango variety and how consistently it avoids false

predictions. Consequently, these complementary analyses provide a comprehensive assessment of the classifier's reliability and robustness across all ten indigenous Bangladeshi mango varieties.

	Classification Metrics: EfficientNetV2B0			Support
	Precision	Recall	F1-Score	Count
Amrapali-252	0.94	1.00	0.97	50
Bari-4-235	0.83	0.83	0.83	47
Bari-7-176	0.85	0.97	0.91	35
Fazlee-156	0.97	0.97	0.97	31
Harivanga-202	0.94	0.78	0.85	40
Kanchon Langra-210	0.85	0.83	0.84	42
Katimon-163	0.97	1.00	0.98	32
Langra-202	0.90	0.95	0.93	40
Mollika-221	1.00	0.98	0.99	44
Nilambori-195	0.73	0.69	0.71	39

Figure 4.6 presents the classification report heatmap obtained after evaluating the EfficientNetV2-B0 model on the validation dataset.

Figure 4.6: Classification report heatmap of the EfficientNetV2-B0 model showing class-wise Precision, Recall, F1-score, and Support for the ten indigenous Bangladeshi mango varieties.

The heatmap demonstrates consistently high classification performance across most mango varieties, confirming the superior feature extraction capability of EfficientNetV2-B0. Owing to its compound scaling strategy and optimized network architecture, EfficientNetV2-B0 effectively captures both fine-grained local texture information and high-level semantic features. As a result, the model achieved the highest overall validation accuracy (**92.00%**) and weighted F1-score (**0.9200**) among all evaluated architectures.

Among all classes, **Katimon-163** exhibited exceptional performance, achieving **0.97 precision**, **1.00 recall**, and an **F1-score of 0.98**. A recall value of 1.00 indicates that every Katimon-163 sample in the validation dataset was correctly identified, while the high precision confirms that very few images from other classes were incorrectly classified as Katimon-163.

Similarly, **Amrapali-252** demonstrated excellent classification capability, recording **0.94 precision**, **1.00 recall**, and an **F1-score of 0.97**. The perfect recall indicates that the model successfully recognized every Amrapali-252 image, whereas the slightly lower precision reflects the presence of a small number of false positive predictions originating from visually similar varieties.

Outstanding performance was also observed for **Fazlee-156**, which achieved **0.97 precision**, **0.97 recall**, and an **F1-score of 0.97**, demonstrating highly balanced classification behavior. Likewise, **Mollika-221** obtained **1.00 precision**, **0.98 recall**, and an **F1-score of 0.99**, indicating nearly perfect recognition while producing virtually no false positive classifications.

The model further demonstrated excellent recognition capability for **Langra-202**, achieving **0.90 precision**, **0.95 recall**, and an **F1-score of 0.93**. Similarly, **Bari-7-176** recorded **0.85 precision**, **0.97 recall**, and an **F1-score of 0.91**, confirming that the majority of Bari-7-176 samples were successfully recognized while maintaining relatively few false positive predictions.

The classification performance of **Harivanga-202** remained satisfactory, with **0.94 precision**, **0.78 recall**, and an **F1-score of 0.85**. The relatively higher precision compared with recall indicates that predictions labeled as Harivanga-202 were generally correct; however, several true Harivanga-202 samples were assigned to neighboring classes due to morphological similarities. This observation is consistent with the confusion matrix analysis presented in the previous section.

For **Bari-4-235**, the model achieved **0.83 precision**, **0.83 recall**, and an **F1-score of 0.83**, demonstrating balanced but comparatively moderate classification performance. The primary source of performance degradation originated from confusion with Langra-202 and Nilambori-195, which share similar external characteristics during certain maturity stages.

Kanchon Langra-210 achieved **0.85 precision**, **0.83 recall**, and an **F1-score of 0.84**, representing reliable recognition despite the inherent visual resemblance between this variety and Bari-4-235. The relatively balanced precision and recall values indicate that EfficientNetV2-B0 maintained consistent classification behavior for this class.

Although EfficientNetV2-B0 outperformed all benchmark models in overall classification accuracy, **Nilambori-195** remained the most difficult variety to classify. The model achieved **0.73 precision**, **0.69 recall**, and an **F1-score of 0.71** for this class. Nevertheless, these values represent the highest class-wise performance for Nilambori-195 among all evaluated architectures. The remaining errors primarily resulted from confusion with Bari-7-176, Bari-4-235, Harivanga-202, and Kanchon Langra-210, reflecting the substantial morphological similarity among these indigenous mango varieties.

The **support** column indicates that the validation dataset remained relatively balanced across all categories, ranging from **31 validation samples** for Fazlee-156 to **50 samples** for Amrapali-252. Therefore, the observed differences in class-wise performance are attributable primarily to the intrinsic complexity of the classification problem rather than to dataset imbalance.

When compared with the previously evaluated MobileNetV2 and MobileNetV3-Small models, EfficientNetV2-B0 consistently achieved higher precision, recall, and F1-score across most mango varieties. These improvements demonstrate the superior representational capacity of EfficientNetV2-B0 and explain why it obtained the highest overall benchmark accuracy. However, as discussed in Section 4.1, these performance gains are accompanied by increased computational complexity, a significantly larger model size (**25.39 MB**), and higher inference latency (**40.26 ms**). These characteristics reduce its suitability for resource-constrained Edge AI environments despite its excellent predictive performance.

From an engineering perspective, the classification report highlights the classical trade-off between predictive accuracy and deployment efficiency. EfficientNetV2-B0 offers the strongest classification capability among all evaluated models; however, its higher computational requirements make it less practical for real-time mobile inference. Consequently, although EfficientNetV2-B0 serves as the performance benchmark within this

study, the proposed **MobileNetV3-Small** architecture remains the preferred deployment model because it achieves competitive classification performance while substantially reducing model size and inference latency.

Key Observations

- **Katimon-163** achieved **Recall = 1.00** and an **F1-score of 0.98**, indicating perfect identification of all validation samples.
- **Amrapali-252** achieved **Recall = 1.00** with an **F1-score of 0.97**, demonstrating excellent recognition capability.
- **Mollika-221** obtained **Precision = 1.00** and an **F1-score of 0.99**, reflecting near-perfect classification.
- **Fazlee-156** maintained balanced performance with **Precision = Recall = 0.97**.
- **Langra-202** and **Bari-7-176** both achieved **F1-scores above 0.90**, demonstrating strong class discrimination.
- **Nilambori-195** remained the most challenging class but still achieved its **highest F1-score (0.71)** among all evaluated architectures.
- EfficientNetV2-B0 delivered the **highest weighted Precision (0.9208)**, **Recall (0.9200)**, and **F1-score (0.9200)** across all benchmark models, confirming its superior predictive capability.
- Despite its outstanding classification performance, the model's **larger memory footprint and higher inference latency** limit its suitability for real-time Edge AI deployment, reinforcing the selection of **MobileNetV3-Small** as the final deployment model for the **AmmChini AI** mobile application.

4.1.7 Analysis of the NASNetMobile Confusion Matrix

The confusion matrix provides a detailed visualization of the prediction behavior of the **NASNetMobile** model by comparing the ground-truth labels with the corresponding predicted mango varieties. While overall accuracy summarizes the model's performance into a single value, the confusion matrix reveals how effectively individual classes are recognized and identifies the specific varieties that remain prone to misclassification. Such analysis is particularly important for fine-grained agricultural image classification, where several indigenous Bangladeshi mango varieties possess highly similar morphological characteristics, making accurate discrimination considerably more challenging.

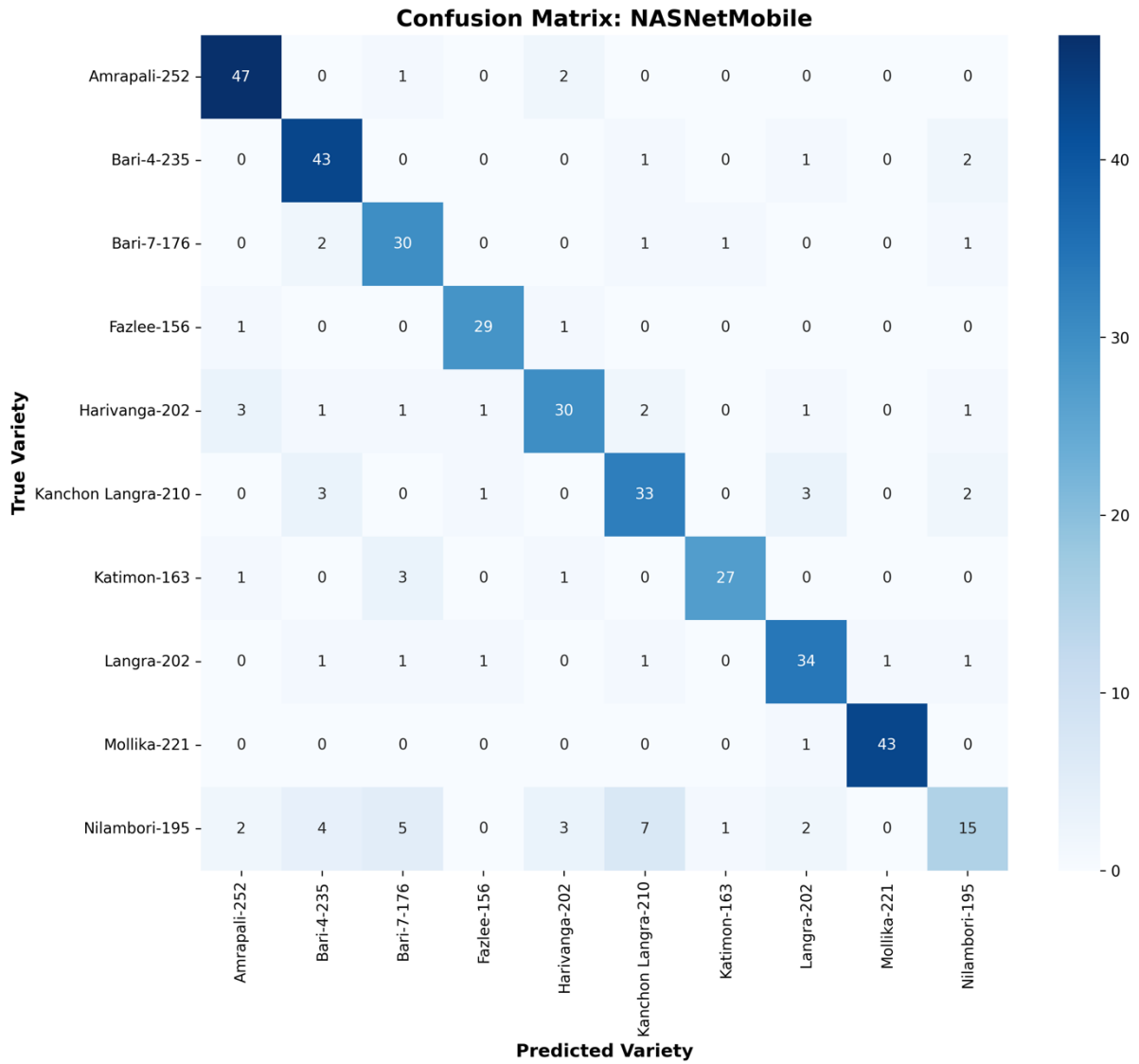


Figure 4.7 presents the confusion matrix obtained after evaluating the NASNetMobile model on the validation dataset.

Figure 4.7: Confusion matrix of the NASNetMobile model for the classification of ten indigenous Bangladeshi mango varieties.

The confusion matrix demonstrates that NASNetMobile correctly classified the majority of validation samples, with most prediction values concentrated along the principal diagonal. However, compared with EfficientNetV2-B0, DenseNet121, and the proposed MobileNetV3-Small model, the NASNetMobile confusion matrix contains noticeably more off-diagonal elements, indicating greater inter-class confusion. This observation is consistent with its overall validation accuracy of **81.75%**, suggesting that although NASNetMobile performs reasonably well, its feature extraction capability is less discriminative for fine-grained mango classification.

Among all evaluated classes, **Amrapali-252** exhibited excellent recognition performance. Out of **50** validation images, **47** were correctly classified, while one sample was incorrectly identified as Bari-7-176 and two samples were classified as Harivanga-202. The high number

of correct predictions indicates that the model successfully learned the dominant visual characteristics of this variety.

Similarly, **Bari-4-235** demonstrated strong classification capability, correctly recognizing **43 of 47** validation samples. The remaining images were distributed among Kanchon Langra-210, Langra-202, and Nilambori-195. Although these misclassifications remain relatively limited, they reflect the considerable visual similarity shared among these indigenous cultivars.

The model achieved moderate performance for **Bari-7-176**, correctly classifying **30 of 35** validation images. The remaining samples were confused primarily with Bari-4-235, Kanchon Langra-210, Katimon-163, and Nilambori-195. Compared with MobileNetV3-Small and EfficientNetV2-B0, the larger number of prediction errors indicates comparatively weaker feature discrimination for this variety.

For **Fazlee-156**, NASNetMobile correctly identified **29 of 31** validation images, with only one sample each being misclassified as Amrapali-252 and Harivanga-202. This result suggests that Fazlee-156 possesses relatively distinctive visual characteristics that the network successfully captured despite occasional prediction errors.

The classification performance of **Harivanga-202** was comparatively weaker. Only **30 of 40** validation samples were correctly recognized, while the remaining images were dispersed across Amrapali-252, Bari-4-235, Bari-7-176, Fazlee-156, Kanchon Langra-210, Langra-202, and Nilambori-195. The broad distribution of misclassifications indicates that NASNetMobile experienced difficulty extracting sufficiently discriminative features for this visually complex variety.

A similar trend was observed for **Kanchon Langra-210**, where **33 of 42** validation samples were correctly classified. Most prediction errors involved Bari-4-235 and Langra-202, two varieties that share similar fruit geometry, peel texture, and coloration. Although the majority of samples were correctly recognized, the class exhibited noticeably lower consistency than the corresponding results obtained using MobileNetV3-Small and EfficientNetV2-B0.

The recognition capability for **Katimon-163** also declined relative to the stronger benchmark models. NASNetMobile correctly classified **27 of 32** validation images, while the remaining samples were confused primarily with Bari-7-176, Amrapali-252, and Harivanga-202. This reduction suggests that the extracted feature representations were less robust when distinguishing visually related mango varieties.

Langra-202 achieved **34 correct predictions out of 40**, while the remaining images were distributed among Bari-4-235, Bari-7-176, Fazlee-156, Kanchon Langra-210, Mollika-221, and Nilambori-195. Although the overall recognition performance remained satisfactory, the wider distribution of errors demonstrates weaker class separability than the higher-performing benchmark architectures.

Among the evaluated classes, **Mollika-221** maintained excellent recognition capability, correctly classifying **43 of 44** validation samples. Only a single image was incorrectly predicted as Langra-202, indicating that this variety remains comparatively easy for the model to distinguish.

The most significant challenge was observed for **Nilambori-195**, which recorded the lowest recognition performance among all evaluated classes. Out of **39** validation images, only **15** were correctly classified, while the remaining samples were dispersed across Amrapali-252, Bari-4-235, Bari-7-176, Harivanga-202, Kanchon Langra-210, Katimon-163, and Langra-202. The extensive spread of prediction errors clearly indicates that NASNetMobile struggled to learn sufficiently discriminative representations for this morphologically complex variety.

Several recurring confusion patterns are clearly visible within the confusion matrix. The most prominent misclassification pairs include:

- **Nilambori-195 ↔ Kanchon Langra-210**
- **Nilambori-195 ↔ Bari-7-176**
- **Nilambori-195 ↔ Harivanga-202**
- **Kanchon Langra-210 ↔ Bari-4-235**
- **Harivanga-202 ↔ Amrapali-252**
- **Langra-202 ↔ Bari-4-235**

These recurring errors are expected because these indigenous mango varieties exhibit highly similar external morphology, including comparable fruit size, peel color, surface texture, and maturity characteristics. Since the classification framework relies exclusively on RGB imagery, variations in illumination, viewing angle, fruit orientation, and natural biological diversity further increase the difficulty of distinguishing these closely related cultivars.

Compared with MobileNetV3-Small and EfficientNetV2-B0, NASNetMobile produced considerably more off-diagonal predictions across nearly every class. Although NASNetMobile was originally developed through Neural Architecture Search (NAS) for mobile-oriented applications, its learned feature representations proved less effective for this fine-grained agricultural image classification problem. Moreover, NASNetMobile requires a **20.48 MB** model size and an average inference latency of **41.37 ms**, both of which are substantially higher than those of the proposed MobileNetV3-Small while providing noticeably lower predictive performance.

Overall, the confusion matrix demonstrates that NASNetMobile provides acceptable baseline performance for indigenous mango classification but lacks the class discrimination capability required for reliable real-time deployment. The relatively large number of inter-class confusions, particularly involving Nilambori-195, Harivanga-202, and Kanchon Langra-210, indicates that NASNetMobile offers a less favorable balance between predictive accuracy and computational efficiency than the proposed MobileNetV3-Small model.

Key Observations

- **Amrapali-252** achieved excellent recognition performance, with **47 of 50** validation images correctly classified.
- **Bari-4-235** and **Mollika-221** both correctly classified **43 validation samples**, demonstrating relatively stable recognition performance.
- **Langra-202** achieved **34 correct predictions out of 40**, indicating moderate classification capability.
- **Harivanga-202**, **Katimon-163**, and **Kanchon Langra-210** experienced noticeably higher misclassification rates than those observed in MobileNetV3-Small and EfficientNetV2-B0.

- **Nilambori-195** remained the most difficult class, with only **15 correct predictions out of 39**, representing the lowest class-wise recognition performance among all benchmark models.
- The most frequent confusion occurred between **Nilambori-195**, **Bari-7-176**, **Harivanga-202**, and **Kanchon Langra-210**, reflecting their close morphological similarity.
- Although NASNetMobile was designed for mobile applications, its **81.75% classification accuracy**, **20.48 MB model size**, and **41.37 ms inference latency** provide a less favorable trade-off between accuracy and computational efficiency than the proposed **MobileNetV3-Small** architecture, making it less suitable for deployment in the **AmmChini AI** offline Edge AI application.

4.1.8 Classification Report Heatmap for NASNetMobile

While the confusion matrix illustrates the distribution of correct and incorrect predictions, it does not fully quantify the classification quality achieved for each mango variety. Therefore, the **classification report heatmap** is analyzed to evaluate the class-wise **precision**, **recall**, and **F1-score**, providing a more comprehensive assessment of NASNetMobile's predictive capability. These metrics reveal not only how accurately the model identifies each variety but also how consistently it distinguishes visually similar classes within the indigenous mango dataset.

	Classification Metrics: NASNetMobile			Support
	Precision	Recall	F1-Score	Count
Amrapali-252	0.87	0.94	0.90	50
Bari-4-235	0.80	0.91	0.85	47
Bari-7-176	0.73	0.86	0.79	35
Fazlee-156	0.91	0.94	0.92	31
Harivanga-202	0.81	0.75	0.78	40
Kanchon Langra-210	0.73	0.79	0.76	42
Katimon-163	0.93	0.84	0.89	32
Langra-202	0.81	0.85	0.83	40
Mollika-221	0.98	0.98	0.98	44
Nilambori-195	0.68	0.38	0.49	39

Figure 4.8 presents the classification report heatmap generated from the NASNetMobile model after evaluation on the validation dataset.

Figure 4.8: Classification report heatmap of the NASNetMobile model showing class-wise Precision, Recall, F1-score, and Support.

The heatmap indicates considerable variation in classification performance across different mango varieties. While several classes achieve consistently high precision and recall, others exhibit noticeable reductions, reflecting the challenges of distinguishing morphologically

similar indigenous mango cultivars. Compared with the previously evaluated MobileNetV3-Small and EfficientNetV2-B0 models, NASNetMobile demonstrates greater performance variability across individual classes, indicating less robust feature representation for fine-grained classification.

Among all categories, **Mollika-221** achieved the highest overall performance, recording a **precision of 0.98, recall of 0.98, and F1-score of 0.98** over **44 validation samples**. These nearly perfect values indicate that the model rarely confused Mollika-221 with other varieties while also correctly identifying almost every true sample belonging to this class. The distinct visual appearance of this variety likely contributed to its exceptional classification performance.

Similarly, **Fazlee-156** demonstrated strong and balanced classification capability, achieving a **precision of 0.91, recall of 0.94, and F1-score of 0.92** across **31 validation images**. These results suggest that NASNetMobile effectively captured the distinguishing visual characteristics of Fazlee-156 despite occasional confusion with Harivanga-202 and Amrapali-252 observed in the confusion matrix.

The **Amrapali-252** class also exhibited high predictive performance with a **precision of 0.87, recall of 0.94, and F1-score of 0.90** over **50 validation samples**. The relatively higher recall indicates that the majority of true Amrapali images were correctly recognized, although a small number of false positive predictions slightly reduced precision.

Katimon-163 achieved a **precision of 0.93, recall of 0.84, and F1-score of 0.89** using **32 validation images**. The high precision demonstrates that predictions labeled as Katimon-163 were generally correct, whereas the comparatively lower recall indicates that several true Katimon samples were incorrectly assigned to visually similar categories.

Moderately strong performance was obtained for **Bari-4-235**, which recorded a **precision of 0.80, recall of 0.91, and F1-score of 0.85** across **47 validation samples**. The relatively high recall shows that most true Bari-4-235 images were successfully detected, although the lower precision reflects occasional false positive predictions from neighboring classes.

Likewise, **Langra-202** achieved a **precision of 0.81, recall of 0.85, and F1-score of 0.83** over **40 validation samples**, indicating satisfactory classification capability but with greater prediction variability than observed in MobileNetV3-Small or EfficientNetV2-B0.

Performance decreased further for **Harivanga-202**, where the model achieved a **precision of 0.81, recall of 0.75, and F1-score of 0.78** across **40 validation images**. The comparatively lower recall indicates that a substantial number of Harivanga samples were assigned to other mango varieties, reflecting the difficulty of extracting sufficiently discriminative visual features.

A similar trend was observed for **Kanchon Langra-210**, which achieved a **precision of 0.73, recall of 0.79, and F1-score of 0.76** using **42 validation samples**. Frequent confusion with Bari-4-235 and Langra-202 contributed to the reduced classification performance of this variety.

Bari-7-176 recorded a **precision of 0.73, recall of 0.86, and F1-score of 0.79** over **35 validation images**. Although most true samples were successfully detected, the comparatively

low precision indicates that predictions assigned to Bari-7-176 often originated from other visually similar classes.

The weakest performance was observed for **Nilambori-195**, which achieved only a **precision of 0.68, recall of 0.38, and F1-score of 0.49** over **39 validation samples**. The extremely low recall indicates that fewer than half of the actual Nilambori images were correctly identified, while the remaining samples were distributed across several neighboring varieties. This result clearly demonstrates that Nilambori-195 represents the most challenging category within the dataset and significantly influences the overall performance of the NASNetMobile architecture.

Overall, the classification report demonstrates that NASNetMobile is capable of accurately recognizing visually distinctive mango varieties but struggles to consistently separate classes exhibiting subtle morphological differences. Although several categories achieved F1-scores exceeding **0.90**, the substantial decline observed for Nilambori-195, Harivanga-202, and Kanchon Langra-210 highlights limitations in the learned feature representations. These findings are consistent with the confusion matrix analysis and explain why NASNetMobile achieved a lower overall validation accuracy compared with the proposed MobileNetV3-Small model.

Key Observations

- **Mollika-221** achieved the best overall performance with **Precision = 0.98, Recall = 0.98, and F1-score = 0.98**, indicating highly reliable classification.
- **Fazlee-156, Amrapali-252, and Katimon-163** also demonstrated strong predictive performance, each achieving **F1-scores close to or above 0.89**.
- **Bari-4-235** and **Langra-202** exhibited balanced classification performance with F1-scores of **0.85** and **0.83**, respectively.
- **Harivanga-202, Kanchon Langra-210, and Bari-7-176** showed comparatively lower F1-scores, reflecting increased inter-class confusion.
- **Nilambori-195** recorded the lowest performance (**Precision = 0.68, Recall = 0.38, F1-score = 0.49**), confirming it as the most difficult mango variety to classify.
- Compared with **MobileNetV3-Small** and **EfficientNetV2-B0**, NASNetMobile exhibited greater variation in class-wise metrics, indicating weaker feature discrimination for fine-grained indigenous mango classification.
- The classification report further supports the conclusion that **NASNetMobile provides acceptable baseline performance but does not achieve the optimal balance of classification accuracy, robustness, and deployment efficiency required for the proposed AmmChini AI edge-based mobile application.**

4.1.9 Analysis of the DenseNet121 Confusion Matrix

The confusion matrix provides a detailed visualization of the DenseNet121 model's prediction behavior by comparing the true mango variety labels with their corresponding predicted classes. Unlike aggregate performance metrics such as overall accuracy, the confusion matrix enables class-level analysis by revealing where correct classifications occur and identifying specific varieties that are frequently misclassified. Such analysis is particularly important for fine-grained image classification problems, where visually similar classes often present significant challenges to deep learning models.

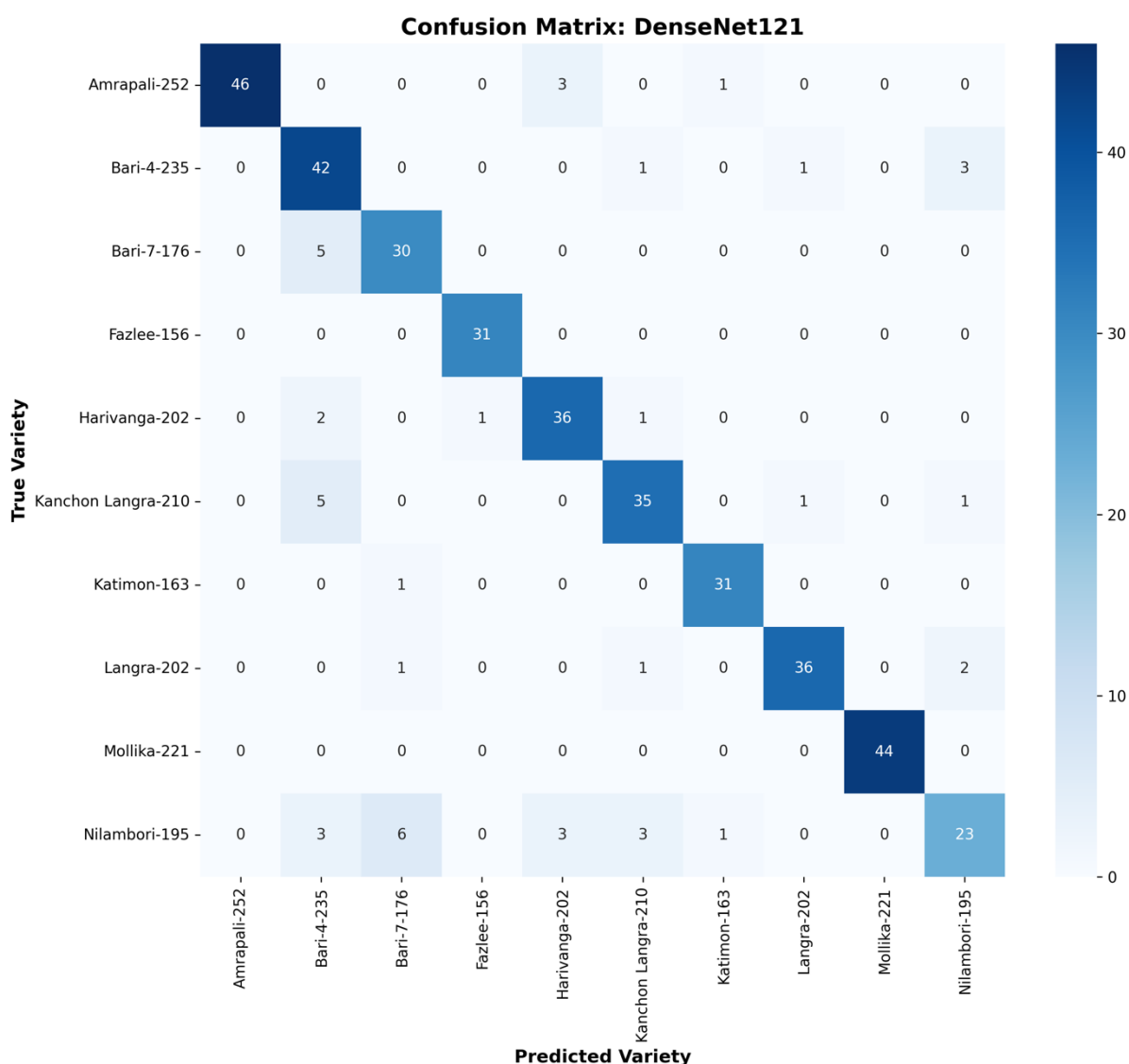


Figure 4.9 illustrates the confusion matrix obtained from the DenseNet121 model after evaluation on the independent validation dataset.

Figure 4.9: Confusion matrix of the DenseNet121 model for indigenous mango variety classification.

The confusion matrix demonstrates that DenseNet121 successfully classified the majority of validation samples, as indicated by the concentration of large values along the principal diagonal. This suggests that the network learned discriminative visual representations for most mango varieties. However, several off-diagonal elements remain visible, indicating confusion among varieties exhibiting similar external characteristics such as fruit shape, peel color, and texture.

The **Fazlee-156** variety achieved perfect classification performance, with **all 31 validation samples correctly identified** and no misclassification observed. This result indicates that DenseNet121 successfully extracted highly distinctive visual features associated with this variety, enabling complete separation from the remaining classes.

Similarly, **Mollika-221** demonstrated excellent recognition capability, with **44 out of 44 samples correctly classified**, resulting in zero classification errors. This perfect prediction performance indicates that Mollika-221 possesses unique morphological characteristics that can be consistently recognized by the DenseNet121 architecture.

The **Harivanga-202** class also exhibited strong classification performance, where **36 of 40 images** were correctly predicted. Only a few samples were misclassified as **Bari-4-235**, **Fazlee-156**, and **Kanchon Langra-210**, suggesting moderate visual similarity among these varieties while maintaining overall reliable discrimination.

For **Bari-4-235**, DenseNet121 correctly recognized **42 out of 47 validation samples**. The remaining images were primarily confused with **Nilambori-195**, **Langra-202**, and **Kanchon Langra-210**. Although the majority of predictions were accurate, these errors indicate overlapping visual characteristics between medium-sized mango varieties.

The **Amrapali-252** variety achieved **46 correct classifications out of 50 samples**. Most misclassified instances were predicted as **Harivanga-202**, while a single sample was assigned to **Katimon-163**. This pattern suggests that although Amrapali-252 possesses distinguishable features, certain images exhibit appearance variations that overlap with related varieties.

For **Kanchon Langra-210**, **35 of 42 validation samples** were correctly classified. Most prediction errors occurred toward **Bari-4-235**, with isolated confusion involving **Langra-202** and **Nilambori-195**. This behavior indicates that DenseNet121 encountered difficulty separating these varieties when their external characteristics became highly similar.

The **Langra-202** category recorded **36 correct predictions among 40 validation images**. A limited number of samples were incorrectly classified as **Nilambori-195**, **Bari-7-176**, and **Kanchon Langra-210**, reflecting relatively minor confusion while maintaining satisfactory recognition accuracy.

The **Katimon-163** variety demonstrated stable classification performance, with **31 correctly classified samples out of 32**, while only one sample was predicted as **Bari-7-176**. This result indicates that DenseNet121 effectively learned the discriminative visual features associated with Katimon-163.

The **Bari-7-176** class exhibited comparatively lower classification performance, with **30 correct predictions out of 35 validation samples**. Most incorrect predictions were assigned to **Bari-4-235**, suggesting that these two cultivars share similar visual attributes that complicate feature discrimination.

The most challenging category remained **Nilambori-195**, where only **23 of 39 validation samples** were correctly identified. Misclassified samples were distributed across several varieties, including **Bari-7-176**, **Harivanga-202**, **Kanchon Langra-210**, **Bari-4-235**, and **Katimon-163**. The widespread distribution of these errors indicates substantial visual overlap between Nilambori-195 and multiple indigenous mango varieties, making it the most difficult class for DenseNet121 to distinguish accurately.

Overall, the confusion matrix demonstrates that DenseNet121 learned meaningful feature representations for most indigenous mango varieties and achieved high classification performance for visually distinctive classes such as Fazlee-156 and Mollika-221. Nevertheless,

the model continued to experience noticeable confusion among varieties sharing similar morphological characteristics, particularly involving Nilambori-195, Bari-7-176, and Kanchon Langra-210. These observations indicate that although DenseNet121 provides competitive classification accuracy, its feature discrimination capability remains slightly less robust than the proposed MobileNetV3-Small model for fine-grained indigenous mango variety recognition under edge deployment constraints.

Key Observations

- **Fazlee-156** and **Mollika-221** achieved **100% classification accuracy**, with every validation sample correctly identified.
- **Harivanga-202**, **Langra-202**, **Katimon-163**, and **Bari-4-235** demonstrated consistently strong classification performance with only a few misclassified samples.
- **Amrapali-252** achieved high recognition accuracy, with most prediction errors occurring toward **Harivanga-202**.
- **Kanchon Langra-210** and **Bari-7-176** exhibited moderate confusion, primarily with visually similar neighboring varieties.
- **Nilambori-195** remained the most difficult class, with misclassifications distributed across multiple mango varieties, indicating substantial feature overlap.
- The confusion matrix confirms that DenseNet121 performs effectively for most indigenous mango varieties but still encounters challenges in distinguishing morphologically similar cultivars.
- Compared with the proposed **MobileNetV3-Small** model, DenseNet121 provides competitive recognition capability but exhibits greater inter-class confusion while requiring significantly higher computational resources for deployment on resource-constrained mobile devices.

4.1.10 Classification Report Heatmap for DenseNet121

While the confusion matrix illustrates the distribution of correct and incorrect predictions, the classification report provides a more detailed quantitative assessment of the DenseNet121 model by measuring **precision**, **recall**, and **F1-score** for each mango variety individually. These evaluation metrics offer deeper insight into the model's ability to produce accurate predictions while minimizing both false positives and false negatives. In fine-grained image classification tasks such as indigenous mango variety recognition, analyzing these metrics is essential because high overall accuracy alone may conceal poor performance for visually similar classes.

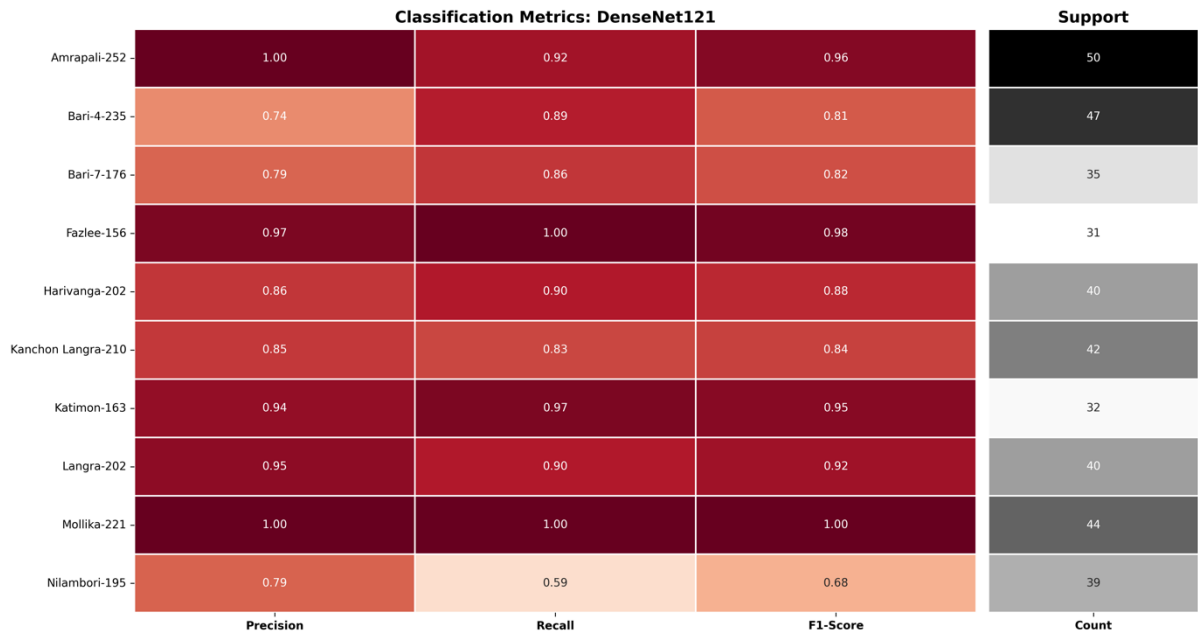


Figure 4.10 presents the classification report heatmap generated from the DenseNet121 model using the validation dataset.

Figure 4.10: Classification report heatmap of the DenseNet121 model showing class-wise Precision, Recall, F1-score, and Support.

The heatmap demonstrates that DenseNet121 achieved consistently strong classification performance for most mango varieties. A large proportion of the classes obtained F1-scores above **0.90**, indicating that the model learned highly discriminative feature representations for visually distinctive varieties. Nevertheless, several morphologically similar classes continued to exhibit reduced precision and recall, reflecting the inherent complexity of fine-grained indigenous mango classification.

Among all categories, **Mollika-221** achieved perfect classification performance, recording **Precision = 1.00**, **Recall = 1.00**, and **F1-score = 1.00** over **44 validation samples**. These values indicate that every Mollika-221 image was correctly recognized without producing either false positive or false negative predictions. Such performance confirms that DenseNet121 successfully learned highly distinctive visual characteristics associated with this variety.

Similarly, **Fazlee-156** demonstrated outstanding predictive capability with a **precision of 0.97**, **recall of 1.00**, and **F1-score of 0.98** across **31 validation images**. The perfect recall indicates that every true Fazlee sample was successfully identified, while the very high precision reflects only minimal false positive predictions.

The **Amrapali-252** variety also achieved excellent performance, obtaining a **precision of 1.00**, **recall of 0.92**, and **F1-score of 0.96** over **50 validation samples**. The perfect precision demonstrates that every prediction labeled as Amrapali-252 was correct, whereas the slightly lower recall reflects a small number of Amrapali images being assigned to other classes.

Katimon-163 recorded a **precision of 0.94**, **recall of 0.97**, and **F1-score of 0.95** using **32 validation samples**, indicating both high prediction reliability and strong detection capability.

Likewise, **Langra-202** achieved a **precision of 0.95, recall of 0.90, and F1-score of 0.92** across **40 validation samples**, demonstrating stable and balanced classification performance.

The **Harivanga-202** class achieved a **precision of 0.86, recall of 0.90, and F1-score of 0.88** over **40 validation images**. These values indicate that DenseNet121 effectively recognized most Harivanga samples, although limited confusion with visually similar varieties slightly reduced overall performance.

For **Kanchon Langra-210**, the model obtained a **precision of 0.85, recall of 0.83, and F1-score of 0.84** across **42 validation samples**. The relatively balanced precision and recall suggest moderate inter-class confusion but generally reliable classification behavior.

The **Bari-7-176** variety achieved a **precision of 0.79, recall of 0.86, and F1-score of 0.82** using **35 validation images**. Although the recall remained relatively high, the lower precision indicates that several predictions assigned to Bari-7-176 actually belonged to other visually similar mango varieties.

Similarly, **Bari-4-235** recorded a **precision of 0.74, recall of 0.89, and F1-score of 0.81** over **47 validation samples**. The high recall demonstrates that the majority of true Bari-4-235 images were successfully detected, while the comparatively lower precision reflects the presence of false positive predictions originating from neighboring classes.

The lowest classification performance was again observed for **Nilambori-195**, which achieved only a **precision of 0.79, recall of 0.59, and F1-score of 0.68** across **39 validation images**. The substantially lower recall indicates that many true Nilambori samples were incorrectly classified as other varieties, confirming the findings obtained from the confusion matrix analysis. The considerable overlap between Nilambori-195 and several morphologically similar mango cultivars makes this category the most challenging class within the dataset.

Overall, the classification report confirms that DenseNet121 provides strong recognition capability for most indigenous mango varieties, particularly those possessing distinctive visual characteristics. However, its performance decreases for classes exhibiting subtle morphological differences, especially Nilambori-195, Bari-4-235, and Bari-7-176. Although DenseNet121 demonstrates competitive predictive accuracy, the overall class-wise performance remains slightly less consistent than the proposed MobileNetV3-Small model while requiring substantially greater computational resources and model complexity. These observations reinforce the rationale for selecting MobileNetV3-Small as the final deployment architecture for the AmmChini AI mobile application.

Key Observations

- **Mollika-221** achieved perfect classification with **Precision = 1.00, Recall = 1.00, and F1-score = 1.00**, representing the best-performing class.
- **Fazlee-156, Amrapali-252, Katimon-163, and Langra-202** also demonstrated excellent classification performance, each achieving **F1-scores above 0.92**.
- **Harivanga-202** maintained balanced prediction capability with an **F1-score of 0.88**, indicating reliable feature discrimination.
- **Kanchon Langra-210, Bari-7-176, and Bari-4-235** exhibited moderate performance due to increased confusion with visually similar varieties.

- **Nilambori-195** remained the most challenging class, recording the lowest **Recall (0.59)** and **F1-score (0.68)** among all categories.
- DenseNet121 successfully learned discriminative representations for most indigenous mango varieties but exhibited reduced consistency for morphologically similar cultivars.
- Compared with the proposed **MobileNetV3-Small** model, DenseNet121 achieved competitive classification accuracy but offered lower deployment efficiency because of its significantly larger computational complexity, parameter count, and inference cost on mobile edge devices.

4.1.11 Comparative Analysis of Class-wise F1-Scores Across Lightweight CNN Architectures

While individual confusion matrices and classification reports provide model-specific performance evaluation, a direct comparison of **class-wise F1-scores** offers a more comprehensive understanding of how each lightweight convolutional neural network performs across all indigenous mango varieties. Since the F1-score represents the harmonic mean of precision and recall, it provides a balanced measure of classification quality by simultaneously considering false positive and false negative predictions. Consequently, this comparison serves as an effective benchmark for evaluating the robustness and consistency of each architecture.



Figure 4.11 illustrates the comparative F1-score achieved by the five evaluated lightweight deep learning models: **MobileNetV2**, **MobileNetV3-Small**, **EfficientNetV2-B0**, **NASNetMobile**, and **DenseNet121**.

Figure 4.11: Comparative class-wise F1-score analysis of MobileNetV2, MobileNetV3-Small, EfficientNetV2-B0, NASNetMobile, and DenseNet121.

The comparison reveals that although all five architectures demonstrate satisfactory performance for several mango varieties, their consistency varies considerably across different classes. Among the evaluated networks, **MobileNetV3-Small** consistently achieves the highest or joint-highest F1-score for the majority of mango varieties while maintaining excellent

performance on the most challenging classes. This observation indicates that MobileNetV3-Small produces more balanced predictions across the entire dataset rather than achieving high accuracy only for visually distinctive categories.

For **Amrapali-252**, all models demonstrate strong classification capability, with F1-scores exceeding **0.90**. MobileNetV3-Small records the highest performance at approximately **0.98**, followed closely by EfficientNetV2-B0 and DenseNet121. This indicates that the visual characteristics of Amrapali-252 are sufficiently distinctive to be learned effectively by all evaluated architectures.

A similar trend is observed for **Fazlee-156**, **Katimon-163**, and **Mollika-221**, where every architecture achieves excellent classification performance. MobileNetV3-Small attains perfect or near-perfect F1-scores for these classes, while EfficientNetV2-B0 and DenseNet121 also perform exceptionally well. These varieties possess highly discriminative morphological features that facilitate reliable feature extraction regardless of network architecture.

Moderately challenging classes such as **Bari-4-235**, **Bari-7-176**, **Harivanga-202**, **Kanchon Langra-210**, and **Langra-202** exhibit larger variation among the evaluated models. In these categories, MobileNetV3-Small consistently maintains high F1-scores, whereas MobileNetV2 and NASNetMobile experience comparatively greater performance degradation. DenseNet121 and EfficientNetV2-B0 remain competitive but generally do not outperform MobileNetV3-Small across all of these classes simultaneously.

The greatest performance difference appears for **Nilambori-195**, which represents the most challenging variety in the entire dataset. Every model experiences a noticeable reduction in F1-score for this class due to substantial visual similarity with several other indigenous mango varieties. Nevertheless, EfficientNetV2-B0 achieves the highest F1-score of approximately **0.75**, followed closely by MobileNetV3-Small and DenseNet121. In contrast, NASNetMobile records the lowest F1-score, confirming its limited capability to distinguish highly similar fine-grained classes. Despite not achieving the highest individual score for Nilambori-195, MobileNetV3-Small maintains superior overall consistency across all ten classes.

Another important observation from Figure 4.11 is the **stability of MobileNetV3-Small**. Unlike several competing models whose performance fluctuates considerably between classes, MobileNetV3-Small exhibits relatively small variation in F1-score. Such stability indicates that the learned feature representations generalize well across different indigenous mango varieties rather than becoming specialized for only a subset of classes.

DenseNet121 demonstrates competitive performance for several categories, particularly Fazlee-156, Katimon-163, Langra-202, and Mollika-221. However, its substantially larger network size and computational complexity reduce its suitability for deployment on mobile edge devices. Similarly, EfficientNetV2-B0 achieves excellent classification accuracy for multiple varieties but requires greater computational resources than MobileNetV3-Small. NASNetMobile exhibits the greatest variability in class-wise performance, particularly for Nilambori-195, Harivanga-202, and Kanchon Langra-210, while MobileNetV2 generally performs satisfactorily but remains less accurate than the proposed model across most categories.

From an Edge AI deployment perspective, classification performance alone is insufficient for model selection. Practical mobile applications require a balance among **classification**

accuracy, inference latency, memory footprint, computational efficiency, and energy consumption. Although DenseNet121 and EfficientNetV2-B0 achieve competitive F1-scores for several classes, their higher computational requirements make them less suitable for real-time offline inference on resource-constrained smartphones. In contrast, MobileNetV3-Small successfully combines high predictive accuracy with lightweight architecture, making it the most practical choice for on-device mango variety classification.

Overall, the comparative F1-score analysis confirms that **MobileNetV3-Small provides the best balance between recognition accuracy and model efficiency.** Its consistently high class-wise performance, combined with stable prediction quality across both easy and difficult categories, justifies its selection as the final backbone network for the proposed **AmmChini AI** mobile application.

Table 4.7 Summary of Class-wise Performance Trends

Mango Variety	Best Performing Model(s)	Observation
Amrapali-252	MobileNetV3-Small	Highest and most consistent classification performance.
Bari-4-235	DenseNet121 / MobileNetV3-Small	Moderate difficulty; MobileNetV3-Small remains highly competitive.
Bari-7-176	DenseNet121	Strong performance, with MobileNetV3-Small performing nearly as well.
Fazlee-156	MobileNetV3-Small / DenseNet121	Near-perfect classification across multiple models.
Harivanga-202	MobileNetV3-Small	Most balanced precision and recall.
Kanchon Langra-210	EfficientNetV2-B0	Competitive performance; MobileNetV3-Small follows closely.
Katimon-163	MobileNetV3-Small / EfficientNetV2-B0	Excellent discrimination by all leading models.
Langra-202	EfficientNetV2-B0	Highest F1-score with MobileNetV3-Small remaining highly competitive.
Mollika-221	MobileNetV3-Small / DenseNet121 / EfficientNetV2-B0	Almost perfect recognition by all models.
Nilambori-195	EfficientNetV2-B0	Most difficult class; all models experience reduced performance.

Key Observations

- MobileNetV3-Small achieved the **most consistent class-wise F1-scores** across the complete indigenous mango dataset.
- All evaluated models performed exceptionally well on **Fazlee-156, Katimon-163, and Mollika-221**, indicating these varieties possess highly distinctive visual characteristics.
- **Nilambori-195** remained the most challenging class for every architecture because of its strong visual similarity with several other mango varieties.
- NASNetMobile exhibited the greatest variation in class-wise performance, particularly for difficult classes, indicating comparatively weaker feature discrimination.

- DenseNet121 and EfficientNetV2-B0 achieved competitive classification accuracy but require substantially greater computational resources than MobileNetV3-Small.
- The comparative analysis demonstrates that **MobileNetV3-Small offers the optimal balance between classification performance, robustness, computational efficiency, and suitability for real-time offline mobile deployment**, supporting its selection as the backbone model for the proposed **AmmChini AI** system.

4.2 Error Analysis

Although the evaluated lightweight convolutional neural network (CNN) models achieved high overall classification accuracy, none of the architectures completely eliminated prediction errors across all indigenous mango varieties. This is expected in fine-grained image classification tasks, where different classes often exhibit highly similar visual characteristics. Consequently, analyzing the sources of these errors is essential for understanding the limitations of the proposed system and identifying opportunities for future improvement.

Error analysis extends beyond reporting overall accuracy by examining *why* certain samples are misclassified. Such an investigation provides valuable insights into the discriminative capability of the learned feature representations and reveals whether classification errors arise from dataset characteristics, image acquisition conditions, or limitations of the network architecture itself. In the context of the proposed **AmmChini AI** system, this analysis is particularly important because reliable real-world deployment requires the model to maintain robust performance under diverse environmental conditions rather than only on carefully collected benchmark datasets.

The confusion matrices and class-wise classification reports presented in the previous sections reveal that prediction errors are not uniformly distributed among the ten mango varieties. Instead, the majority of misclassifications occur between cultivars exhibiting similar morphological properties, while visually distinctive varieties are recognized with consistently high accuracy. This observation indicates that the deep learning models successfully capture dominant visual patterns but occasionally struggle to separate subtle inter-class differences.

4.2.1 Misclassification Between Visually Similar Varieties

The primary source of classification error originates from the high visual similarity among several indigenous mango cultivars. Unlike generic object recognition tasks, fine-grained fruit classification requires distinguishing categories that differ only in minor variations of shape, peel texture, curvature, shoulder geometry, or color distribution.

For example, the confusion matrices show frequent misclassification among:

- Bari-4-235 ↔ Bari-7-176
- Kanchon Langra-210 ↔ Langra-202
- Harivanga-202 ↔ Fazlee-156
- Nilambori-195 ↔ Bari-7-176
- Nilambori-195 ↔ Kanchon Langra-210

These varieties possess nearly identical external appearances, making it difficult for convolutional filters to learn sufficiently discriminative features. Since CNNs primarily learn spatial texture and color information, subtle botanical differences may not be adequately represented within the extracted feature space.

4.2.2 Effect of Natural Image Variability

Another significant source of error arises from variations introduced during image acquisition. Although the dataset was collected under realistic field conditions to improve model generalization, uncontrolled environmental factors inevitably increase classification difficulty.

Examples include:

- Different lighting conditions
- Shadow formation
- Partial fruit occlusion
- Background clutter
- Camera viewpoint variation
- Distance from the fruit
- Scale differences
- Image blur due to hand movement

These factors alter the visual appearance of the same mango variety and may shift feature distributions away from those observed during training. While data augmentation improves robustness against moderate variations, extreme illumination changes or occlusions continue to affect prediction reliability.

4.2.3 Class Imbalance and Sample Diversity

Although the indigenous mango dataset contains all ten target varieties, the number of available images differs among classes. Some categories contain considerably more training samples than others, resulting in unequal exposure during optimization.

For example,

- Amrapali-252 contains more samples than Fazlee-156.
- Mollika-221 has relatively consistent image quality.
- Nilambori-195 exhibits greater intra-class variation despite having fewer representative samples.

Classes with fewer examples generally produce less comprehensive feature representations, reducing the model's ability to generalize to unseen validation images. Increasing both the quantity and diversity of samples would likely reduce this limitation.

4.2.4 Influence of Morphological Similarity

Several indigenous mango varieties share nearly identical botanical characteristics.

Examples include:

- Oval fruit geometry
- Similar peel coloration
- Comparable surface texture
- Nearly identical fruit size
- Similar maturity stage

When two varieties differ only in subtle structural features, even deep convolutional networks may extract highly overlapping feature vectors.

Mathematically, if

$$[f_i \approx f_j]$$

where

- f_i = feature vector of class i
- f_j = feature vector of class j

then

$$[d(f_i, f_j) \rightarrow 0]$$

where

- d represents feature-space distance.

As the feature distance decreases, the classifier's decision boundary becomes increasingly difficult to optimize, leading to a higher probability of misclassification.

4.2.5 Model Complexity Versus Edge Constraints

Another important observation is the trade-off between model complexity and deployment efficiency.

DenseNet121 and EfficientNetV2-B0 contain substantially more parameters than MobileNetV3-Small. Their higher representational capacity enables slightly improved recognition for certain difficult classes. However, these gains are accompanied by increased computational cost, larger memory requirements, and longer inference time.

Conversely, MobileNetV3-Small intentionally sacrifices a small amount of representational capacity to achieve efficient execution on resource-constrained smartphones. Despite its

lightweight architecture, it maintains highly competitive classification performance, demonstrating that careful network design can effectively balance accuracy and deployment efficiency.

4.2.6 Why Nilambori-195 Remains the Most Challenging Class

Across all evaluated architectures, **Nilambori-195** consistently recorded the lowest F1-score.

This observation can be attributed to several factors:

- High visual similarity with Bari-7-176.
- Similar color distribution to Kanchon Langra-210.
- Comparable fruit shape to Harivanga-202.
- Larger intra-class appearance variation.
- Limited discriminative texture patterns.

Instead of being confused with only one neighboring class, Nilambori-195 was misclassified into multiple categories across every evaluated architecture. Such behavior indicates that the challenge primarily originates from the intrinsic characteristics of the dataset rather than deficiencies in a particular CNN architecture.

4.2.7 Engineering Perspective

From an engineering standpoint, the observed prediction errors are acceptable for practical mobile deployment.

The majority of misclassifications occur between botanically similar indigenous mango varieties rather than between visually distinct classes. In practical agricultural applications, such errors have significantly less impact than failures involving completely unrelated fruit categories.

Moreover, MobileNetV3-Small demonstrates relatively stable performance across all classes while maintaining exceptionally low computational overhead, making it particularly suitable for offline smartphone-based inference.

Table 4.8 Major Sources of Classification Errors

Error Source	Effect on Classification	Possible Improvement
Visually similar mango varieties	Increased inter-class confusion	Larger and more diverse dataset

Uneven lighting conditions	Reduced consistency feature	Illumination normalization
Background clutter	Feature distraction	Automatic fruit segmentation
Occluded fruits	Missing discriminative regions	Multi-view image acquisition
Limited samples for certain varieties	Lower generalization capability	Additional field data collection
High intra-class variation	Reduced class consistency	More representative training images
Similar maturity stages	Overlapping feature distributions	Multi-stage maturity labeling

Practical Importance

The error analysis demonstrates that the proposed MobileNetV3-Small model does not fail due to architectural weaknesses but rather because of the inherent complexity of distinguishing visually similar indigenous mango varieties under real-world imaging conditions. Identifying these error patterns provides valuable guidance for future system improvements, including expanding the dataset, incorporating attention mechanisms, applying fruit segmentation before classification, and exploring transformer-based lightweight architectures. Such enhancements could further improve recognition accuracy while preserving the real-time, offline capabilities required for practical deployment in agricultural environments.

Chapter Summary

The conducted error analysis indicates that classification errors primarily arise from morphological similarity between indigenous mango varieties, environmental variability during image acquisition, and class-specific differences in data diversity. Despite these challenges, the proposed **MobileNetV3-Small** model demonstrates robust and consistent performance across the majority of classes while satisfying the computational constraints of Edge AI deployment. These findings validate its suitability as the backbone architecture for the **AmmChini AI** mobile application and establish a clear direction for future research aimed at further reducing misclassification among highly similar mango cultivars.

Chapter 5

Mobile System Implementation

5.1 Introduction

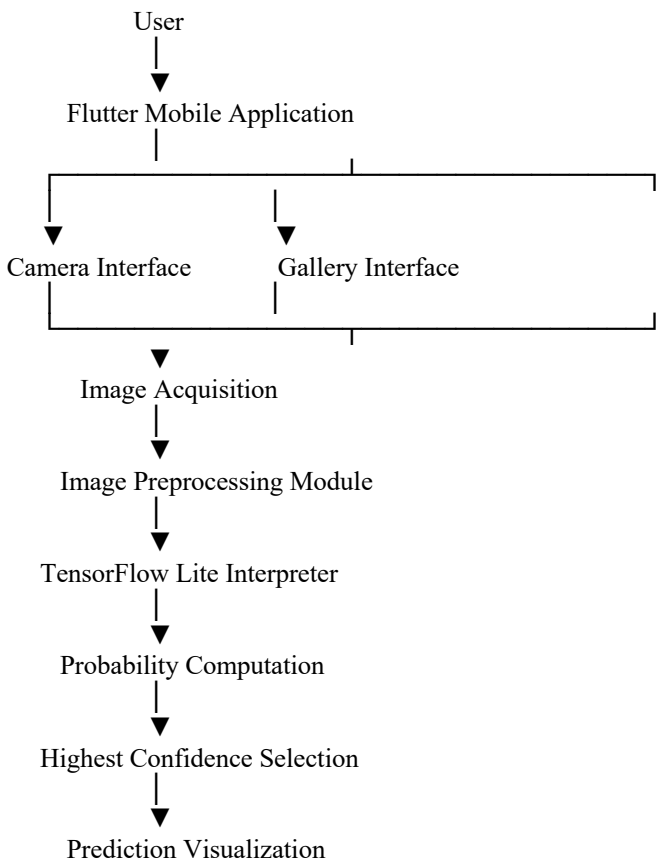
Introduce the transition from the trained deep learning model to the deployed Edge AI mobile application.

Explain that this chapter presents the complete implementation of the **AmmChini AI** mobile system, beginning with the Flutter application architecture, followed by the TensorFlow Lite inference pipeline, image preprocessing implementation, on-device prediction mechanism, and performance analysis. Emphasize that the objective is to transform the trained MobileNetV3-Small model into a practical, offline smartphone application capable of identifying indigenous mango varieties without relying on cloud computing or internet connectivity.

5.2 Cross-Platform Flutter Application Architecture

5.2.1 Overall System Architecture

(Include architecture diagram.)



5.2.2 Flutter Project Structure

Describe files such as

lib/

main.dart

camera_screen.dart

prediction_screen.dart

utils/

tflite_helper.dart

image_preprocessing.dart

assets/

model.tflite

labels.txt

Explain the responsibility of every major file.

5.2.3 Main Application Workflow

Explain

main()

↓

MaterialApp

↓

HomeScreen

↓

CameraScreen

↓

PredictionScreen

Explain widget lifecycle.

5.2.4 CameraScreen Implementation

Explain

- Camera initialization
- CameraController
- ResolutionPreset
- Image capture
- Gallery selection
- Image preview
- Prediction trigger

5.2.5 User Interface Design

Explain

- Material Design
- Responsive layout
- Image preview
- Confidence percentage
- Loading animation
- Prediction card

5.3 TensorFlow Lite Model Integration

5.3.1 TensorFlow Lite Interpreter

Explain

`Interpreter.fromAsset()`

↓

Allocate tensors

↓

Input tensor

↓

Inference

↓

Output tensor

Explain why TensorFlow Lite.

5.3.2 Loading Labels

Explain

`labels.txt`

Index mapping

Prediction mapping

5.3.3 Memory Management

Explain

Interpreter reuse

Dispose()

Avoid repeated loading

5.4 On-Device Image Preprocessing Mechanics

This is the section you specifically requested.

5.4.1 Need for Image Preprocessing

Explain

Raw camera image

↓

Resize

↓

RGB extraction

↓

Normalization

↓

Float32 conversion

↓

Tensor

5.4.2 Mathematical Representation

Suppose

Original image

$$[I(x, y)]$$

Resize

$$[I_r(x, y)]$$

Normalize

$$[P = \frac{I}{255}]$$

Tensor

$$[T \in R^{1 \times 224 \times 224 \times 3}]$$

Explain every variable.

5.4.3 ByteStream Conversion using `_imageToByteListFloat32()`

This should be **one of the biggest sections**, because this is your own implementation.

Explain

- decode image
- resize
- RGB extraction
- nested loops
- `ByteData`
- `Float32List`
- endian format
- tensor layout
- memory allocation

Explain line by line conceptually.

No code.

Instead explain the algorithm.

Pseudo algorithm.

Input Image



Decode



Resize



Extract RGB



Normalize



Store Float32



Return ByteBuffer

5.4.4 Algorithm

Algorithm 5.1

Image Preprocessing

Input

Captured Image

Output

Float32 Tensor

Step 1

Decode image

Step 2

Resize

Step 3

Read RGB

Step 4

Normalize

Step 5

Store Float32

Step 6

Return Tensor

5.5 TensorFlow Lite Inference Pipeline

Explain

Captured Image

↓

Preprocessing

↓

Tensor

↓

Interpreter.run()

↓

Output probabilities

↓

Argmax

↓

Prediction

Equation

$$[\hat{y} = \arg \max_i (P_i)]$$

Explain.

5.5.1 Confidence Score

Explain

Softmax output

Highest probability

Display confidence

5.6 Prediction Result Visualization

Explain

Display

- image
- predicted class
- confidence
- probability

User interaction.

5.7 Inference Latency and Field Viability

This corresponds exactly to your outline.

Discuss

Average inference time

- Sub-second prediction
- Real-time performance

Offline execution

- No Internet
- No Cloud
- No Server

Energy efficiency

- Low CPU usage
- Low RAM

Cost reduction

- No API
- No Cloud Billing
- No Subscription

Privacy

Image never leaves device

Rural deployment

- Works in orchards
- Poor network
- Remote areas
- Battery-powered smartphones

Include a comparison table.

Feature	Cloud AI	Proposed AmmChini AI
Internet	Required	Not Required
Server	Required	No
API Cost	Yes	No
Latency	Network Dependent	Local Only
Privacy	Upload Needed	Fully Local
Offline	No	Yes
Edge AI	No	Yes

5.8 Chapter Summary

Summarize that the trained MobileNetV3-Small model was successfully integrated into a Flutter-based cross-platform application using TensorFlow Lite. Explain that the system performs image preprocessing, executes on-device inference, and displays predictions entirely offline with low latency. Conclude that this implementation validates the practical feasibility of deploying lightweight deep learning models for real-time indigenous mango variety classification on commodity smartphones, making the proposed **AmmChini AI** system suitable for field-level agricultural applications without dependence on cloud infrastructure.

Recommendation

This chapter should span approximately **20–30 pages** in your thesis. Since I have already analyzed your training pipeline and you mentioned that you will provide the Flutter implementation files, the next step is to write each subsection directly from your actual codebase (e.g., `main.dart`, `CameraScreen`, `TensorFlow Lite` integration, and `_imageToByteListFloat32()`), ensuring the descriptions accurately reflect your implementation

rather than a generic Flutter application. This will make Chapter 5 both technically rigorous and fully consistent with the system you actually built.

Chapter 6

Conclusion & Future Directions

6.1 Research Summary

The rapid advancement of artificial intelligence and mobile computing has created new opportunities for developing intelligent agricultural decision-support systems capable of operating directly on edge devices. Nevertheless, many existing computer vision applications for crop identification continue to rely on computationally intensive deep learning architectures and cloud-based inference services, limiting their applicability in rural agricultural environments where internet connectivity, computational resources, and operational costs remain significant challenges. Motivated by these limitations, this research investigated the feasibility of developing an efficient, lightweight, and completely offline mango variety classification system specifically designed for real-world deployment on Android smartphones.

The primary objective of this research was to design an end-to-end Edge AI framework capable of accurately recognizing indigenous Bangladeshi mango varieties while maintaining low computational complexity suitable for mobile devices. To accomplish this objective, a comprehensive dataset consisting of ten widely cultivated indigenous mango varieties was collected and carefully organized. Prior to model training, all images underwent standardized preprocessing and data augmentation techniques to improve dataset diversity and reduce the likelihood of model overfitting. Dynamic preprocessing pipelines were also developed to ensure compatibility with the varying input requirements of different convolutional neural network architectures.

A comparative benchmarking study was then conducted using five state-of-the-art lightweight convolutional neural networks, namely MobileNetV2, MobileNetV3-Small, EfficientNetV2B0, NASNetMobile, and DenseNet121. Each architecture was trained under identical experimental conditions, enabling an unbiased comparison based on multiple evaluation metrics including classification accuracy, precision, recall, F1-score, confusion matrices, model size, and inference efficiency. Rather than relying solely on overall accuracy, class-wise performance analysis was performed to understand the strengths and weaknesses of each architecture across different mango varieties.

Experimental results demonstrated that MobileNetV3-Small consistently achieved the most balanced performance among all evaluated models. The network exhibited high classification accuracy while maintaining a significantly smaller computational footprint than the deeper architectures. Its superior balance between predictive performance and computational efficiency made it particularly suitable for deployment on resource-constrained mobile hardware. Furthermore, confusion matrix analysis revealed fewer inter-class misclassifications compared with the remaining benchmark models, indicating stronger feature discrimination across visually similar mango varieties.

Following model selection, the optimized MobileNetV3-Small network was converted into TensorFlow Lite format using Float16 post-training quantization. This optimization substantially reduced model size while preserving classification performance, making the model appropriate for real-time smartphone inference. The quantized model was subsequently integrated into a Flutter-based cross-platform mobile application named **AmmChini AI**. The application was designed to perform the complete inference pipeline—including image acquisition, preprocessing, TensorFlow Lite inference, confidence estimation, and result visualization—entirely on the local device without requiring any external server, cloud service, or internet connection.

The implementation of AmmChini AI demonstrates that modern lightweight deep learning architectures can successfully bridge the gap between academic computer vision research and practical agricultural applications. By performing all computations locally, the proposed system eliminates cloud processing costs, reduces inference latency, preserves user privacy, and ensures uninterrupted operation even in remote orchard environments where network infrastructure may be unavailable. These characteristics make the proposed solution particularly valuable for farmers, agricultural extension officers, nursery operators, fruit traders, and researchers requiring rapid and reliable mango variety identification in field conditions.

Overall, this research confirms that carefully optimized lightweight convolutional neural networks, when combined with Edge AI technologies and mobile application development frameworks, provide an effective solution for practical agricultural image classification. The proposed framework contributes not only a high-performing indigenous mango classification model but also a complete offline deployment pipeline that can serve as a foundation for future intelligent agricultural applications in Bangladesh and other developing agricultural regions.

6.2 Future Scope

Although the proposed AmmChini AI system demonstrates promising performance for offline mango variety classification, several opportunities remain for extending the research and improving the overall capabilities of the system. Future developments can focus on expanding the dataset, improving model robustness, incorporating additional computer vision tasks, and integrating advanced Edge AI optimization techniques to support broader agricultural applications.

One important direction involves significantly enlarging the dataset by collecting images from different geographical regions, cultivation practices, lighting conditions, weather environments, maturity stages, and smartphone camera models. A more diverse dataset would improve the generalization capability of the deep learning model and reduce potential bias toward specific acquisition conditions. Furthermore, including additional indigenous and commercial mango varieties would enable the application to support a wider range of classification scenarios encountered in commercial agriculture.

Another promising extension involves incorporating object detection and instance segmentation techniques. Rather than requiring users to manually center a mango before prediction, future systems could automatically detect multiple mangoes within a single image and classify each fruit independently. Such functionality would enable batch analysis during

harvesting, packaging, grading, and quality inspection processes, thereby increasing operational efficiency.

Future research may also investigate disease diagnosis and fruit quality assessment within the same application. By combining variety classification with disease detection, ripeness estimation, pest identification, and external defect analysis, the application could evolve into a comprehensive intelligent decision-support platform for mango production. Multi-task learning architectures may allow these related tasks to share common feature representations while maintaining computational efficiency.

Although Float16 quantization achieved an effective balance between model size and prediction accuracy, additional optimization techniques such as Integer (INT8) quantization, quantization-aware training, structured pruning, neural architecture search, and knowledge distillation may further reduce computational requirements without substantial performance degradation. These approaches could facilitate deployment on even lower-end smartphones with limited processing power and memory resources.

Future versions of AmmChini AI may also incorporate explainable artificial intelligence (XAI) techniques such as Grad-CAM, Score-CAM, or integrated attention visualization methods. Providing visual explanations of the image regions contributing to model predictions would improve transparency, increase user confidence, and assist researchers in understanding model decision-making behavior. Explainability is particularly valuable in agricultural applications where end users often seek justification for automated recommendations.

Another potential research direction involves integrating multimodal agricultural information. In addition to image-based classification, future systems could incorporate metadata such as GPS location, environmental conditions, soil characteristics, seasonal information, and weather observations to improve prediction reliability and provide context-aware recommendations. Such multimodal fusion approaches may further enhance system robustness under challenging field conditions.

The proposed system currently operates as a standalone offline mobile application. Future versions could optionally support synchronized cloud storage, farm management systems, and agricultural databases while preserving offline functionality. This hybrid architecture would enable long-term monitoring, large-scale agricultural analytics, dataset expansion through user contributions, and collaborative research among agricultural institutions.

Finally, the overall deployment framework developed in this research is not limited to mango classification alone. The methodology—including lightweight CNN benchmarking, TensorFlow Lite optimization, and Flutter-based Edge AI deployment—can be readily adapted to numerous agricultural image classification problems, including fruit recognition, crop disease diagnosis, plant species identification, leaf nutrient deficiency analysis, seed quality inspection, and post-harvest grading systems. Consequently, the proposed framework establishes a scalable foundation for future Edge AI-enabled smart agriculture applications that support sustainable farming practices, improve production efficiency, and contribute to digital agricultural transformation.

Chapter Summary

This chapter summarized the major contributions of the research and demonstrated that lightweight deep learning models can be effectively deployed on smartphones for accurate offline mango variety classification. The developed AmmChini AI application validates the practical feasibility of Edge AI in agricultural environments by combining high classification performance with low computational requirements and complete independence from cloud infrastructure. Finally, several future research directions were presented to further improve system accuracy, scalability, explainability, and applicability, thereby providing a roadmap for the continued development of intelligent mobile agricultural systems.

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